Integral Calculus

Section 1. Introduction

The first section of this book dealt with *Differential Calculus*, literally the rules for differences. We learned how to differentiate various functions and how to apply this to various modeling problems. For example, one of the exercises we had was to show that a hanging chain (catenary) satisfied the differential equation

$$\frac{d^2y}{dx^2} = \frac{w}{H}\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

A later exercise had you show that

$$y = \frac{H}{w}\left(\frac{e^{\frac{wx}{H}} + e^{-\frac{wx}{H}}}{2}\right)$$

in fact, satisfied this differential equation. All this is well and good, but you might have been wondering at the time where we pulled this function out of to differentiate. This is what integral calculus is about. In general, given a function $y = y(x)$, we can differentiate it to obtain the differential dy. Integration is the process of starting with a differential and finding the function that was differentiated to obtain it.

For example, if $y = x^2$ then $dy = 2x\,dx$. It stands to reason that if we integrate $dy = 2x\,dx$, we should end up with $y = x^2$. There is a little bit of a complication here, as differentiating $y = x^2 + c$ yields $dy = 2x\,dx$ for any constant c. It turns out that this is the only such complication; if $df = dg$, then f and g must differ by a constant. To see why, notice that if $df = dg$ then $d(f - g) = df - dg = 0$. Dividing by dx so we can think of this as a slope, we have $\frac{d(f-g)}{dx} = 0$. What kind of curve would have a slope which is constantly 0? A moment's thought will suggest that this curve must be a horizontal line Indeed, if it was not a horizontal line, then somewhere it should have a nonzero slope. Such an argument can be made rigorous, but for now, we will go with this intuitive understanding.

With this place, we will introduce some notation. The notation is motivated by the fact that integration (putting together, summing) is the inverse process of differentiation (taking a difference). With this in mind, the integral of a differential dy is denoted by

$$\int dy$$

where the "stylized S" really means a sum of differentials (differences). There are a couple of things to note about this notation. First, WE ONLY INTEGRATE DIFFERENTIALS. In other words, whatever you are integrating must be a differential. For example, $\int 2x\,dx$ makes sense, $\int 2x$ does not.

Second, notice we did not write $\int dy = y$. This is because $\int dy$ has an infinite number of possibilities. Fortunately, as we mentioned before, these all differ by constants. Thus we can write something like

$$\int 2x\,dx = x^2 + c$$

where c is an arbitrary constant. This represents all of the possible functions whose derivative is $2x$, so with that in mind $\int 2x\,dx$ is referred to as the indefinite integral or anti-derivative of $2x$. These names

come from the foundational switch from differentials to limits and derivatives, but we still want to emphasize that we integrate differentials, not functions (so the dx is not optional).

Section 2: Rules for Integration

Just as Differential Calculus is literally the rules for (infinitely small) differences, Integral Calculus is the rules for summing \up these differences. Not surprisingly, each differentiation rule has its integration rule counterpart. The following table illustrates some of these rules.

Differentiation Rule	Integration Rule
$d(constant) = 0$	$\int 0\, dx = c,$ where c is a constant
$d(f+g) = df + dg$	$\int df + dg = \int d(f+g) = f + g + c$ $= \int df + \int dg$
If a is a constant, then $d(ay) = a\, dy$	$\int ady = \int d(ay) = ay + c = a\int dy$
$d(x^{n+1}) = (n+1)x^n dx$	$\int x^n\ dx = \frac{x^{n+1}}{n+1} + c, \qquad n \neq -1$
$d(\ln x) = \frac{1}{x}dx = x^{-1}dx$	$\int x^{-1}dx = \ln\lvert x\rvert + c$
$d(e^x) = e^x dx$	$\int e^x dx = e^x + c$
$d(\sin x) = \cos x\ dx$	$\int \cos x\ dx = \sin x + c$
$d(\cos x) = -\sin x$	$\int \sin x\ dx = -\cos x + c$
$d(\tan x) = \sec^2 x\ dx$	$\int \sec^2 x\ dx = \tan x + c$
$d(\sec x) = \sec x \tan x\, dx$	$\int \sec x \tan x\, dx = \sec x + c$
$d(\arcsin x) = \frac{1}{\sqrt{1-x^2}}\ dx$	$\int \frac{1}{\sqrt{1-x^2}}\ dx = \arcsin x + c$
$d(\arctan x) = \frac{1}{1+x^2}\ dx$	$\int \frac{1}{1+x^2}\ dx = \arctan x + c$
$d(\operatorname{arcsec} x) = \frac{1}{x\sqrt{x^2-1}}\ dx$	$\int \frac{1}{x\sqrt{x^2-1}}\ dx = \operatorname{arcsec} x + c$

Table 1

This table is not exhaustive, as every differentiation performed gives rise to a corresponding integration formula. Conspicuously absent are the product rule, quotient rule, and chain rule. The analog of the quotient rule is rather limited in its applicability and typically is not used. The analogs of the others are quite useful and will be covered later. Notice that the analog of the power ruler (line 4) makes sense verbally. To differentiate x to a power, you multiply by the exponent and then subtract one from the exponent. To reverse the process, you reverse each step and reverse the order of the steps (much as you put your socks on and then your shoes and to reverse it you take off your shoes first and then your socks). Thus, you add one to the exponent and then divide by the exponent. Notice that the rule will not work if

the exponent is $n = -1$ as you cannot divide by zero. This is where the natural logarithm fills in the gap (line 5). While we are at line five, notice that we have inserted an absolute value.

$$\int x^{-1} dx = \ln|x| + c$$

We hadn't brought it up before, but the logarithm of a negative number is not a real number. It turns out that inserting an absolute value allows us to extend the natural logarithm function to negative numbers while still maintaining the same differentiation rule

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Problem

One way to define the absolute value function is as follows

$$|x| = \sqrt{x^2}$$

where this signifies the non-negative square root. For example, $|3| = \sqrt{3^2} = 3$ and $|-3| = \sqrt{(-3)^2} = 3$.

Show that with this definition of absolute value, we still obtain the formula

$$d(\ln|x|) = \frac{1}{x} dx$$

Note: You still must be careful that you do not try to write the logarithm of a negative number.

A common practice is to write $\int x^{-1} dx = \ln x + c$, even though this is only valid for $x > 0$. We will adopt this practice, and resort to an absolute value only when there is an issue about x being negative.

Problem

It looks like we broke our own rule in lines two and three in the above table of integration rules. After all,

$$\int df + \int dg = f + c_1 + g + c_2$$

where c_1 and c_2 are arbitrary constants, and

$$a \int dy = a(y + c) = ay + ac$$

Explain why what we have in the table is still legitimate.

In light of the above problem, practitioners often will perform the integration of a differential applying whatever tools are appropriate and then combining any arbitrary constants into one constant at the end. For example, consider

$$\begin{aligned}
&\int \left(5x^2 - 3\sqrt{x} + \frac{2}{1+x^2}\right)\, dx \\
&= 5\int x^2\, dx - 3\int x^{\frac{1}{2}}\, dx + 2\int \frac{1}{1+x^2}\, dx \\
&= 5\left(\frac{x^3}{3}\right) - \frac{3\left(x^{\frac{3}{2}}\right)}{\frac{3}{2}} + 2\arctan x + c \\
&= \frac{5}{3}x^3 - 2x^{\frac{3}{2}} + 2\arctan x + c
\end{aligned}$$

As with differentiation rules, when you get more comfortable with the integration rules, you will probably do these in your head. Again, developing this fluency this is why you need to practice.

Drill

Integrate the following.

We have made a good start, but we are still a bit away from solving differential equations such as

$$\frac{d^2y}{dx^2} = \frac{w}{H}\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

A start might be to let $z = \frac{dy}{dx}$, so we have

$$\frac{dz}{dx} = \frac{w}{H}\sqrt{1+z^2}$$

$$\frac{1}{\sqrt{1+z^2}}\, dz = \frac{w}{H}\, dx$$

$$\int \frac{1}{\sqrt{1+z^2}}\, dz = \int \frac{w}{H}\, dx$$

The right-hand side of this doesn't pose any problems since $\frac{w}{H}$ is a constant. However, the left-hand side is another matter, since it doesn't quite fit with anything in our table. We will learn techniques to handle such situations, but first let's look at some applications of integration which utilize what we know.

Section 3. First application – projectile motion.

We did this in the differential calc book, but I think it is worth doing again, emphasizing the integral aspect. Here is what I propose.

We actually examined projectile motion in Section 4.4 of the differential calculus portion of this book, but we will look at it again to put in our integration aspect and to use it as a springboard for adding resistance. With this in mind, we come back to the basic question we started with back then. Suppose you throw a ball straight up into the air. Of course, it will go up and come back down. For the moment, let's ignore air resistance. How high will the ball go? How long will it take the ball to hit the ground? What will be its impact velocity? Suppose you throw the ball up with twice the initial speed. Will the ball go twice as high? As we pointed out before, these were the types of questions that interested scientists like Galileo Galilei (1564-1642) who was one of the early scientists to systematically apply mathematics to science, specifically projectile motion. Through Galileo, we know that acceleration due to gravity is constant. Let's look more into projectile motion. This fits into our discussion on integral calculus since we are given information about the acceleration and must integrate twice to recover the position of the ball. Specifically, let's say that the ball is thrown upward with an initial velocity of 19.6 meters per second (roughly 44 miles per hour) and is released at a point 2 meters above the ground. How high will the ball go? When will it hit the ground? What will be the impact velocity?

To answer these questions, we must translate this problem into a mathematics problem. With this in mind, suppose the height of the ball is given by $y = y(t)$ where y is measured in meters and t is measured in seconds. Then we have the following information

$$\frac{d^2y}{dt^2} = -9.8\frac{m}{s^2}, \qquad \left.\frac{dy}{dt}\right|_{t=0} = 19.6\frac{m}{s}, \quad y(0) = 2\,m$$

Notice that the acceleration is constant (we are only looking a gravity and ignoring air resistance) and is negative (we've set up y so that up is the positive direction and down is the negative direction).

With this in mind, we want to find the following quantities.

$$y \text{ when } \frac{dy}{dt} = 0 \text{ (how high)}, \quad t \text{ when } y = 0 \text{ (when it hits the ground)}, \quad \left.\frac{dy}{dt}\right|_{y=0} \text{ (impact velocity)}$$

For convenience, let $v = \frac{dy}{dt}$ denote velocity, and $a = \frac{dv}{dt} = \frac{d^2y}{dt^2}$ denote acceleration. Starting with our constant acceleration $a = -9.8$, we need to integrate to find v and y. Once we find these, then we can answer the questions posed.

$$\begin{gathered} a = \frac{dv}{dt} = -9.8 \\ dv = -9.8\,dt \\ \int dv = \int -9.8\,dt \\ v + c_1 = -9.8t + c_2 \\ v = -9.8t + (c_2 - c_1) \end{gathered}$$

Since c_1 and c_2 are arbitrary constants, then we will rewrite $c = c_2 - c_1$ and notice that c is an arbitrary constant. From this point on, we will automatically combine arbitrary constants whenever we can without saying so. To find what c is in this case, we utilize the fact that $v(0) = 19.6$. Thus

$$\begin{gathered} 19.6 = v(0) = -9.8(0) + c \\ 19.6 = c \end{gathered}$$

and so our velocity is given by

$$v = -9.8t + 19.6$$

We now integrate this to find the ball's position.

$$\frac{dy}{dt} = -9.8t + 19.6$$
$$dy = (-9.8t + 19.6)dt$$
$$\int dy = \int (-9.8t + 19.6)dt$$
$$y = -9.8\left(\frac{t^2}{2}\right) + 19.6t + c$$

To find the constant c now, we utilize that $y(0) = 2$.

$$2 = y(0) = -4.9(0)^2 + 19.6(0) + c$$
$$2 = c$$

Thus the ball's position is given by

$$y = -4.9t^2 + 19.6t + 2$$

Once we have these three equations of motion (acceleration, velocity, position) we can answer any question about the ball's motion. In general, we suggest you find these three equations before trying to answer any questions about motion.

To find the maximum height, we set $v = 0$.

$$0 = -9.8t + 19.6$$
$$t = 2\ sec$$
$$y(2) = -4.9(2^2) + 19.6(2) + 2 = 21.6\ meters$$

so the ball will reach a height of 21.6 meters.

To find the impact velocity, we first need to find when the ball hits the ground, or t when $y = 0$.

$$0 = y = -4.9t^2 + 9.6t + 2$$
$$t = \frac{-9.6 \pm \sqrt{9.6^2 - 4(-4.9)2}}{2(-4.9)}$$
$$t = 4.1\ \ or\ \ t = -0.1$$

Since the negative time would represent time in the past, we can say that the ball will hit the ground when $t = 4.1\ sec$. Thus the impact velocity is

$$v(4.1) = -9.8(4.1) + 19.6 = -20.58\ m/s$$

The negative answer makes sense as the ball is traveling in the negative direction (given how we set up the y axis).

The following problems were done in the differential calculus section of this book, but we will include them here to get you more used to the idea and notation of integration.

Problem

Suppose we throw a ball straight up with an initial velocity of v_0 and then throw it up with an initial velocity of $2v_0$? Will the ball travel twice as high (from the release point)? Again, we will ignore air resistance.

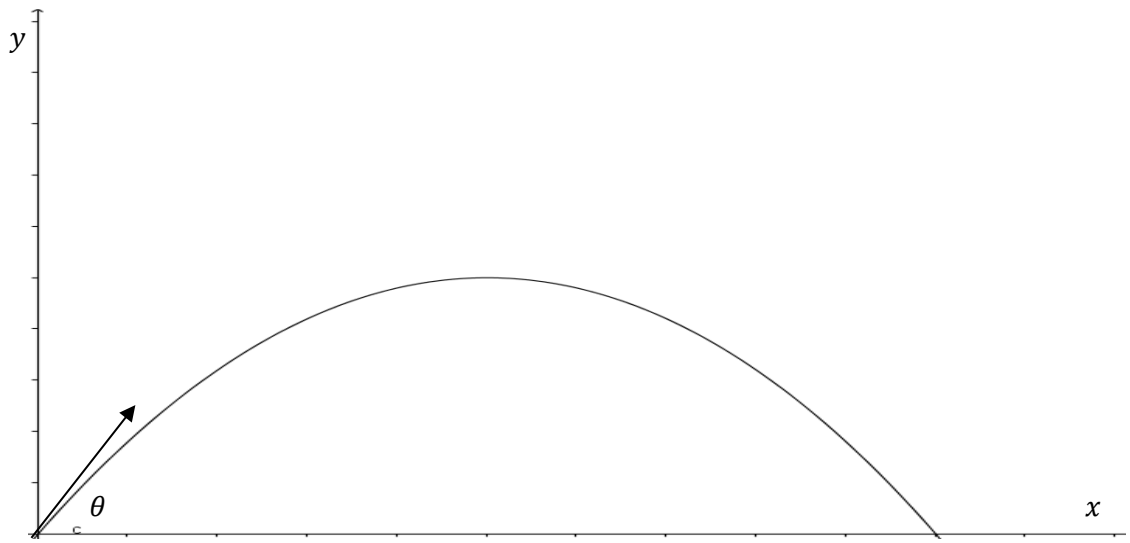
Drill

Include some with negative initial velocity

Of course, the same principles will work in two dimensions as with a single dimension. All we do is treat the horizontal and vertical directions separately. With this in mind, the horizontal acceleration would be zero.

Problem

Consider the following projectile launched from the origin, with an initial velocity of v_0 and angle of elevation θ.



Assuming that there is no air resistance and the acceleration due to gravity is $-g$

1. Show that the trajectory is given by the parabola $y = -\frac{g}{2(v_0 \cos\theta)^2}x^2 + (\tan\theta)x$.

2. Use the equation in question 1 to show that the range of the projectile is given by $\frac{v_0^2}{g}\sin(2\theta)$.

3. Show that the maximum range occurs when $\theta = \frac{\pi}{4}$.

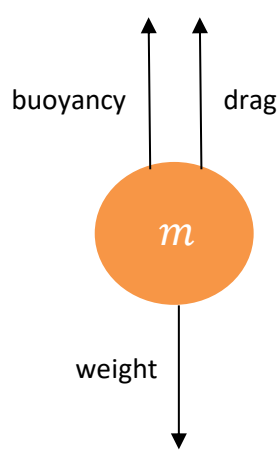
4. On Feb. 6, 1971, astronaut Alan Shepard pulled out a makeshift six-iron he smuggled on board Apollo 14 and hit two golf balls on the lunar surface, becoming the first, and only, person to play golf anywhere other than earth. Assuming that the gravity on the moon is 1/6 that of earth and that Shepard can hit the ball identically on the earth and the moon, and ignoring air resistance on the earth, would the ball go 6 times farther before it touches the ground on the moon than on the earth? Justify your answer.

Adding Resistance – This Looks Like a Job for Substitution!

Up to now, we have ignored any sort of air resistance. We will start to consider resistance (drag) in such problems. It turns out that modeling projectile motion with drag is a more complicated matter. In general, resistance in a medium is modeled to be a function of velocity. You've probably experienced this if you held your hand out a car window while it was moving. The faster the car went, the more force you felt on your hand. Likewise, the faster you moved your hand under water, the more force you felt on it. Many things affect the drag (speed, viscosity of the medium, shape and size of the object, turbulence). To keep a model simple, for a medium with a relatively high viscosity and a relatively small object we will assume that the drag is proportional to the velocity of the object. An example of this is a grain of sand falling in water. For a medium with a relatively low viscosity, such as air, and a relatively large object, the drag is assumed to be proportional to the square of the velocity. This would be the model to use for a baseball falling in the air.

To get a really accurate picture, one could use both a linear term and a quadratic term in the drag, but we will keep it simple at this point and keep these separate. We will look at the simpler model of linear drag and look at quadratic drag a little later and re-examine a ball falling in air.

Example: To look at linear drag, let's look at the case of a grain of sand of mass m descending in water. To model this, let $y(t)$ be distance the sand has fallen (so the positive y axis is pointing downward), with $y(0) = 0$ representing the surface of the water. If we draw a diagram of the sand, there are three forces we need to consider: the weight of the object, the buoyancy of the object, and the drag.



If we denote the acceleration due to gravity by g, then the weight of the sand is mg. The drag, which we assume is linear, is proportional to the velocity, so we will denote this by $-Rv$, where R is a constant and

$v = \frac{dy}{dt}$ is the velocity. Notice that this force is negative as it points up (the negative direction). The buoyancy is really just the upward force that water would apply if the sand wasn't there. That is, it is the weight of the water that the grain of sand displaces. This is just $m_w g$, where m_w is the mass of the water displaced.

Problem

a. Utilizing that $force = mass \cdot acceleration$, show that the velocity of the grain of sand must satisfy the equations

$$m\frac{dv}{dt} = (m - m_w)g - Rv$$
$$v(0) = v_0$$

Let $M = m - m_w$ and note that $M > 0$ (why?). Show that the above leads to the differential equation

$$\frac{dv}{Rv - Mg} = -\frac{1}{m}dt \qquad v(0) = v_0$$

b. Use the substitution $z = Rv - Mg$, to rewrite the above equation as

$$\frac{\frac{1}{R}dz}{z} = -\frac{1}{m}dt$$

and show that

$$v = \frac{Mg}{R} + \left(v_0 - \frac{Mg}{R}\right)e^{-\frac{R}{m}t}$$

c. Compute $\lim_{t\to\infty} v$. This will be the terminal velocity of the sand settling in the water. Notice that this should be independent of v_0. How does your answer for terminal velocity depend on $M = m - m_w$ and R? Does this make sense physically?

The above problem utilized the technique of substitution and is the integration analog of the chain rule, which itself was the technique of substitution applied to differentiation. As with differentiation, this did not actually compute the integral, but made it simpler to apply formulas and techniques with which we are already familiar. This technique of substitution is very powerful and allows us to integrate differentials which are not in our table. We already saw one example in our previous problem. For another, consider that in the table, we had $\int \sin x \; dx$ and $\int \cos x \; dx$, but we did not have $\int \tan x \; dx$. Now we have the tools to do this one.

$$\int \tan x \; dx = \int \frac{\sin x}{\cos x} \; dx$$

If we make the substitution $z = \cos x$, then $dz = -\sin x\ dx$ and so $dx = -\frac{dz}{\sin x}$.

Thus

$$\begin{aligned}\int \tan x\ dx &= \int \frac{\sin x}{\cos x}\ dx \\ &= \int \frac{\sin x}{z}\left(-\frac{dz}{\sin x}\right) = -\int \frac{dz}{z} = -\ln z + c \\ &= -\ln(\cos x) + c = \ln\left(\frac{1}{\cos x}\right) + c = \ln(\sec x) + c\end{aligned}$$

Problem

Compute $\int \cot x\ dx$ in a manner similar to the derivation above.

The beauty of a substitution is that there is no such thing as a wrong substitution. For example, in the above computation of $\int \tan x\ dx$, we could have let $z = x$, so that $dz = dx$, and we would have $\int \tan x\ dx = \int \tan z\ dz$, which though correct, would not have been very helpful. Alternatively, we could have let $u = \tan x$ so that $du = \sec^2 x\ dx$, and so

$$\int \tan x\ dx = \int \frac{u\ du}{\sec^2 x} = \int \frac{u}{1 + \tan^2 x}\ du = \int \frac{u}{1 + u^2}\ du$$

This is a doable integral, but this seems worse than the original strategy. Hey! If we originally had the integral $\int \frac{u}{1+u^2} du$, we could utilize the same substitution, to transform this into the integral $\int \tan x\ dx$, which we now know how to do. This would be entirely correct, but alas, it is doing this integral the hard way. We could compute $\int \frac{u}{1+u^2}\ du$, more easily by letting $y = 1 + u^2$, so that $dy = 2u\ du$, and so

$$\int \frac{u}{1 + u^2}\ du = \int \frac{u}{y}\left(\frac{dy}{2u}\right) = \frac{1}{2}\int \frac{dy}{y} = \frac{1}{2}\ln y + c = \ln\sqrt{1 + u^2} + c$$

Drill

Show that both of these approaches yield the same result. We ignored the negative square root. Does this mess anything up?

The lesson in all of this is that the good news is that you can do any substitution you wish and as long as it is performed correctly, it is not wrong. The bad news is that you can do any substitution you wish, and as long as it is performed correctly, it is not wrong, but it may not be useful. Sometimes they can not only lead to harder integrals, but ones where there is nothing you can really do (at least readily).

For example, consider

$$\int x^2\sqrt{x^3+1}\,dx$$

We could let $u = x^2$ so that $du = 2x\,dx$. This would lead to

$$\int x^2\sqrt{x^3+1}\,dx = \int u\sqrt{u^{\frac{3}{2}}+1}\left(\frac{du}{2x}\right) = \frac{1}{2}\int \frac{u\sqrt{u^{\frac{3}{2}}+1}}{x}\,du$$

Notice that we can "bring the 1/2 outside the integral" because it is a constant, but we cannot do the same with $\frac{1}{x}$, because it is not a constant (it depends on u) and that is not a legal move. We could rewrite $x = \sqrt{u}$, so that

$$\int x^2\sqrt{x^3+1}\,dx = \frac{1}{2}\int \frac{u\sqrt{u^{\frac{3}{2}}+1}}{\sqrt{u}}\,du = \frac{1}{2}\int \sqrt{u}\sqrt{u^{\frac{3}{2}}+1}\,du = \frac{1}{2}\int \sqrt{u^{\frac{5}{2}}+u}\;du$$

This is all correct, but leads to an integral far worse than the original, especially if you notice that x^2 is "almost" the derivative of x^3+1. It is off by a factor of 3, but as we've seen, constants can be brought outside of the integral, so this is not really a problem. So let's make $y = x^3+1$, so that $dy = 3x^2dx$. Thus

$$\int x^2\sqrt{x^3+1}\,dx = \int x^2\sqrt{y}\left(\frac{dy}{3x^2}\right) = \frac{1}{3}\int y^{\frac{1}{2}}\,dy = \frac{1}{3}\left(\frac{y^{\frac{3}{2}}}{\frac{3}{2}}\right) + c = \frac{2}{9}(x^3+1)^{\frac{3}{2}} + c$$

So what's the definitive answer on a substitution? Unfortunately, there is none. Basically, you can try anything. If it works, great! If it doesn't, don't do something incorrectly to try and force it. Try something else. Something you can look for is parts of the integrand (the thing you are integrating) which are differentials of other parts of the integrand. But there are no guarantees. This is what makes integration harder than differentiation, much as long division is harder than multiplication. But try something! As you practice, you will gain more experience, so make sure you start looking for things that you have seen before. And Practice, Practice, Practice!

Drill Substitutions.

Problem

Back in Section 4.5 the differential calculus section of this book, we had the following problem

Problem #100: Find the pattern

We know that it is not generally true that $a^b = a \cdot b$ even though there are certain exceptions, like $a = b = 1$, $a = 4$ and $b = 1/2$, or $a = b = 2$. In the same way, even though the Product Rule makes it very clear that

$$\frac{\mathrm{d}(y \cdot z)}{\mathrm{d}x} \neq \frac{\mathrm{d}y}{\mathrm{d}x} \cdot \frac{\mathrm{d}z}{\mathrm{d}x}. \tag{4.8}$$

there are certain pairs of functions which are exceptions; for which Equation (4.8) *is* true.

For example, show that for each of the following it is true that $\frac{\mathrm{d}(y \cdot z)}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}x} \cdot \frac{\mathrm{d}z}{\mathrm{d}x}$.

(a) **(i)** $y = x$ $z = \frac{1}{1-x}$ **(ii)** $y = x^2$ $z = \frac{1}{(2-x)^2}$ **(iii)** $y = x^3$ $z = (3-x)^{-3}$

(b) Find the general pattern in part (a).

(c) Those pairs of functions which fit the pattern you found in part (b) are not the only exceptional pairs. Can you find others?

Let's explore this problem in more detail. Here we will use prime notation for aesthetic reasons.

Problem: a. Suppose $f'(x) \neq f(x)$, show that if

$$(fg)'(x) = f'(x)g'(x)$$

then

$$g(x) = e^{\int (1/(1-f/f')) \, dx}$$

b. Apply this to the function $f(x) = x^n$ and compare this to the results in Problem # 60 above.

c. Suppose $f(x) = e^x$. What would $g(x)$ be in this case?

Problem

We've integrated the trigonometric functions $\sin x$, $\cos x$, $\tan x$, $\cot x$. Now we will deal with secant and cosecant.

a. Consider the following "trick"

$$\int \sec x \; dx = \int \sec x \frac{\sec x + \tan x}{\tan x + \sec x} \, dx = \int \frac{\sec^2 x + \sec x \tan x}{\tan x + \sec x} \, dx$$

Do you recognize a substitution that can be used? Use this to integrate $\sec x$

b. Use a trick similar to that in part a to compute

$$\int \csc x \; dx$$

The Opposite of the Product Rule — Integration by Parts

Substitution is a powerful tool, but it is not a "cure all." For example, consider

$$\int x \cos x \ dx$$

You can try, but there is no substitution that will really get you anywhere. Unfortunately,

$$\int x \cos x \ dx \neq \int x \, dx \int \cos x \ dx$$

so that is out of the question. (It doesn't work for differentiation, what would make you think that the integral of a product would ever equal the product of the integrals?)

Since the product rule is what we use to differentiate a product, then perhaps its analog would be of use here. In general, we have the product rule as

$$d(uv) = udv + vdu$$

Rearranging this, we have

$$udv = d(uv) - vdu$$
$$\int u \, dv = \int (d(uv) - v \, du) = \int d(uv) - \int v \, du = uv - \int v \, du$$

This formula $\int u \, dv = uv - \int v \, du$ is called the integration by parts formula and is the opposite of the product rule and is just what we need for the previous integral. Specifically, if we let $u = x$ and $dv = \cos x \, dx$, then $du = dx$ and $v = \int \cos x \ dx = \sin x$ (don't worry about the arbitrary constant here for the moment). Substituting these into our integration by parts formula, $\int u \, dv = uv - \int v \, du$, we have

$$\int x \cos x \ dx = x \sin x - \int \sin x \ dx = x \sin x — \cos x + c = x \sin x + \cos x + c$$

At this point, we should take note of a few things. First, notice that this was not a substitution; we did not end up with an integral with u's or v's in it. These were introduced just to keep track of the integration by parts formula. Along that vein, notice that we needed to use the entire integrand as opposed to a substitution where we looked for parts of the integrand which were differentials of other parts of the integrand.

Second, notice that like a substitution, the integration by parts technique does not compute the integral. Rather, it replaces that integral with, hopefully, an easier integral. This means that as long as the integration by parts technique is applied correctly, it is not wrong; it just might not be helpful. For example, in

$$\int x \cos x \ dx$$

we could have let $u = \cos x$ and $dv = x \, dx$. This would give $du = -\sin x \ dx$ and $v = x^2/2$. Substituting these into the integration by parts formula, we get

$$\int x \cos x \ dx = \frac{x^2}{2} \cos x + \frac{1}{2} \int x^2 \sin x \ dx$$

Everything is correct, but we have an integral that is worse than before. So you will need to develop some experience with using this technique as well.

Lastly, notice that we suppressed the arbitrary constant in v. We will see that this will not alter the result.

Problem

a. In performing integration by parts on $\int x \cos x \ dx$, suppose we let $u = x$ and $dv = \cos x \ dx$. As before, $du = dx$, but suppose this time that we write $v = \sin x + a$ where a is an arbitrary constant. Show that we will obtain the same integral as before.

b. Show that in general, if we let $\int dv = v + a$ in the integration by parts formula, then we will obtain the same result as before, namely $\int u \, dv = uv - \int v \, du$.

Problem

Go back and notice in our table of integrals that we had $\int e^x dx = e^x + c$, but we had no analog for $\ln x$. Perform integration by parts on

$$\int \ln x \ dx$$

with $u = \ln x$ and $dv = dx$ to compute this integral. (Note: Rather than trying to memorize this integral, just remember to use integration by parts.)

As with differentiation techniques and formulas, techniques such as integration by parts and substitution are performed in concert, utilizing whatever technique that is appropriate at the time. As an example of this consider

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Problem

Perform integration by parts and then substitution to compute

$$\int \arcsin x \ dx \quad \text{and} \quad \int \arctan x \, dx$$

[We will apply integration by parts to $\int \operatorname{arcsec} x \ dx$ a bit later, when we've developed some more techniques.]

Just as integration by parts can be used with other techniques, it can be used with itself multiple times.

Problem

Compute

$$\int x^2 e^x dx$$

by utilizing integration by parts twice.

As a word of advice, if you do perform integration by parts twice, be sure not to switch the roles of "u" and "v". This is not incorrect, but will lead you back to where you started as indicated below

$$\int u\, dv = uv - \int v\, du = uv - \left[vu - \int u\, dv\right] = \int u\, dv$$

Before we send you off to practice, we will look at one more "trick" one could do with integration by parts. We will also illustrate a way to keep track of u, dv, du, v

Example: Consider

$$\int e^x \sin x\ dx$$

If we let $u = e^x$ and $dv = \sin x\ dx$, then

$$\begin{array}{cc} u = e^x & dv = \sin x\ dx \\ du = e^x\, dx & v = -\cos x \end{array}$$

so

$$\int e^x \sin x\ dx = -e^x \cos x + \int e^x \cos x\ dx$$

We can apply integration by parts again, being careful not to "switch the roles of u and v."

$$\begin{array}{cc} u = e^x & dv = \cos x\ dx \\ du = e^x\, dx & v = \sin x \end{array}$$

so

$$\int e^x \sin x\ dx = -e^x \cos x + \int e^x \cos x\ dx$$

$$\int e^x \sin x\ dx = -e^x \cos x + \left[e^x \sin x - \int e^x \sin x\ dx\right] = -e^x \cos x + e^x \sin x - \int e^x \sin x\ dx$$

It looks like we went full circle and came back to where we started. However, if we let $I = \int e^x \sin x\ dx$, we really have the equation

$$I = -e^x \cos x + e^x \sin x - I$$

and we can solve for I (remembering to put in our arbitrary constant c).

$$2I = -e^x \cos x + e^x \sin x$$

$$\int e^x \sin x\ dx = I = \frac{-e^x \cos x + e^x \sin x}{2} + c$$

So basically, when you are integrating you can use any mathematically legal trick you wish. As long as you do that, what you are doing is not wrong. It may not help, but it is not wrong. Again experience helps you to recognize fruitful paths to take. So again, Practice, Practice, Practice!

Drill Integration by parts and sprinkle in some substitutions

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Trigonometric Substitutions!?

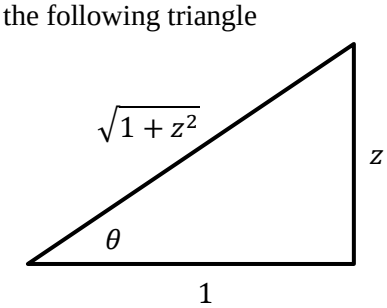
Recall that earlier in this section we encountered

$$\int \frac{1}{\sqrt{1+z^2}}\,dz$$

in trying to determine the solution to the catenary. We actually have the tools to solve this type of integral once we notice that (surprisingly!) we would like to use the substitution $z = \tan\theta$. As strange as it sounds to take a perfectly good integral and purposely insert trigonometry into it, this is just the ticket here. Since we have the trigonometric identity $\tan^2\theta + 1 = \sec^2\theta$, then this substitution is just the thing to rid us of the square root, which poses more of an issue than the trigonometry does. Specifically, if we let $z = \tan\theta$, then $dz = \sec^2\theta\ d\theta$, and so we have

$$\int \frac{1}{\sqrt{1+z^2}}\,dz = \int \frac{\sec^2\theta}{\sqrt{1+\tan^2\theta}}\,d\theta = \int \frac{\sec^2\theta}{\sec\theta}\,d\theta = \int \sec\theta\ d\theta = \ln(\tan\theta + \sec\theta) + c$$

The trick now is to put things back in terms of z. We have $\tan\theta = z$, but what about $\sec\theta$? We could certainly utilize the trigonometric identity from before, but a more general way to do this is to draw a triangle. The beauty of this is that we don't know what the angle θ is, nor do we really care. All we need is that $\tan\theta = \frac{z}{1}$. This leads to the following triangle



With this triangle, not only can we obtain the aforementioned $\tan\theta = \frac{z}{1}$ and $\sec\theta = \frac{\sqrt{1+z^2}}{1}$ which is what we need in the problem, but we could also obtain the following if we needed it.

$$\sin\theta = \frac{z}{\sqrt{1+z^2}}\ ,\cos\theta = \frac{1}{\sqrt{1+z^2}}\ ,\cot\theta = \frac{1}{z}\ ,\csc\theta = \frac{\sqrt{1+z^2}}{z}$$

We really urge you to do this (DRAW A TRIANGLE) instead of relying on memorizing trigonometric identities which actually are derived from the triangle. This gives us that the differential equation

$$\int \frac{1}{\sqrt{1+z^2}}\, dz = \int \frac{w}{H}\, dx$$

is satisfied by

$$\ln\left(z+\sqrt{1+z^2}\right) = \frac{w}{H}x + c$$

Problem

a. Utilize that $z = \frac{dy}{dx}$ for the catenary and that $z(0) = \left.\frac{dy}{dx}\right|_{x=0} = 0$ (this was the low point on the hanging chain) to show that $c = 0$ and so

$$\frac{dy}{dx} = z = \frac{1}{2}\left(e^{\frac{w}{H}x} - e^{-\frac{w}{H}x}\right)$$

b. Integrate the result in part a to obtain

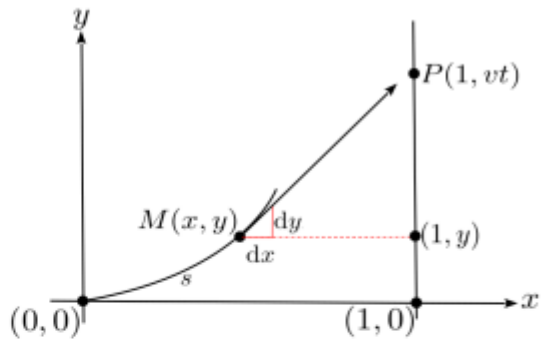
$$y = \frac{H}{w}\left(\frac{e^{\frac{wx}{H}} + e^{-\frac{wx}{H}}}{2}\right) + c$$

which is the equation of the catenary as stated earlier.

For another example where a trigonometric substitution helps our cause, recall Problem # 252 form Section 6.3 of the differential calculus portion.

Problem #252:

Suppose a rocket R travels up the line $x = 1$ at a constant speed v. As the rocket passes through the point $(1, 0)$, a missile M is fired from the origin directly at the rocket. Assume that the missile travels at a speed which is $\frac{3}{2}$ times the speed of the rocket and is always aimed directly at the rocket. At time t the missile is at the point $M(x, y)$ and the rocket is at the point $R(1, vt)$ We want to find the path the missile will follow. The diagram at the right shows the situation at time t.



(a) Find s if s denotes the length of the the missile's path at time t.

(b) Show that the missile's path must satisfy the IVP

$$\frac{dy}{dx} = \frac{\frac{2s}{3} - y}{1 - x}, \quad y(0) = 0.r$$

(**Hint:** Find a triangle which is proportional to the differential triangle shown.)

(c) Apply Euler's Method to fill in the following table.

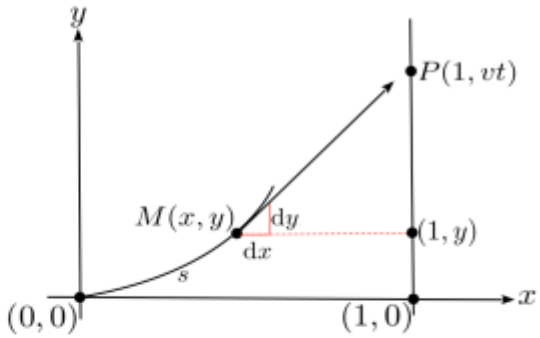
n	x_n	$y_n = y_{n-1} + \left(\frac{dy}{dx}\Big\|_{x_{n-1}}\right) \cdot (x_n - x_{n-1})$	s	n	x_n	$y_n = y_{n-1} + \left(\frac{dy}{dx}\Big\|_{x_{n-1}}\right) \cdot (x_n - x_{n-1})$	s
0	0	0	0	5	0.5		
1	0.1			6	0.6		
2	0.2			7	0.7		
3	0.3			8	0.8		
4	0.4			9	0.9		

As you can see, this problem had you utilize Euler's Method to approximate what this trajectory would be. At that point, we did not have the tools to consider an analytical solution. Later, in Section 10.2, we had the related problem where we actually provided an analytic solution.

Problem #443:

In Problem #252 we assumed that rocket R starting is traveling vertically up the line $x = 1$ at a constant speed v. When the rocket reaches the point $(1, 0)$, a missile M is fired from the origin directly at the rocket.

Assuming that the missile is always aimed directly at the rocket, and that it travels at a speed which is[4] k times the speed of the rocket ($k > 1$) we showed in Problem #252 that the curve described by the path the missile will satisfy the IVP:



$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\frac{s}{k} - y}{1 - x}, \quad y(0) = 0. \tag{10.12}$$

where s denotes the length of the the missile's path at time t.

(a) Use IVP (10.12) to show that the missile's path must satisfy the (second order) differential equation

$$(1 - x)\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} = \frac{1}{k}\frac{\mathrm{d}s}{\mathrm{d}x} = \frac{1}{k}\sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}$$

with the initial conditions: $y(0) = 0$ and $\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=0} = 0$.

(b) Show that

$$y = \frac{1}{2}\left(\frac{k}{k+1}(1-x)^{\frac{k+1}{k}} - \frac{k}{k-1}(1-x)^{\frac{k-1}{k}}\right) + \frac{k}{k^2 - 1}$$

satisfies the differential equation and initial conditions in part (a).

(c) Find how long it takes for the missile to catch the rocket for $k = 3$, $k = 2$, and $k = 1.1$. Does this make sense to you physically? Explain

This still begs the question, "Where how was this formula obtained?" A trigonometric substitution provides the key.

Problem

a. Differentiate the above equation to show that the missile's path must satisfy the differential equation

$$(1 - x)\frac{d^2 y}{dx^2} = \frac{1}{k}\frac{ds}{dx} = \frac{1}{k}\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

with the initial conditions $y(0) = 0$, $\frac{dy}{dx}(0) = 0$.

b. Let $z = \frac{dy}{dx}$, so that the differential equation in part a becomes

$$\frac{1}{\sqrt{1+z^2}}\,dz = \frac{1}{k}\cdot\frac{1}{1-x} \qquad z(0) = 0$$

Utilize the trigonometric substitution we used in the solution of the catenary to show

$$z = \frac{1}{2}\left[(1-x)^{-\frac{1}{k}} - (1-x)^{\frac{1}{k}}\right]$$

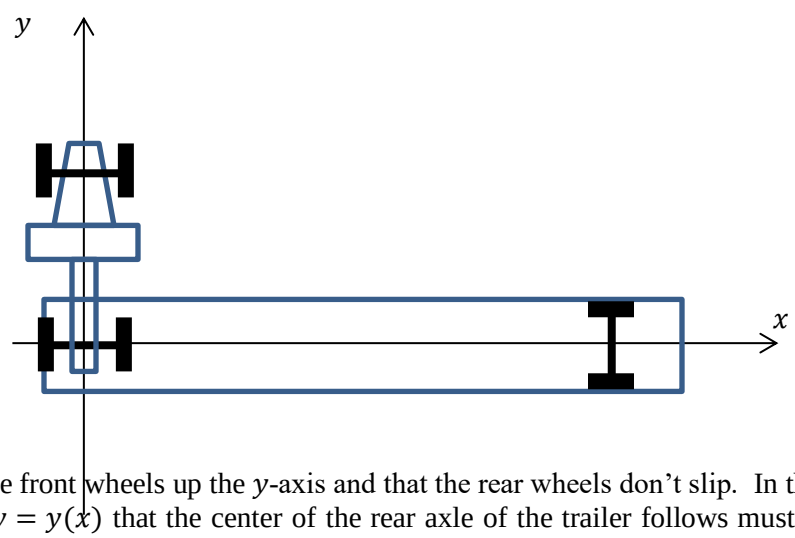
c. Use the fact that $z = \frac{dy}{dx}$ and $y(0) = 0$ to show that

$$y = \frac{k}{2}\left[\frac{(1-x)^{\frac{k+1}{k}}}{k+1} - \frac{(1-x)^{\frac{k-1}{k}}}{k-1}\right] + \frac{k}{k^2-1}$$

d. How far has the plane gone when the missile reaches it? What happens as $k \to 1$?

Of course, the tangent function is not the only trigonometric function at our disposal. Recall the Tractrix Problem (Problem # 251 and Problem # 442)

The following is the top view of a tractor-trailer. Initially, the center of the rear axle of the tractor is at the origin and the center of the rear axle of the trailer is at the point (1,0).



Suppose the tractor pulls the front wheels up the y-axis and that the rear wheels don't slip. In the problem you had to show the path $y = y(x)$ that the center of the rear axle of the trailer follows must satisfy the equations

$$\frac{dy}{dx} = -\frac{\sqrt{1-x^2}}{x}$$

$$y(1) = 0$$

Again, you utilized Euler's method to approximate the graph of this curve and then showed that a given formula solved it. Now we can see where that came from.

Problem

a. Find the equation $y = y(x)$ for the tractrix. [Hint: After you separate the variables, you can substitute $x = \sin\theta$ in your integral.]

b. Graph your answer and see how it compares to the graph plotted back in Problem 51 in Section 3.2

Again, having trigonometry involved was a better option than having the square root. Here, we utilized the trigonometric identity $\sin^2\theta + \cos^2\theta = 1$. A surprising number of integrals involve terms such as $\sqrt{1-x^2}, \sqrt{1+x^2}$, or $\sqrt{x^2-1}$. In these cases, it is often advantageous to utilize a trigonometric substitution, using the identities $\cos\theta = \sqrt{1-\sin^2\theta}, \sec\theta = \sqrt{1+\tan^2\theta}$, or $\tan\theta = \sqrt{\sec^2\theta - 1}$ to remove the square root, which typically is more a concern than the trigonometry. However, you should remember that there are other techniques at your disposal, so you need to be judicious about what you utilize. For example, consider

$$\int \frac{x}{\sqrt{1+x^2}}\,dx$$

Problem

a. Utilize a trigonometric substitution to compute the above integral.

b. Utilize a non-trigonometric substitution to compute the above integral.

c. Which did you find more appealing to do?

Problem

As you recall from our original integration table, we had

$d(\arcsin x) = \dfrac{1}{\sqrt{1-x^2}}\,dx$	$\displaystyle\int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin x + c$
$d(\arctan x) = \dfrac{1}{1+x^2}\,dx$	$\displaystyle\int \frac{1}{1+x^2}\,dx = \arctan x + c$

$d(\operatorname{arcsec} x) = \dfrac{1}{x\sqrt{x^2-1}}\,dx$	$\displaystyle\int \frac{1}{x\sqrt{x^2-1}}\,dx = \operatorname{arcsec} x + c$

In case you forget these integrals, don't worry, you can use a trigonometric substitution to obtain them.

a. Use the substitution $x = \sin\theta$ to compute
$$\int \frac{1}{\sqrt{1-x^2}}\,dx$$
b. Use the substitution $x = \tan\theta$ to compute
$$\int \frac{1}{1+x^2}\,dx$$
c. Use the substitution $x = \sec\theta$ to compute
$$\int \frac{1}{x\sqrt{x^2-1}}\,dx$$

As we mentioned, trigonometry can be used for integrals involving $\sqrt{1-x^2}, \sqrt{1+x^2}$, or $\sqrt{x^2-1}$. What if we had

$$\int \frac{dx}{\sqrt{4+x^2}}$$

Rather than learning a set of rules, we can utilize some algebra to transform it into something we know. The trick is that we really want a 1 where the 4 is. We can't just change it, but we can factor it out.

$$\int \frac{dx}{\sqrt{4+x^2}} = \int \frac{dx}{\sqrt{4\left(1+\frac{x^2}{4}\right)}} = \frac{1}{2}\int \frac{dx}{\sqrt{1+\left(\frac{x}{2}\right)^2}}$$

Now we can let $\frac{x}{2} = \tan\theta$, and proceed as before.

Similarly,

$$\int \frac{dx}{\sqrt{5-x^2}} = \int \frac{dx}{\sqrt{5\left(1-\frac{x^2}{5}\right)}} = \frac{1}{\sqrt{5}}\int \frac{dx}{\sqrt{1-\left(\frac{x}{\sqrt{5}}\right)^2}}$$

and we can let $\frac{x}{\sqrt{5}} = \sin\theta$.

We can even handle something like

$$\int \frac{dx}{\sqrt{x^2-6x}}$$

by completing the square first.

$$\int \frac{dx}{\sqrt{x^2-6x}} = \int \frac{dx}{\sqrt{x^2-6x+9-9}} = \int \frac{dx}{\sqrt{(x-3)^2-9}} = \frac{1}{3}\int \frac{dx}{\sqrt{\left(\frac{x-3}{3}\right)^2-1}}$$

Now we can let $\frac{x-3}{3} = \sec\theta$. As long as the expression inside the radical is a quadratic, we can manipulate it algebraically to produce something of the form $\sqrt{1+\left(\frac{x-b}{a}\right)^2}$, $\sqrt{1-\left(\frac{x-b}{a}\right)^2}$, or $\sqrt{\left(\frac{x-b}{a}\right)^2-1}$. From there, we can use the appropriate trigonometric substitution.

Earlier we computed terminal velocity when considering linear drag. Now we will compare this to terminal velocity when the drag is proportional to the square of the velocity. An example of this would be a ball of mass M falling in air. (We can include buoyancy again, but it is typically negligent so we will discount it). Again, to model this, let $y(t)$ be distance the ball has fallen (so the positive y axis is pointing downward), with $y(0) = 0$ representing the initial position of the ball. In a similar analysis to before, we denote the acceleration due to gravity by g, so the weight of the ball is Mg. The drag, which we assume is now quadratic, is proportional to the square of the velocity, so we will denote this by $-Rv^2$, where R is a constant and $v = \frac{dy}{dt}$ is the velocity. For simplicity, we will assume $v(0) = 0$.

Problem

a. Utilizing that $force = mass \cdot acceleration$, show that the velocity of the ball must satisfy the equations

$$M\frac{dv}{dt} = Mg - Rv^2 \qquad v(0) = 0$$

Show that the above leads to the differential equation

$$\frac{dv}{1-\frac{R}{Mg}v^2} = gdt \qquad v(0) = 0$$

b. Use the substitution $\sqrt{\frac{R}{Mg}}v = \sin\theta$ to show that

$$v = \sqrt{\frac{Mg}{R}}\left[\frac{1-e^{-2gt}}{1+e^{-2gt}}\right]$$

c. Compute $\lim_{t\to\infty} v$. Compare this terminal velocity to what we obtained before for linear drag.

Drill Problems

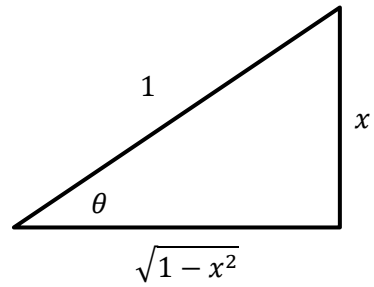
The trigonometric substitutions you've dealt with so far have dealt with square roots. How about something like

$$\int \frac{dx}{(1-x^2)^{\frac{3}{2}}}$$

If we utilize the substitution $x = \sin\theta$, then $dx = \cos\theta\, d\theta$, so

$$\int \frac{dx}{(1-x^2)^{\frac{3}{2}}} = \int \frac{\cos\theta}{\left(\sqrt{1-\sin^2\theta}\right)^3}\, d\theta = \int \frac{\cos\theta}{\cos^3\theta}\, d\theta = \int \sec^2\theta\ d\theta = \tan\theta + c$$

Utilizing the fact that $\sin\theta = \frac{x}{1}$, we get the triangle



so

$$\int \frac{dx}{(1-x^2)^{\frac{3}{2}}} = \tan\theta + c = \frac{x}{\sqrt{1-x^2}} + c$$

Fortunately, we were still able to handle this, but what if we started with the integral

$$\int \frac{dx}{(x^2-1)^{\frac{3}{2}}}$$

Utilizing the substitution $x = \sec\theta$ would lead to the integral

$$\int \frac{\sec\theta}{\tan^2\theta}\, d\theta$$

What do we do with this one? We can rewrite this as

$$\int \frac{\sec\theta}{\tan^2\theta}\, d\theta = \int \frac{(1/\cos\theta)}{(\sin^2\theta\,/\cos^2\theta)}\, d\theta = \int \frac{\cos\theta}{\sin^2\theta} d\theta = \int \cot\theta\csc\theta\ d\theta = -\csc\theta + c$$

This underscores the need to integrate a number of integrals involving trigonometric functions. Again, it is a trial and error kind of thing, but there are some standard tricks to learn. It is probably best to learn these tricks by looking at some specific examples, but as you do, focus on the technique being used and how it can be utilized on similar integrals.

Example 1.

$$\int \sin^3\theta\cos^2\theta\ d\theta$$

Since sine is to an odd power, it makes sense to "save one of the sines" and convert everything else to cosine, utilizing the identity $\sin^2\theta = 1 - \cos^2\theta$, and then applying a substitution.

$$\int \sin^3\theta\cos^2\theta\ d\theta = \int \sin\theta\sin^2\theta\cos^2\theta\ d\theta = \int (1-\cos^2\theta)\cos^2\theta\sin\theta\ d\theta$$

Letting $z = \cos\theta$, $dz = -\sin\theta\ d\theta$, so we have

$$\int \sin^3\theta\cos^2\theta\ d\theta = \int (1-z^2)z^2(-dz) = \int (z^4 - z^2)dz = \frac{z^5}{5} - \frac{z^3}{3} + c = \frac{\cos^5\theta}{5} - \frac{\cos^3\theta}{3} + c$$

Notice that if cosine had been to an odd power, you could have done the same trick, focusing on saving one cosine as part of the differential, and converting the rest to sine. Notice also that there would have been nothing "wrong" with doing the following.

$$\int \sin^3\theta\cos^2\theta\ d\theta = \int \sin^3\theta\cos\theta\cos\theta\ d\theta = \int \sin^3\theta\sqrt{1-\sin^2\theta}\ \cos\theta\, d\theta$$

Letting $x = \sin\theta$, $dx = \cos\theta\ d\theta$ would give

$$\int \sin^3\theta\cos^2\theta\ d\theta = \int x^3\sqrt{1-x^2}\ dx$$

which is potentially a harder integral than the original (though you could do it by parts). Remember that we introduced trigonometry to handle such integrals.

Problem

Consider

$$\int \sin^3\theta\cos^3\theta\ d\theta$$

a. Do this integral by "saving" one of the cosines and converting everything else into sines.

b. Do this integral by "saving" one of the sines and converting everything else into cosines.

c. Did you obtain the same result? Explain.

Example 2

What happens if you don't have an odd power of sine or cosine? For example, suppose you have

$$\int \sin^2\theta\cos^2\theta\ d\theta$$

Saving a sine or cosine would lead to an integrand with a square root, which we are trying to avoid, if we can help it. In this case, there are a couple of trigonometric identities that can help

$$\sin^2\theta = \frac{1}{2}[1-\cos 2\theta] \qquad \cos^2\theta = \frac{1}{2}[1+\cos 2\theta]$$

Utilizing these, we have

$$\int \sin^3\theta\cos^2\theta\ d\theta = \frac{1}{4}\int(1-\cos 2\theta)(1+\cos 2\theta)d\theta = \frac{1}{4}\int(1-\cos^2 2\theta)\ d\theta = \frac{1}{4}\int \sin^2 2\theta\ d\theta$$

Utilizing this "half angle" identity again, we obtain

$$\frac{1}{4}\int \sin^2 2\theta\ d\theta = \frac{1}{8}\int(1-\cos 4\theta)d\theta = \frac{1}{8}\left(\theta - \frac{1}{4}\sin 4\theta\right) + c$$

Problem in Context

Show that the same result would have occurred if we used the double angle identity

$$\sin 2\theta = 2\sin\theta\cos\theta$$

Example 3.

$$\int \sec^4\theta\tan^2\theta\ d\theta$$

In this example, we can save a $\sec^2\theta$ as part of the differential and utilize the identity $\sec^2\theta = 1+\tan^2\theta$ to change everything else into an expression in tangent. Since the secant is to an even power this again avoids square roots, which is the general idea.

$$\int \sec^4\theta\tan^2\theta\ d\theta = \int \sec^2\theta\tan^2\theta\sec^2\theta\, d\theta = \int(1+\tan^2\theta)\tan^2\theta\sec^2\theta\, d\theta$$

Letting $y = \tan\theta$, we have $dy = \sec^2\theta\ d\theta$, so we obtain

$$\int(1+y^2)y^2\, dy = \int(y^2+y^4)dy = \frac{y^3}{3} + \frac{y^5}{5} + c = \frac{\tan^3\theta}{3} + \frac{\tan^5\theta}{5} + c$$

Example 4.

If tangent occurs to an odd power, we can save a tangent and a secant as part of the differential and use the same identity to convert everything else to secant.

$$\int \sec^3\theta\tan^3\theta\ d\theta = \int \sec^2\theta\tan^2\theta\sec\theta\tan\theta\, d\theta = \int \sec^2\theta\,(\sec^2\theta - 1)\sec\theta\tan\theta\, d\theta$$

Letting $u = \sec\theta$, $du = \sec\theta\tan\theta\ d\theta$, so we have

$$\int u^2(u^2-1)du = \int(u^4-u^2)du = \frac{u^5}{5} - \frac{u^3}{3} + c = \frac{\sec^5\theta}{5} - \frac{\sec^3\theta}{3} + c$$

Problem

Consider

$$\int \sec^4\theta \tan^3\theta \ d\theta$$

a. Compute this integral by saving a $\sec^2\theta$ and converting the rest into an expression in tangent.

b. Compute this integral by saving a $\sec\theta\tan\theta$ and converting the rest into an expression in secant.

c. Verify that the two answers you obtained are really the same.

Problem

Previously, we computed $\int \tan\theta \ d\theta$ by rewriting $\tan\theta = \frac{\sin\theta}{\cos\theta}$. Alternatively, compute this integral by rewriting it as

$$\int \tan\theta \ d\theta = \int \frac{\sec\theta\tan\theta}{\sec\theta} \ d\theta$$

Do we get a different answer? Explain.

The same tricks we applied with secant and tangent would apply equally well to integrals involving cosecant and cotangent utilizing the identity $\csc^2\theta = 1 + \cot^2\theta$. Of course, we have not covered every type of integral that can occur and you can always find integrals that will take other tricks. We've only provided some basics that you can build on as you gain experience. So practice, practice, practice!

Drill Problems

1. Trig integrals
2. More trig substitutions

XXXXXXXXXXXXXXXXXXXXXXXXXXX

Back to Logistic Growth: Partial Fractions.

Recall that in the graphing section of this book (Section 6.4) we had an example for logistic growth

$$\frac{dy}{dt} = .01y\left(1 - \frac{y}{100}\right) = .0001y(100 - y)$$

In that chapter, we analyzed this growth rate to draw a qualitative graph of $y = y(t)$, but we really could not produce specific values of t for where things happened, because we didn't have a formula for $y(t)$. Separating the variables is no problem

$$\frac{dy}{y(100-y)} = .0001\, dt$$

To integrate the left-hand side, we could expand the denominator to complete the square and apply the appropriate trigonometric substitution.

As we've mentioned before, all mathematical models are simplifications of reality, and typically are modified to take into account more complexities. For example, consider a modification of the logistic model above

$$\frac{dy}{dt} = .01y\left(1 - \frac{y}{100}\right)\left(\frac{y}{10} - 1\right) = .00001y(100-y)(y-10)$$

In this case, 10 is called the *minimum viability level of the population.* (Why?) Solving this would require solving

$$\int \frac{dy}{y(100-y)(y-10)}$$

which is not really conducive to a trigonometric substitution.

Here, we will examine a technique which does not involve trigonometry and is really algebraic in nature. It will also be useful for examining integrals such as the two above. The trick to try and separate the fraction into "partial fractions" which will be easier to integrate. For example, consider $\frac{1}{y(100-y)}$. Unfortunately, we cannot just expand the denominator and divide this into two fractions as we could if it was the numerator. This is due to the fact that fractions are added by finding a common denominator, not just adding denominators. However it is reasonable to make an educated guess that such a fraction can be written as

$$\frac{1}{y(100-y)} = \frac{A}{y} + \frac{B}{100-y}$$

for some as yet to be determined A and B. We can see if this guess pans out by combining the partial fractions to obtain the original.

$$\frac{1}{y(100-y)} = \frac{A}{y} + \frac{B}{100-y} = \frac{A(100-y) + By}{y(100-y)} = \frac{100A + (B-A)y}{y}$$

This says that

$$\begin{aligned} B - A &= 0 \\ 100A &= 1 \end{aligned}$$

so that $A = \frac{1}{100}$ and $B = A = \frac{1}{100}$. Hence

$$\frac{1}{y(100-y)} = \frac{1/100}{y} + \frac{1/100}{100-y}$$

Back to our integration problem, we would obtain

$$\int \frac{dy}{y(100-y)} = \int \left(\frac{\frac{1}{100}}{y} + \frac{\frac{1}{100}}{100-y} \right) dy = \frac{1}{100}\left[\int \frac{dy}{y} + \int \frac{dy}{100-y}\right]$$
$$= \frac{1}{100}[\ln y - \ln(100-y)] + c$$

Problem in Context

a. Use the integral provided above to obtain an equation for y in the logistic growth model

$$\frac{dy}{dt} = .01y\left(1 - \frac{y}{100}\right), \quad y(0) = y_0$$

for the case where $0 < y < 100$

b. Solve the same problem for the case where $y > 100$. [This is a spot where you need to look at the absolute value in the logarithm. We told you it comes up occasionally.]

End of PIC

Now that we've shown you an example where decomposing a fraction of polynomials (called a rational function) into a sum of partial fractions is practical, let's look into a systematic way of doing this algebraically. All of this hinges on the following theoretical fact about polynomials.

Suppose we have two polynomials $p(x)$ and $q(x)$ which have no common factors (other than constants, which are considered trivial. For example, $p(x) = 2\left(\frac{p(x)}{2}\right), q(x) = 2\left(\frac{q(x)}{2}\right)$, which would make 2 a trivial common factor of $p(x)$ and $q(x)$.) Then there are polynomials $r(x)$ and $s(x)$ such that

$$r(x) \cdot p(x) + s(x) \cdot q(x) = 1$$

In other words, there is some (linear) combination of p and q which produces 1. For example, notice that

$$(x+a) - (x+b) = a - b$$

so if $a \neq b$, then

$$\frac{1}{a-b}(x+a) + \frac{-1}{a-b}(x+b) = 1$$

We determined this combination in a somewhat ad-hoc way, but there is a systematic way to do this involving no more than long division of polynomials. We won't delve into this, but will adopt a more (educated) guess and check method. Notice that in the above example

$$\frac{1}{(x+a)(x+b)} = \frac{\frac{1}{a-b}(x+a) + \frac{-1}{a-b}(x+b)}{(x+a)(x+b)} = \frac{\frac{1}{a-b}}{x+b} + \frac{\frac{-1}{a-b}}{x+a} = \frac{\frac{-1}{a-b}}{x+a} + \frac{\frac{1}{a-b}}{x+b}$$

In general, if we have

$$r(x) \cdot p(x) + s(x) \cdot q(x) = 1$$

then

$$\frac{1}{p(x) \cdot q(x)} = \frac{r(x) \cdot p(x) + s(x) \cdot q(x)}{p(x) \cdot q(x)} = \frac{s(x)}{p(x)} + \frac{r(x)}{q(x)}$$

Notice that if $t(x)$ is any polynomial, then

$$\frac{t(x)}{p(x) \cdot q(x)} = \frac{t(x) \cdot s(x)}{p(x)} + \frac{t(x) \cdot r(x)}{q(x)} = \frac{A(x)}{p(x)} + \frac{B(x)}{q(x)}$$

where $A(x) = t(x) \cdot s(x)$ and $B(x) = t(x) \cdot r(x)$ are two polynomials.

To summarize all of this, we have, in theory, that if $p(x)$ and $q(x)$ are two polynomials with no non-trivial common factors, then for any polynomial $t(x)$, there are polynomials $A(x)$ and $B(x)$ such that

$$\frac{t(x)}{p(x) \cdot q(x)} = \frac{A(x)}{p(x)} + \frac{B(x)}{q(x)}$$

To extend this idea further, if $p(x), q(x), r(x)$ have no nontrivial factors, then there are polynomials $A(x), B(x), C(x)$ with

$$\frac{t(x)}{p(x) \cdot q(x) \cdot r(x)} = \frac{A(x)}{p(x)} + \frac{B(x)}{q(x)} + \frac{C(x)}{r(x)}$$

and this can be extended to any number of factors in the denominator. Again, this can be proven theoretically, and this partial fractions decomposition can be obtained systematically, but we will adopt a guess and check method to find A, B, C. To make our guess a little more educated, the following fact can be employed. You can make peace with this in your own closet.

If the degree of $t(x)$ is less than the degree of $p(x) \cdot q(x) \cdot r(x)$, then A, B, C can be chosen with $\deg A < \deg p$, $\deg B < \deg q$, $\deg C < \deg r$.

Example: Recall we had

$$\int \frac{dy}{y(100-y)(y-10)}$$

To compute this we will find the partial fractions decomposition of $\frac{1}{y(100-y)(y-10)}$. We make the educated guess

$$\frac{1}{y(100-y)(y-10)} = \frac{A}{y} + \frac{B}{100-y} + \frac{C}{y-10}$$

and determine what A, B, C are. This can be done in a number of ways, but the most direct (and labor intensive) way to do this is to combine the terms in the right-hand side and compare coefficients.

$$\frac{1}{y(100-y)(y-10)} = \frac{A(100-y)(y-10) + By(y-10) + Cy(100-y)}{y(100-y)(y-10)}$$

$$= \frac{A(-y^2+110y-100) + B(y^2-10y) + C(100y-y^2)}{y(100-y)(y-10)}$$

$$= \frac{(-A+B-C)y^2 + (110A-10B+100C)y + (-1000A)}{y(100-y)(y-10)}$$

Since the fractions must be equal and the denominators are the same, then we can set the numerators equal.

$$1 = 0y^2 + 0y + 1 = (-A+B-C)y^2 + (110A-10B+100C)y + (-1000A)$$

$$\begin{gathered} -A+B-C = 0 \\ 110A - 10B + 100C = 0 \\ -1000A = 1 \end{gathered}$$

Solve this anyway you wish, but you should get

$$A = -\frac{1}{1000}, \quad B = \frac{1}{9000}, \quad C = \frac{1}{900}$$

Thus

$$\frac{1}{y(100-y)(y-10)} = \frac{-1/1000}{y} + \frac{1/9000}{100-y} + \frac{1/900}{y-10}$$

so

$$\int \frac{dy}{y(100-y)(y-10)} = -\frac{1}{1000}\int \frac{dy}{y} + \frac{1}{9000}\int \frac{dy}{100-y} + \frac{1}{900}\int \frac{dy}{y-10}$$

$$= -\frac{1}{1000}\ln y - \frac{1}{9000}\ln(100-y) + \frac{1}{900}\ln(y-10) + c$$

A few Practice Problems with linear terms and no shortcuts.

The practice problems above illustrate that determining the partial fractions decomposition is really algebra and not calculus. Bear in mind that the algebra is tedious, but it is algebra. To eliminate some of the tedium, there are some shortcuts. We will explore one of them here. We had you practice some without the shortcut because one should not follow a shortcut unless he/she is familiar with the non-shortcut. Furthermore, shortcuts don't always pan out; the traditional algebraic method will always work. We'll illustrate this with an example. I took this from your paper on partial fractions.

Example. Suppose we wish to find the partial fractions decomposition of $\frac{3x+1}{(x-1)(x-2)(x-5)}$. As before, we have

$$\frac{3x+1}{(x-1)(x-2)(x-5)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-5} \qquad (*)$$

This leads to

$$3x + 1 = A(x-2)(x-5) + B(x-1)(x-5) + C(x-1)(x-2)$$

Observing that (1) this formula will be true for every real (or complex) value of x and that (2) the factor $x - 1$ appears in two of the terms on the right suggests that we let $x = 1$. In that case we have

$$3(1) + 1 = A(1-2)(1-5)$$

or

$$\frac{3(1)+1}{(1-2)(1-5)} = A.$$

We did not simplify this last formula in order to highlight the following. Observe that we get the same result if we simply "cover up" the factor $(x - 1)$ (Actually it is replacing $x - 1$ with 1.) on the left side of (*) above and set $x = 1$.

$$A = \frac{3(1)+1}{(covered)(1-2)(1-5)} = 1.$$

In precisely the same fashion (setting $x = 2$ and $x = 3$) we get:

$$B = \frac{(3)2+1}{(2-1)(covered)(2-5)} = \frac{-7}{3}$$

,

and

$$C = \frac{(3)5+1}{(5-1)(5-2)(covered)} = \frac{4}{3},$$

so that

$$\frac{3x+1}{(x-1)(x-2)(x-5)} = \frac{1}{x-1} + \frac{-7/3}{x-2} + \frac{4/3}{x-5}.$$

This trick is known as "Heaviside's Cover-Up Method" (HCUM). It is well known, and can be used to find the PFD of a rational function as long as the denominator can be factored into distinct linear factors.

Practice Problems Using HCUM

Expanding Partial Fractions Decompositions

In the previous partial fractions decompositions, we started with a rational function where the degree of the numerator was less than the degree of the denominator and where the factors in the denominator were linear. In this section, we will expand our techniques to encompass other situations.

For example, suppose the degree of the numerator is not less than the degree of the denominator. If this is the case, then we can perform long division first and then find the partial fractions decomposition of the remainder. For example, suppose we have

$$\frac{x^3-2x+1}{x^2+x-6}=\frac{x^3-2x+1}{(x-2)(x+3)}$$

We first perform long division of the original rational function.

$$\frac{x^3-2x+1}{x^2+x-6}=x-1+\frac{5x-5}{x^2+x-6}=x-1+5\left(\frac{x-1}{(x-2)(x+3)}\right)$$

We can now perform a partial fractions decomposition on the remaining fraction.

Practice Exercises with integrals with involving rational functions requiring long division.

What about something like

$$\frac{3x+2}{x^2(x-1)}$$

Since x^2 and $x-1$ have no common (nontrivial) factors, then we can still preform our partial fractions decomposition. Recall that in the decomposition, the numerators in the individual terms can be chosen so that the degrees are less than the degrees in the denominators. This being said, we make the following educated guess as to the form of the partial fractions decomposition.

$$\frac{3x+2}{x^2(x-1)}=\frac{Ax+B}{x^2}+\frac{C}{x-1}$$

We can now recombine this to determine A, B, and C.

$$\frac{3x+2}{x^2(x-1)}=\frac{Ax+B}{x^2}+\frac{C}{x-1}=\frac{(Ax+B)(x-1)+Cx^2}{x^2(x-1)}$$
$$=\frac{Ax^2+Bx-Ax-B+Cx^2}{x^2(x-1)}=\frac{(A+C)x^2+(-A+B)x-B}{x^2(x-1)}$$

This leads to

$$A+C=0$$
$$-A+B=3$$
$$-B=2$$

Solving this (any way you can) we get

$$A=-5, B=-2, C=5$$

so that

$$\frac{3x+2}{x^2(x-1)}=\frac{-5x-2}{x^2}+\frac{5}{x-1}$$

If we wanted to integrate this, we would obtain

$$\int\frac{3x+2}{x^2(x-1)}\,dx=\int\left(\frac{-5x-2}{x^2}+\frac{5}{x-1}\right)dx$$

$$= \int \left(-\frac{5}{x} - \frac{2}{x^2} + \frac{5}{x-1} \right) dx$$
$$= -5\ln x + 2x^{-1} + 5\ln(x-1) + c$$

Notice in this example that we really needed to look at the most general possible polynomial of degree one less than the denominators in the partial fractions decomposition; we would not have obtained the correct answer otherwise.

Problem in Context

Show that if we tried the following decomposition

$$\frac{3x+2}{x^2(x-1)} = \frac{A}{x^2} + \frac{B}{x-1}$$

then no values for A, B would satisfy this. [This says that we really need to use the most general polynomial of degree one less than the denominators in the partial fractions decomposition.]

End of PIC

Suppose we had

$$\int \frac{25}{(x-2)^2(x^2+1)}\, dx$$

Applying our partial fractions decomposition, we get

$$\frac{25}{(x-2)^2(x^2+1)} = \frac{Ax+B}{(x-2)^2} + \frac{Cx+D}{x^2+1}$$
$$= \frac{(Ax+B)(x^2+1) + (Cx+D)(x-2)^2}{(x-2)^2(x^2+1)}$$
$$= \frac{(A+C)x^3 + (B-4C-D)x^2 + (A+4C+4D)x + (B-4D)}{(x-2)^2(x^2+1)}$$

This leads to

$$A + C = 0\,,\ \ B - 4C - D = 0\,,\ \ A + 4C + 4D = 0\,,\quad B - 4D = 25$$

Solving this (any way you can), we get

$$A = -4\,, B = 13\,, C = 4\,, D = -3$$

Getting back to the integral, we have

$$\int \frac{25}{(x-2)^2(x^2+1)}\, dx = \int \frac{-4x+13}{(x-2)^2} dx + \int \frac{4x-3}{x^2+1} dx$$

$$= \int \frac{-4x+13}{(x-2)^2} dx + 4\int \frac{x}{x^2+1} dx - 3\int \frac{1}{x^2+1} dx$$
$$= \int \frac{-4x+13}{(x-2)^2} dx + 2\ln(x^2+1) - 3\arctan x + c$$

Notice that the last wo integrals were pretty straightforward. This first integral is more problematic. There is a little algebraic trick which will allow us to break up that integral even further.

$$\begin{aligned}\int \frac{-4x+13}{(x-2)^2}dx &= \int \frac{-4\big((x-2)+2\big)+13}{(x-2)^2}dx \\ &= \int \frac{-4(x-2)-8+13}{(x-2)^2}dx \\ &= \int \frac{-4(x-2)+5}{(x-2)^2}dx \\ &= -4\int \frac{1}{x-2}\,dx + 5\int \frac{1}{(x-2)^2}dx \\ &= -4\ln(x-2) - 5(x-2)^{-1}\end{aligned}$$

Putting this all together, we finally have

$$\int \frac{25}{(x-2)^2(x^2+1)}\,dx = -4\ln(x-2) - 5(x-2)^{-1} + 2\ln(x^2+1) - 3\arctan x + c$$

The trick that we employed above is something you might want to remember. It allows us to take our partial fractions decomposition a bit further. For example, if we start with

$$\int \frac{x^3}{(2x^2+1)^2}\,dx$$

we could rewrite this as

$$\begin{aligned}\int \frac{x^3}{(2x^2+1)^2}\,dx &= \int \frac{\frac{1}{2}x(2x^2)}{(2x^2+1)^2}\,dx = \int \frac{\frac{1}{2}x(2x^2+1-1)}{(2x^2+1)^2}\,dx \\ &= \frac{1}{2}\int \frac{x}{2x^2+1}\,dx - \frac{1}{2}\int \frac{x}{(2x^2+1)^2}\,dx\end{aligned}$$

Integrating this would be a matter of utilizing the substitution $y = 2x^2+1$. Of course, this could have been done earlier to produce

$$\int \frac{x^3}{(2x^2+1)^2}\,dx = \int \frac{x^2}{(2x^2+1)^2}x\,dx = \int \frac{\frac{1}{2}(y-1)}{y^2}\left(\frac{1}{4}dy\right) = \frac{1}{8}\left[\int \frac{1}{y}\,dy - \int \frac{1}{y^2}\,dy\right]$$

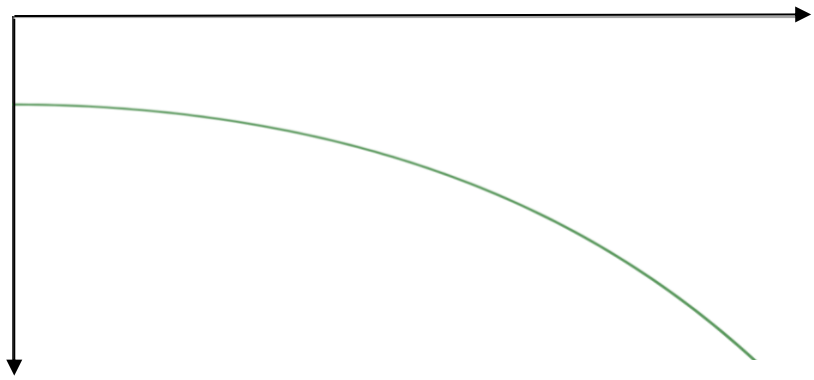
The point is that you have more tools at your disposal to transform integrals you don't know how to do into ones that you recognize.

Before we send off to practice such integrals, we want to present you one more option. You have probably noticed that there is more algebra involved than calculus. Some of it is unavoidable, but there are some shortcuts. For example, finding A, B, C, D in the original partial fractions decomposition required solving 4 equations in 4 unknowns. Again, this was straightforward, but it turns out that there is a shortcut for this similar to the HCUM. We will present it here. Bud – at this point, how much of the paper you wrote should we put in? Will putting it in be too much at this point? Should be some practice exercises first? Should it be appendicized? I'm afraid I've already written too much on partial fractions decompositions. I need your fresh view on this.

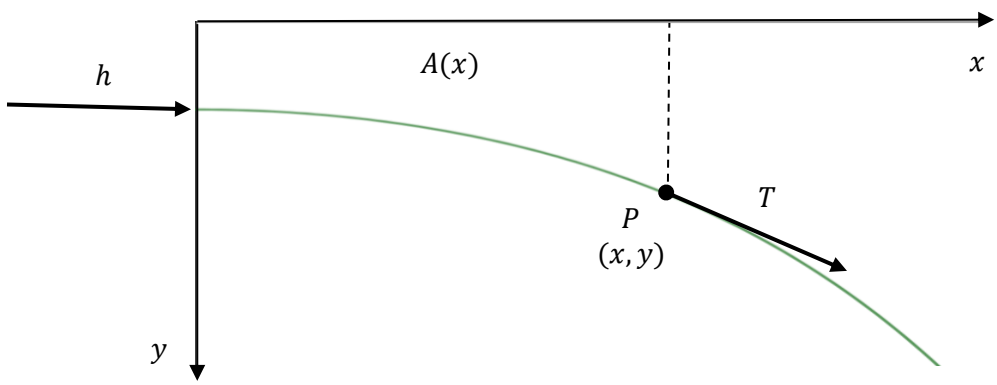
Areas and Integration

Recall, much earlier, that we examined the catenary (hanging chain). We showed that the catenary was not a parabola as Galileo had mentioned. The idea of a catenary was (and is) important in engineering in the construction of arches. Masons before Galileo's time were aware of the fact that the "best" arches were in the shape of an inverted catenary, and they constructed wooden frames for their stone utilizing a hanging chain for the correct shape. This can be seen in the following example.

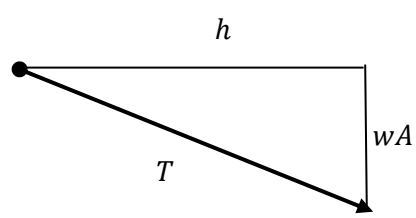
Example: Consider one half of a stone arch bridge drawn below.



We will draw the positive y-axis downward and will focus on the forces at a generic point P with coordinates (x, y). The problem is to find the curve $y = y(x)$ so that the vertical component of the tangential force T at P is equal to the weight of the bridge from 0 to x. If we do this, then that means that the weight of the bridge will be directed toward the base of the bridge. With this in mind, let $A(x)$ denote the area of the side section of the bridge from 0 to x. We will also let h denote the (constant) magnitude of the horizontal force along the length of the bridge and w be the weight density of the stone (per cross sectional area).



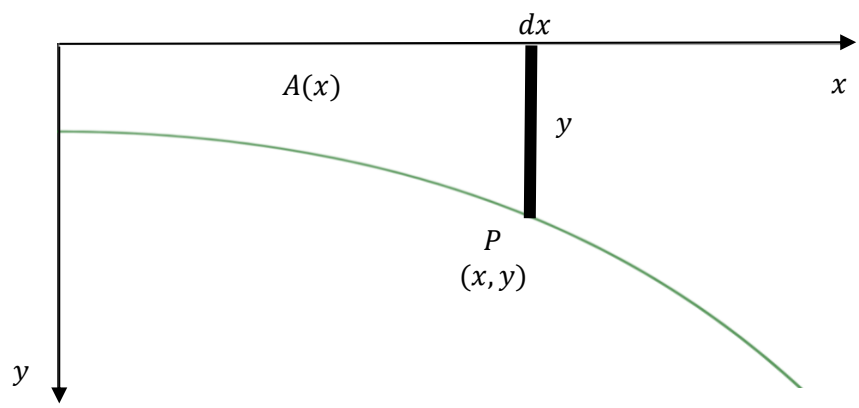
With all of this set up, what we really want is the horizontal component of the force T to be h and the vertical component to be the weight of the bridge from 0 to x, namely, wA. This leads to the following picture.



This leads to the slope of the tangent line at P equaling $\frac{wA}{h}$, so we get the differential equation

$$\frac{dy}{dx} = \frac{w}{h}A$$

The problem here is that we don't know what A is. However, a moment's thought tells us that we know what dA is. If we increase x to $x + dx$, then we can see that $dA = y\,dx$



Thus, if we differentiate the above equation, we get that the arch should satisfy the differential equation

$$\frac{d^2y}{dx^2} = \frac{w}{h}\frac{dA}{dx} = \frac{w}{h}\,y$$

Problem in Context.

a. Show that

$$y = Ae^{\sqrt{\frac{w}{h}}x} + Be^{-\sqrt{\frac{w}{h}}x}$$

satisfies the above differential equation.

b. Show that if $y(0) = \sqrt{\frac{h}{w}}$ and $\left.\frac{dy}{dx}\right|_{x=0} = 0$, then $A = B = \frac{1}{2}\sqrt{\frac{h}{w}}$. Compare this to the equation of the catenary in Problem in Context # 53.

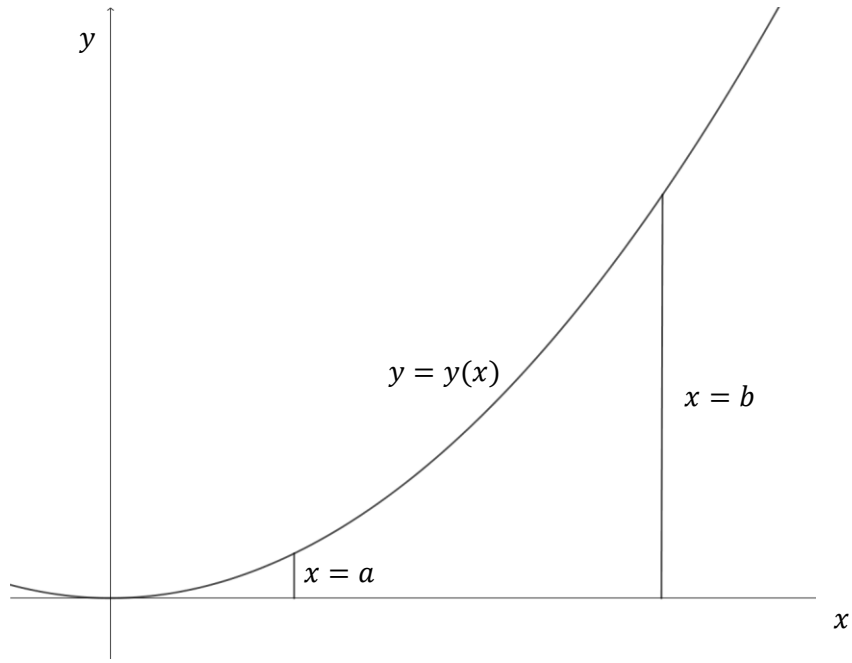
End of PIC

While it is interesting that the above example illustrates the importance of catenaries in engineering, the important thing for us is that $dA = y\,dx$. Thus

$$\int y\,dx = \int dA = A + c$$

which relates integrals to areas. This relationship was known to both Newton and Leibniz, as well as some of their predecessors. It is of such importance that it is known as the **Fundamental Theorem of Calculus.**

To make this a bit more concrete, let's set this up a bit more carefully. Suppose we start with a curve $y = y(x)$ where $y \geq 0$ for $a \leq x \leq b$, and we wish to find the area of the region bounded by the curves $y = y(x), y = 0, x = a, x = b$.



We used the notation $\int y dx$ to denote a sum of differentials and called this integration with the idea that it was the opposite of differentiation and provided a notation for the antiderivative to the function y. In examining a situation like we have with areas above, we will modify our notation a bit to

$$\int_{x=a}^{b} y\, dx$$

to emphasize that we are adding differentials from $x = a$ to $x = b$, so that $\int_{x=a}^{b} y\, dx$ represents a specific number (in this case area), whereas $\int y\, dx$ represents an antiderivative which is a family of functions differing by arbitrary constants. With this in mind, $\int_{x=a}^{b} y\, dx$ is called the definite integral of $y\, dx$ from $x = a$ to $x = b$ and $\int y\, dx$ is called the indefinite integral of $y\, dx$. So, the definite integral is a specific number while the indefinite integral is a family of functions. If using almost the same notation seems confusing at this point, it is brought together by the Fundamental Theorem of Calculus.

Fundamental Theorem of Calculus. Suppose that $\frac{dA}{dx} = y$ (so that A is an antiderivative of y). Then

$$\int_{x=a}^{b} y\, dx = A(b) - A(a)$$

As was mentioned before, this result was known to both Newton and Leibniz and was first publish in a short paper by Leibniz in 1693. The idea is so surprisingly simple and profound, that it warrants an (apocryphal) story. When Leibniz was a young diplomat in Paris, he sought Christian Huygens who, at the time, was one of the most brilliant mathematicians in the world. Leibniz credited much of his mathematical

growth in Paris to Huygens. The idea for summing differences came to Leibniz from a problem given to him by Huygens in 1672, namely to compute the following sum

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\frac{1}{3\cdot 4}+\cdots$$

Leibniz recognized that this could be rewritten as

$$\left(\frac{1}{1}-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)+\cdots$$

and noted that nearly all of the fractions cancelled out leaving $1-\frac{1}{\infty}=1-0=1$.

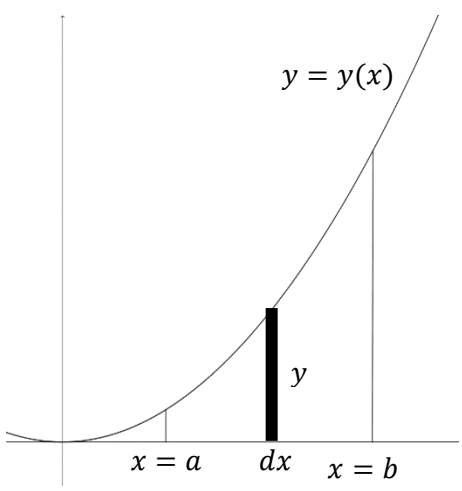
The importance of this was not the result, but the technique, namely that the sum of differences is equal to the difference of the extremes. Applying this same principle to the Fundamental Theorem of calculus, we have

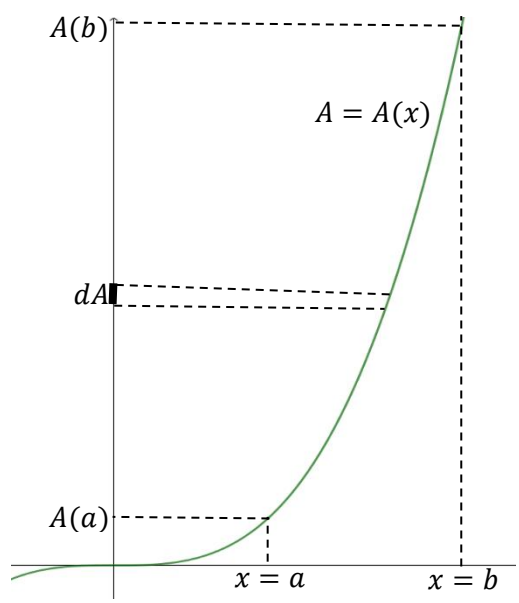
$$\frac{dA}{dx}=y$$

$$y\,dx=dA$$

$$\int_{x=a}^{b} y\,dx=\int_{x=a}^{b} dA=A(b)-A(a)$$

This was so quick, that it probably deserves a picture. Consider the picture below of the two functions y and A related by the fact that $\frac{dA}{dx}=y$.





Notice that since $y\,dx=dA$, then this says that the area of the box with width dx and height y in the first diagram is the same as the length dA of the segment in the second diagram. Of course, these represent

corresponding generic boxes and segments. If we add all of these together, the sum of the areas of the boxes in the first diagram will provide the area of the region; the sum of the lengths of the segments in the second diagram will provide the length of the segment from $A(a)$ to $A(b)$. In other words, we have

$$\int_{x=a}^{b} y\,dx = \int_{x=a}^{b} dA = A(b) - A(a)$$

which is what the Fundamental Theorem of Calculus says.

It is hard to overstate the importance of this result. (Not everything is called a Fundamental Theorem.) In many applications, we will have a need to add all of the differentials which may or may not represent areas. After we set up this sum, computation will bring us back to our old problem of computing antiderivatives. For now, let's just focus on computation of definite integrals via the Fundamental Theorem of Calculus. With the Fundamental Theorem of Calculus in mind, we introduce the following notation. If $\frac{dA}{dx} = y$, then

$$\int_{x=a}^{b} y\,dx = A(x)|_{x=a}^{b} = A(b) - A(a)$$

The symbol $A(x)|_{x=a}^{b}$ is read as "A of x evaluated from $x = a$ to $x = b$" and is the difference between $A(b)$ and $A(a)$. Again, this is purely notational.

Problem in Context: Given our notation for an indefinite integral, the Fundamental Theorem of Calculus says

$$\int_{x=a}^{b} y\,dx = \left(\int y\,dx\right)\Big|_{x=a}^{b}$$

However, there are infinitely many possibilities for $\int y\,dx$. Does this mean that there are infinitely many possibilities for $\int_{x=a}^{b} y\,dx$? Explain.

End of PIC

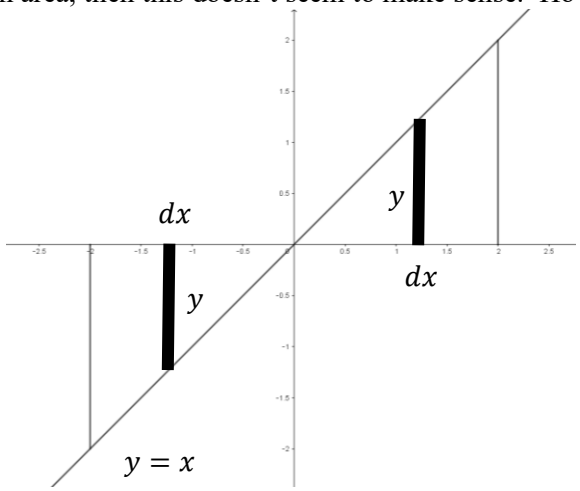
Practice Exercises

Properties of Definite Integrals

While it is true that areas motivated the definite integral and the Fundamental Theorem of Calculus, it is important to remember that we are really talking about a sum of differentials. This helps to explain some anomalies which occur when applying the fundamental theorem. For example, consider

$$\int_{-2}^{2} x\,dx = \frac{x^2}{2}\Big|_{x=-2}^{2} = \frac{2^2}{2} - \frac{(-2)^2}{2} = 0$$

If you only think of this as an area, then this doesn't seem to make sense. However, let's look at a graph.



As you will notice the rectangles under the $x$axis will provide a "negative" area so the total is zero.

The properties of definite integrals mirror that of indefinite integrals. (See Table 1.)

Integration Properties
$\int_{x=a}^{b} 0\, dx = 0$
$\int_{x=a}^{b} f\, dx + g\, dx = \int_{x=a}^{b} f\, dx + \int_{x=a}^{b} g\, dx$
$\int_{x=a}^{b} c\, f\, dx = c \int_{x=a}^{b} f\, dx$
$\int_{x=b}^{a} f\, dx = -\int_{x=a}^{b} f\, dx$
$\int_{x=a}^{a} f\, dx = 0$
$\int_{x=a}^{b} f\, dx + \int_{x=b}^{c} f\, dx = \int_{x=a}^{c} f\, dx$

Table 2

These properties are consistent with the rules for indefinite integrals, which is not surprising in light of the Fundamental Theorem of Calculus. That being said, let's make sense of these properties beyond the fundamental theorem. First and foremost, remember that these integrals are sums of differentials and behave like sums. With this in mind, lets compare the first three properties in the table with their finite sum analogs.

Integration Properties	Sum Properties
$\int_{x=a}^{b} 0\,dx = 0$	$0 + 0 = 0$
$\int_{x=a}^{b} f\,dx + g\,dx = \int_{x=a}^{b} f\,dx + \int_{x=a}^{b} g\,dx$	$(f_1 + g_1) + (f_2 + g_2) = (f_1 + f_2) + (g_1 + g_2)$
$\int_{x=a}^{b} c\,f\,dx = c\int_{x=a}^{b} f\,dx$	$(c\,f_1 + c\,f_2) = c(f_1 + f_2)$

Table 3

The last three properties in Table 2 might seem less obvious. We will make sense of these geometrically, but let's first note that these are consistent with the fundamental theorem of calculus as illustrated in the following examples.

Examples:

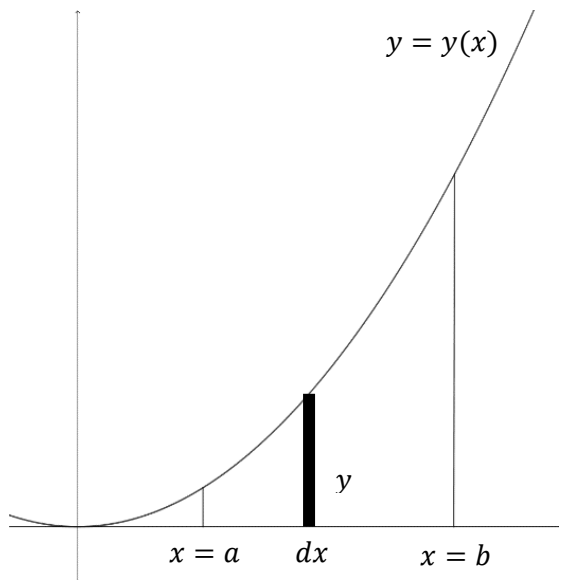
$$\int_{x=b}^{a} x^2 dx = \left.\frac{x^3}{3}\right|_{x=b}^{a} = \frac{a^3}{3} - \frac{b^3}{3} = -\left(\frac{b^3}{3} - \frac{a^3}{3}\right) = -\left(\left.\frac{x^3}{3}\right|_{x=a}^{b}\right) = -\int_{x=a}^{b} x^2 dx$$

$$\int_{x=a}^{a} x^2 dx = \left.\frac{x^3}{3}\right|_{x=a}^{a} = \frac{a^3}{3} - \frac{a^3}{3} = 0$$

$$\int_{x=a}^{b} x^2\,dx + \int_{x=b}^{c} x^2\,dx = \left.\frac{x^3}{3}\right|_{x=a}^{b} + \left.\frac{x^3}{3}\right|_{x=b}^{c}$$

$$= \left(\frac{b^3}{3} - \frac{a^3}{3}\right) + \left(\frac{c^3}{3} - \frac{b^3}{3}\right) = \frac{c^3}{3} - \frac{a^3}{3} = \frac{x^3}{3}\bigg|_{x=a}^{c} = \int_{x=a}^{c} x^2\, dx$$

To see that these properties hold geometrically (without the Fundamental Theorem of Calculus), recall that this really is a sum of differentials. Specifically, we have the following diagram illustrating a generic rectangle that we are summing.



When we consider $\int_{x=a}^{b} y\, dx$ with $a < b$, then the difference dx is a positive change. When we consider $\int_{x=b}^{a} y\, dx$, then the difference dx represents a negative change. Hence, we have

$$\int_{x=b}^{a} y\, dx = -\int_{x=a}^{b} y\, dx$$

Problem in Context:

Use the fact that

$$\int_{x=a}^{a} y\, dx = -\int_{x=a}^{a} y\, dx$$

To show that

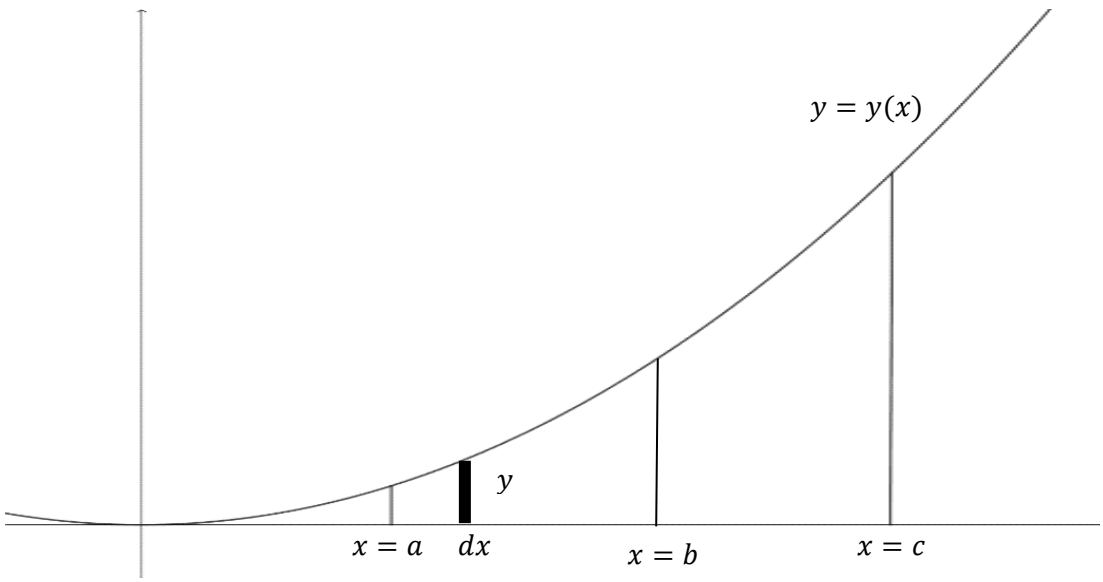
$$\int_{x=a}^{a} y\, dx = 0$$

End of PIC

As for the last property

$$\int_{x=a}^{b} f\,dx + \int_{x=b}^{c} f\,dx = \int_{x=a}^{c} f\,dx$$

Consider the following diagram where $a < b < c$



If we are summing differentials $y\,dx$ from $x = a$ to $x = b$ and summing them from $x = b$ to $x = c$, then this certainly would be the same as summing them from $x = a$ to $x = c$. Thus

$$\int_{x=a}^{b} f\,dx + \int_{x=b}^{c} f\,dx = \int_{x=a}^{c} f\,dx$$

Problem in Context: Show that for the case where $a < c < b$, we still have

$$\int_{x=a}^{b} f\,dx + \int_{x=b}^{c} f\,dx = \int_{x=a}^{c} f\,dx$$

[Hint: We already know that

$$\int_{x=a}^{b} f\,dx = \int_{x=a}^{c} f\,dx + \int_{x=c}^{b} f\,dx$$

Solve for $\int_{x=a}^{c} f\ dx$ and use a previous property.]

End of PIC

Problem in Context: Suppose

$$\int_{x=0}^{5} f\ dx = 3 \qquad \int_{x=0}^{7} f\ dx = 8 \qquad \int_{x=0}^{5} g\ dx = 2$$

Compute

a. $\int_{x=0}^{5} (9f - 4g) dx$

b. $\int_{x=5}^{7} f\ dx$

c. $\int_{x=5}^{0} g\ dx$

d. $\int_{x=0}^{0} f\ dx$

e. $\int_{t=0}^{5} f(t)\ dt$

End of PIC

Part e of the last problem brings up an interesting aspect about definite integrals. Since the final answer is a number, then the variable t is immaterial. For example, we have

$$\int_{x=0}^{1} x^2 dx = \int_{t=0}^{1} t^2 dt = \int_{u=0}^{1} u^2 du = \frac{1}{3}$$

With this in mind, many people call the $x, t,$ or u in the integral a "dummy" variable. This means that you can substitute any letter in and it will not change the results.

Speaking of substitution, all of the techniques we applied when computing indefinite integrals work just as well for definite integrals. We just need to make sure that the limits of integration match.

Example: Consider

$$\int_{x=2}^{3} xe^{x^2}\ dx$$

If we make the substitution $z = x^2$, then $dz = 2x\ dx$, and so

$$\int_{x=2}^{3} xe^{x^2}\ dx = \frac{1}{2} \int_{x=2}^{3} e^z\ dz$$

At this point, we have two options: we can integrate and change the variable back into x to finish the fundamental theorem of calculus, or we can convert everything over to z and not deal with x anymore. We'll show both. In the first approach, we have

$$\frac{1}{2}\int_{x=2}^{3} e^z\,dz = \frac{1}{2}(e^z)|_{x=2}^{3} = \frac{1}{2}e^{x^2}\Big|_{x=2}^{3} = \frac{1}{2}(e^9 - e^4)$$

In the second approach we have that $z = x^2$, so when $x = 2, z = 4$ and when $x = 3, z = 9$, so that

$$\int_{x=2}^{3} xe^{x^2}\,dx = \frac{1}{2}\int_{z=4}^{9} e^z\,dz = \frac{1}{2}(e^z)|_{z=4}^{9} = \frac{1}{2}(e^9 - e^4)$$

Whichever you do is entirely up to you, usually one is not easier than the other. What you cannot do is to substitute an x limit for z or vice versa.

A similar situation occurs in integration by parts.

Example: Computing

$$\int_{x=0}^{\pi} x\sin x\,dx$$

we see that integration by parts seems to be the way to go. If we let $u = x$ and $dv = \sin x\ dx$, then

$$\begin{array}{ll} u = x & dv = \sin x\ dx \\ du = dx & v = -\cos x \end{array}$$

Utilizing the integration by parts formula, we have

$$\int_{x=0}^{\pi} x\sin x\,dx = (-x\cos x)|_{x=0}^{\pi} + \int_{x=0}^{\pi} \cos x\ dx$$
$$= (-\pi\cos\pi + 0\cos 0) + (\sin x)|_{x=0}^{\pi} = (\pi + 0) + (0 - 0) = \pi$$

Practice Problems:

Problem in Context:

a. Suppose that $y = y(x)$ has the property that $y(-x) = y(x)$. Thus (x, y) is on the curve of this function exactly when $(-x, y)$ is on the curve. What would such a curve look like?

Show that in this case

$$\int_{x=-a}^{a} y\,dx = 2\int_{x=0}^{a} y\,dx$$

[Hint: We know that

$$\int_{x=-a}^{a} y(x)\,dx = \int_{x=-a}^{0} y(x)\,dx + \int_{x=0}^{a} y(x)\,dx$$

To show that

$$\int_{x=-a}^{0} y(x)\,dx = \int_{x=0}^{a} y(x)\,dx$$

Use the substitution $z = -x$ in the integral on the left-hand side.]

Does this make sense geometrically? Explain.

b. Suppose that $y = y(x)$ has the property $y(-x) = -y(x)$. Thus (x, y) is on the curve of this function exactly when $(-x, -y)$ is on the curve. What would such a curve look like?

   Show that in this case

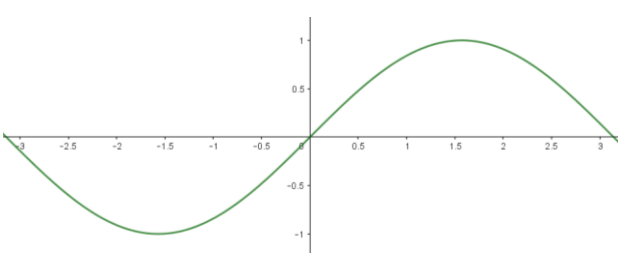
$$\int_{x=-a}^{a} y(x)\,dx = 0$$

Does this make sense geometrically? Explain.

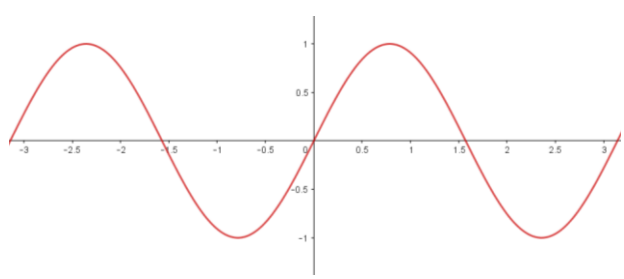
End of PIC

It should be noted that physicists (and mathematicians) make use of these and other symmetry properties with integrals whenever they can. The following is one such situation.

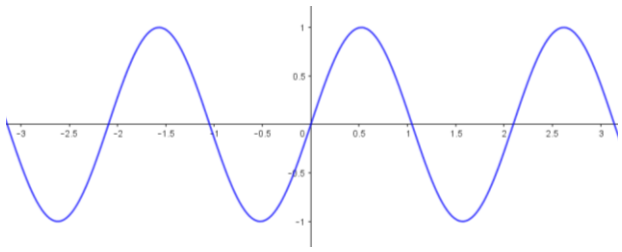
In the theory of acoustics, sine and cosine waves form the "pure tones" from which all other sound waves are formed. Mathematically, we can think of a general sound wave as a function on a closed interval of time (which constitutes on cycle of the sound wave). For an example, we have some pure tones of varying frequencies defined on the interval $[-\pi, \pi]$.



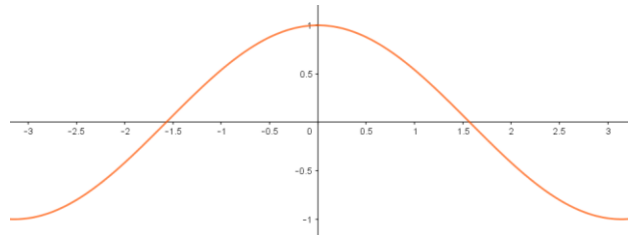
$y = \sin x$



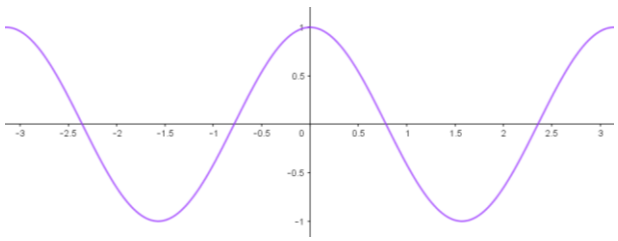
$y = \sin 2x$



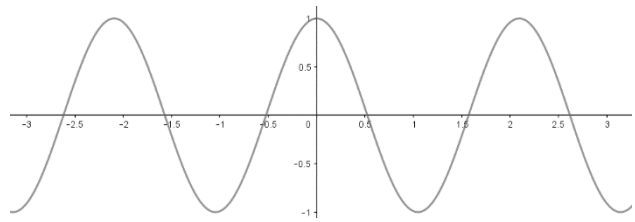
$y = \sin 3x$



$y = \cos x$



$y = \cos 2x$



$y = \cos 3x$

It can be shown (though we won't do it here), that if a function $f(t)$ defined on the interval $[-\pi, \pi]$ can be written as

$$f(t) = a_0 + a_1 \cos t + b_1 \sin t + a_2 \cos 2t + b_2 \sin 2t + a_3 \cos 3t + b_3 \sin 3t + a_4 \cos 4t + b_4 \sin 4t + \cdots$$

then

$$a_0 = \frac{1}{2\pi} \int_{t=-\pi}^{\pi} f(t) \cos(0t)\ dt = \frac{1}{2\pi} \int_{t=-\pi}^{\pi} f(t)\ dt$$

$$a_n = \frac{1}{\pi} \int_{t=-\pi}^{\pi} f(t) \cos nt\ dt, n \geq 1$$

$$b_n = \frac{1}{\pi} \int_{t=-\pi}^{\pi} f(t) \sin nt\ dt, n \geq 1$$

Problem in Context:

a. With the above set up, suppose that f is symmetric about the y axis. That is $f(-t) = f(t)$. Show that in this case, $b_n = 0$ for all n and so f can be written exclusively as a sum of cosine waves.

b. With the above set up, suppose that f is symmetric about the origin. That is $f(-t) = -f(t)$. Show that in this case, $a_n = 0$ for all n and so f can be written exclusively as a sum of sine waves.

c. Given the symmetries of the graphs above, is this surprising? Explain.

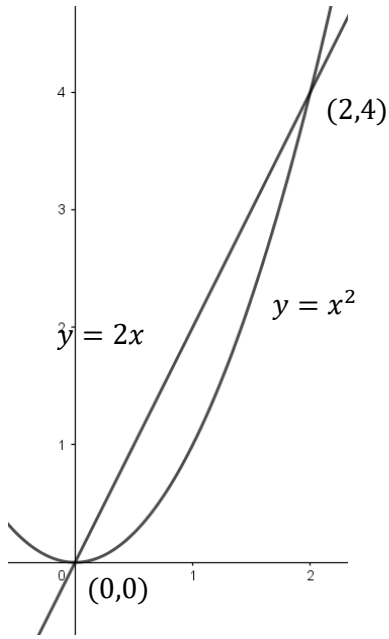
End of PIC

Applications of Definite Integrals

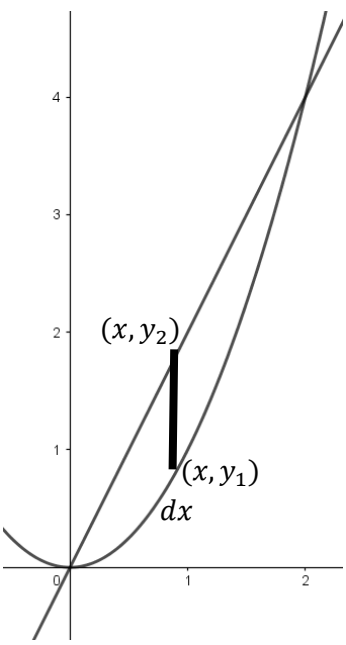
Previously, we put indefinite integrals to work by modeling phenomena with differential equations and antidifferentiating to solve these. We will now put definite integrals to work, but in a slightly different manner. Here we will use a differential to measure some quantity on an infinitely small scale where a simple formula applies. We will "integrate" all of these quantities to give us a total measurement as an integral. Once we have this, then we can calculate this definite integral by antidifferentiation and applying the Fundamental Theorem of Calculus. Sounds easy, huh! As they say, the devil is in the details, but if you follow this basic format and draw and label pictures, you will be surprised by the things you can accomplish. We'll start with the basic problem of computing an area. This is basic enough that there is a tendency to skip all of the steps in setting up the integral, BUT DON'T! Getting into a good work habit when things are relatively straightforward will help when more complicated applications arise, such as computing a volume, or a center of mass, or a moment of inertia, or the kinetic energy of a spinning object. If these applications sound daunting at this point, don't worry. We will see that the same sort of "divide and conquer" strategy applies to a wonderful array of situations. All that changes is what we want to measure. This is where the strategy of setting up things on an infinitely small scale (utilizing pictures) and "integrating" these together will become invaluable.

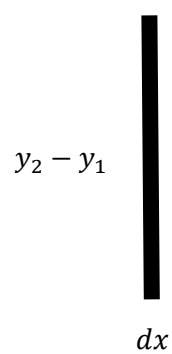
Areas: The simple formula for the area of a rectangle (length times width) provides the basis for determining the area of a more general shape. You will probably notice that we had been using this area model in looking at definite integrals already so this should not be too big a jump. However, do not take a short cut and utilize our general strategy. That is divide the region into infinitely thin rectangles where our simple length times width formula applies, and then "integrate" them into the area of the entire region. Let's demonstrate this with the following example.

Example: Find the area of the following region bounded by the curves $y = x^2$ and $y = 2x$. A graph of this region is below.



Utilizing our strategy, we will draw a single generic rectangle of width dx whose endpoints lie on the bounding curves. We will label the endpoints of this infinitely thin rectangle on the diagram and draw this rectangle on the side with its length and width labeled. You will see that this will make the computation of areas easier (which is the point). **We insist that you follow this routine on every application, no matter how easy it may seem.** This will serve you well as the applications get more complicated.





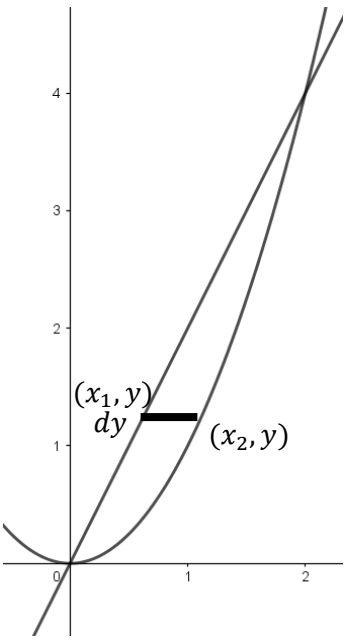
$$Area\ of\ rectangle = (y_2 - y_1)dx$$

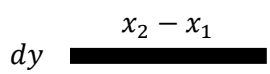
$$Total\ area = \int_{x=0}^{2} (y_2 - y_1)dx$$

This is the set up. Notice that by labeling properly, we could read the area of the rectangle right off the picture. This was the point – to divide the problem into easily computable parts. The integral just added them together. To compute this integral, we need to put everything in terms of one variable. This can done by noticing that $y_2 = 2x$ and $y_1 = x^2$. Thus, we have

$$Total\ area = \int_{x=0}^{2} (y_2 - y_1)dx = \int_{x=0}^{2} (2x - x^2)dx = \left(x^2 - \frac{x^3}{3}\right)\Bigg|_{x=0}^{2} = \left(2^2 - \frac{2^3}{3}\right) - \left(0^2 - \frac{0^3}{3}\right) = \frac{4}{3}$$

Of course, there is nothing special about getting everything in terms of x. We could divide the region into horizontal rectangles and put everything in terms of y. We should obtain the same area. Notice that in the previous set up out two points had the same x coordinate. In this one, they will have the same y coordinate.





$$Area\ of\ rectangle = (x_2 - x_1)dy$$

$$Total\ area = \int_{y=0}^{4} (x_2 - x_1)dy$$

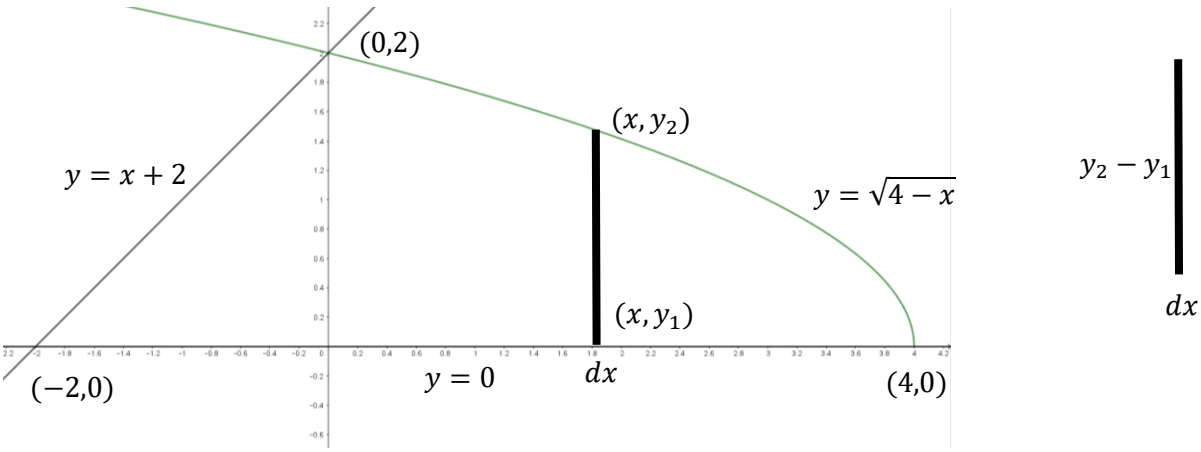
Utilizing the fact that (x_2, y) lies on the curve $y = x^2$and that (x_1, y) lies on the curve $y = 2x$, we get

$$Total\ area = \int_{y=0}^{4} (x_2 - x_1)dy = \int_{y=0}^{4} \left(y^{\frac{1}{2}} - \frac{y}{2}\right)dy = \left(\frac{2y^{\frac{3}{2}}}{3} - \frac{y^2}{4}\right)\Bigg|_{y=0}^{4} = \left(\frac{2(4)^{\frac{3}{2}}}{3} - \frac{4^2}{4}\right) - (0 - 0)$$
$$= \frac{16}{3} - 4 = \frac{4}{3}$$

which is the same answer as before. You can choose whichever way you want; it depends on whether you ultimately want things in terms of x or in terms of y. In this example, one was not much harder than the other. Sometimes the difference in difficulty will help you make the decision. In the next example, we will

set up the problem in both ways and then determine which will be the easiest route to follow. After you have developed some experience, you will often be able to determine this before actually writing anything down.

Example: Compute the area of the following region bounded by $y = 0, y = x + 2$, and $y = \sqrt{4 - x}$. If we want everything in terms of x then we need to draw a generic vertical rectangle of width dx, which we've included in the diagram along with all of the appropriate labeling.



$$Area\ of\ rectangle = (y_2 - y_1)dx$$

$$Total\ area = \int_{x=-2}^{4} (y_2 - y_1)\, dx$$

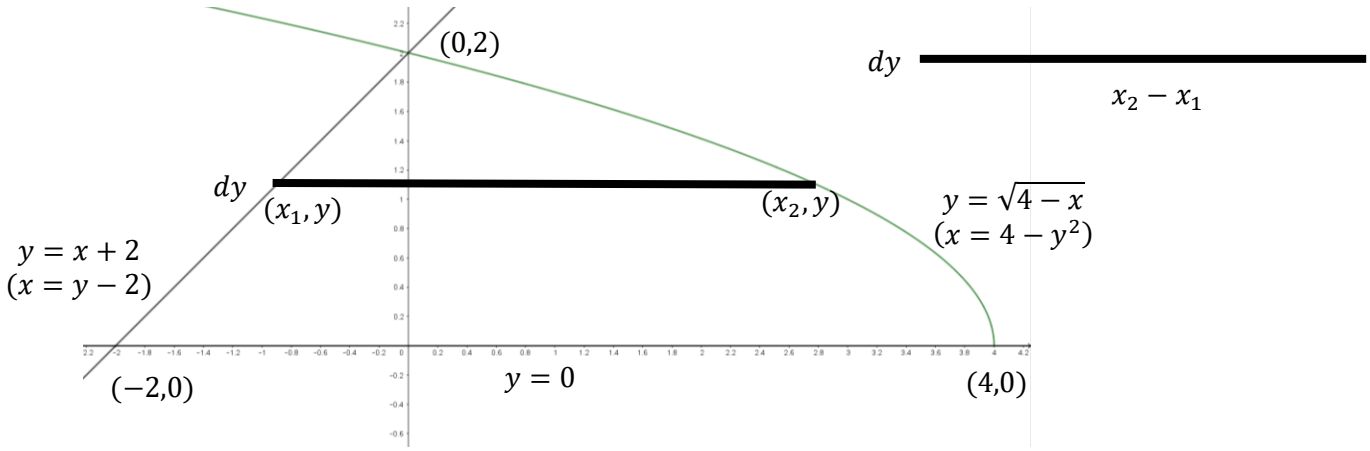
Everything is just as before. The real challenge comes from trying to put everything in terms of x. $y_1 = 0$ on the entire interval, but the formula for y_2 changes. Remember that the rectangle is a generic rectangle and represents only one of the infinitely many rectangles that are being integrated on the interval from $x = -2$ and $x = 4$. On $[-2,0], y_2 = x + 2$, while on $[0,4], y_2 = \sqrt{4 - x}$.

We actually know a way to handle this, utilizing our properties of definite integrals. Specifically, we can do the following

$$Total\ area = \int_{x=-2}^{4} (y_2 - y_1)\, dx = \int_{x=-2}^{0} (y_2 - y_1)\, dx + \int_{x=0}^{4} (y_2 - y_1)\, dx$$

$$= \int_{x=-2}^{0} (x + 2 - 0)\, dx + \int_{x=0}^{4} \left(\sqrt{4 - x} - 0\right) dx$$

Each integral can be handled separately, and the total area is the sum of those two values. This is a perfectly acceptable way to do the problem, and each integral is not too bad to do. But how about if we try a horizontal rectangle and putting everything in terms of y?



$$Area\ of\ rectangle = (x_2 - x_1)dy$$

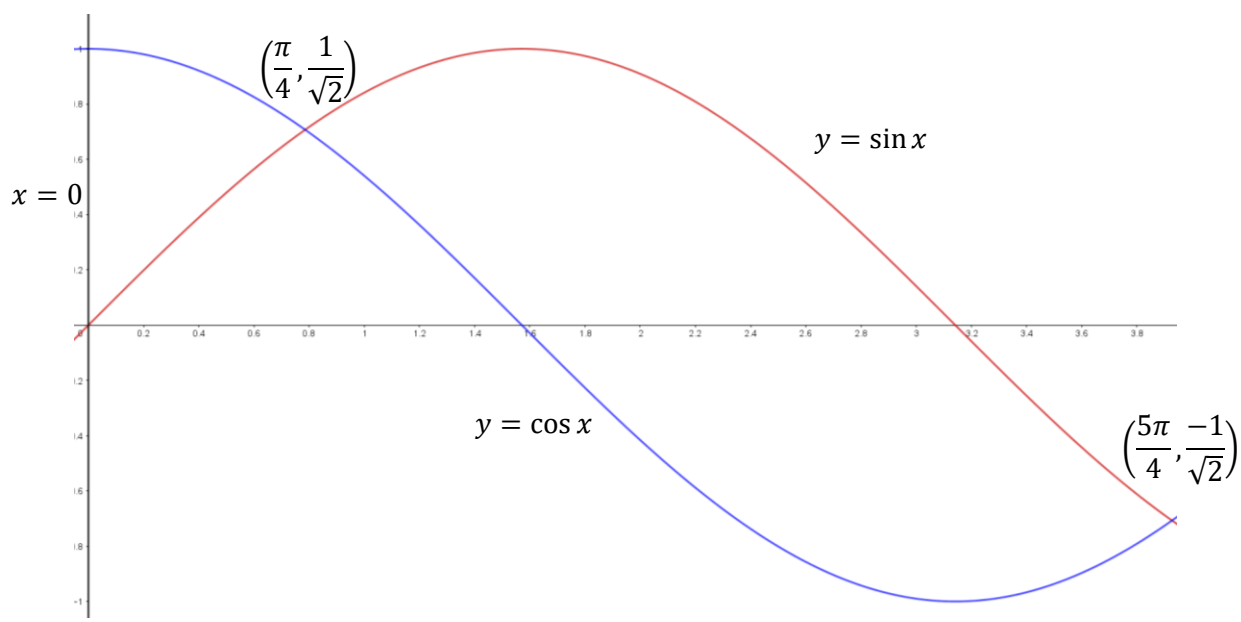
$$Total\ area = \int_{y=0}^{2} (x_2 - x_1)\, dy = \int_{y=0}^{2} \big((4 - y^2) - (y - 2)\big)\, dy$$

Problem in Context: Compute the three integrals in the previous example and verify that you get the same total area done either way.

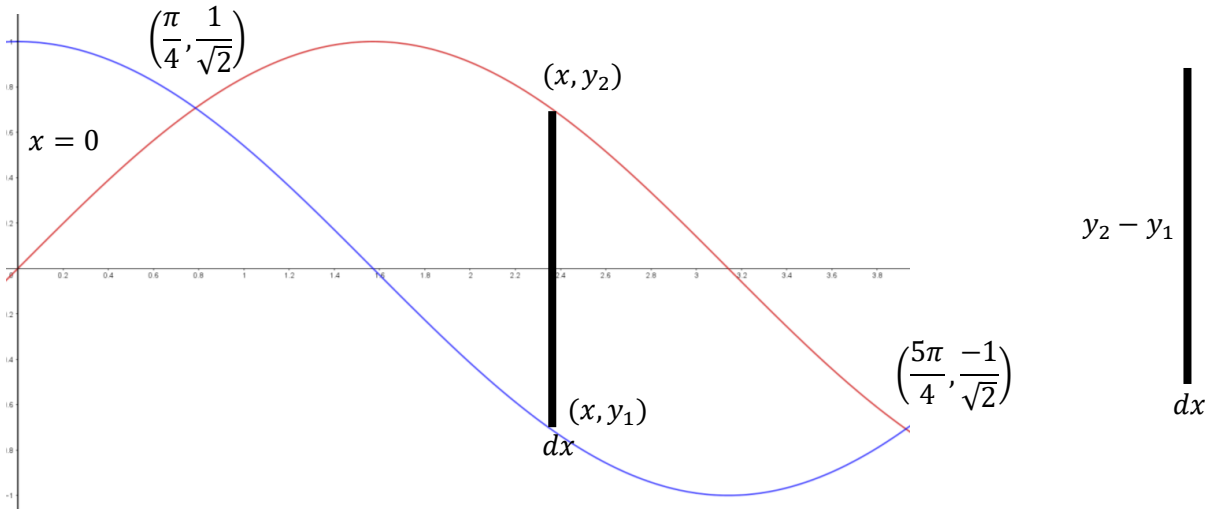
End of PIC

In the previous example, we had an alternative to dividing the interval into subintervals, but this is not always the case.

Example: Compute the following area bounded by the curves $x = 0, y = \sin x, y = \cos x, x = 5\pi/4$.



You should be able to see that drawing a horizontal rectangle and getting things in terms of y is going to be a mess. So, let's put in a vertical rectangle and see what happens. Remember that we are only drawing a single generic rectangle, but it could be positioned anywhere on the interval $[0, 5\pi/4]$.



$$Area\ of\ rectangle = (y_2 - y_1)dx$$

$$Total\ area = \int_{x=0}^{5\pi/4} (y_2 - y_1)dx = \int_{x=0}^{\pi/4} (y_2 - y_1)dx + \int_{x=\pi/4}^{5\pi/4} (y_2 - y_1)dx$$

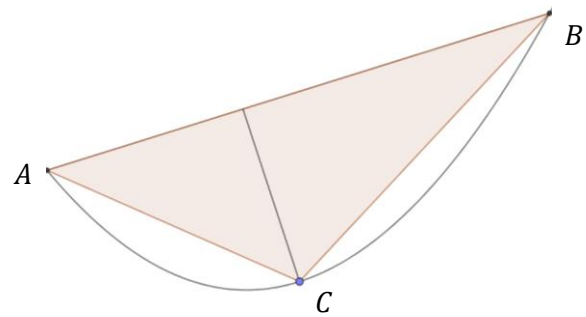
Problem in Context: Finish the previous example to compute the area.

End of PIC

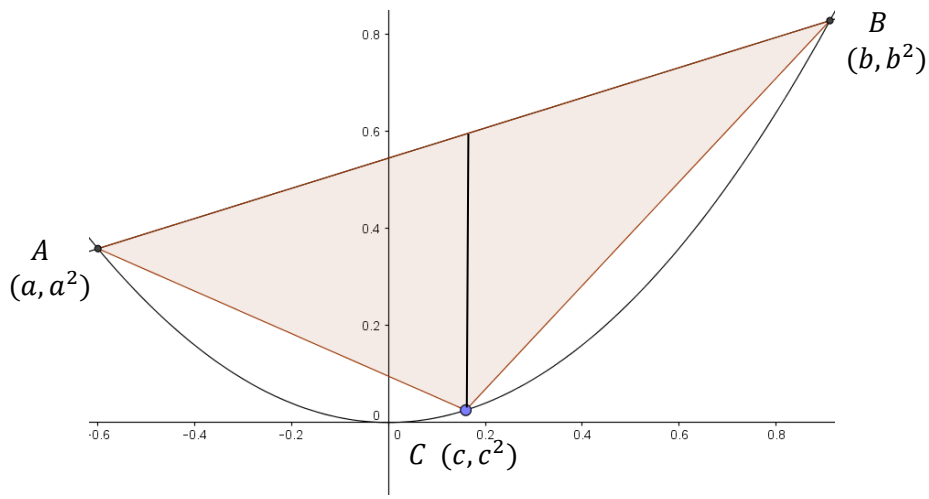
Practice Problems: Compute the areas of the following regions. Make sure you don't skip any steps (drawing and labeling a generic rectangle, etc.)

Problem in Context:

In his treatise, *Quadrature of the Parabola*, Archimedes (287-212 BC) shows that the area of a parabolic segment is $\frac{4}{3}$ the area of its inscribed triangle.



The "inscribed triangle" is the one where the distance from point C to segment $\overline{AB}$ is a maximum. That Archimedes did this without calculus, equations, or coordinate geometry is a tribute to his genius and talent as a mathematician. He didn't even have formulas at his disposal; this is why he stated the area as he did. We will use calculus to confirm Archimedes' result. With this in mind, consider the parabola $y = x^2$, and let $A = (a, a^2),\ B = (b, b^2),\ C = (c, c^2)$.

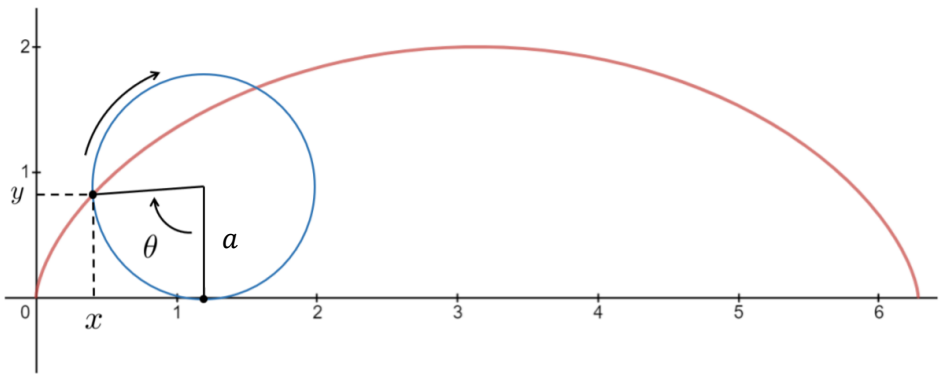


a. Find the area of ΔABC in terms of a, b, c. [Hint: You can do this in many ways, but the "easiest" way might be to draw a vertical line through C and use this to divide the triangle into two triangles (only a suggestion).]

b. Find the coordinates of C which maximizes the distance from C to segment $\overline{AB}$ and verify Archimedes' result that the area of the parabolic segment is $\frac{4}{3}$ the area of ΔABC. [Hint: Again, you can maximize this distance in many ways, but the easiest way might be to notice that the maximum distance (measured along the perpendicular to $\overline{AB}$) occurs when the vertical distance is maximized (again, only a suggestion).]

End of PIC

Recall that in Problem 4.5.5, we looked a cycloid. This was a curve traced out by a point P on a circle as the circle rolls along a straight line.



In that problem, we mentioned that this curve has fascinated mathematicians for a long time. We also had you show that the coordinates of the point P are given by

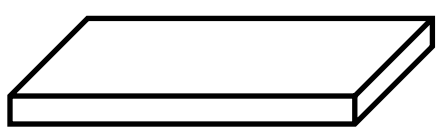
$$x = a(\theta - \sin\theta) \qquad y = a(1 - \cos\theta)$$

Galileo actually gave the curve its name in 1599 (though he was not the first to study it) and attempted to compute the area under one arch. He couldn't accomplish this mathematically, but by weighing pieces of metal, one in the shape of a cycloid and one in the shape of its generating circle, Galileo surmised that the area of the cycloid was approximately three times that of the circle. In 1634, Gilles Personne de Roberval showed that the area under one arch of the cycloid is exactly three times that of its generating circle. This predated the invention of calculus, but he utilized infinitely thin rectangles very much like we have been doing.

Problem in Context: Show that the area under one arch of the above cycloid is $3\pi a^2$, so it is, in fact, three times the area of the generating circle. [Hint: You can set up the area with vertical rectangles just as before, only now get dx in terms of $d\theta$.]

End of PIC

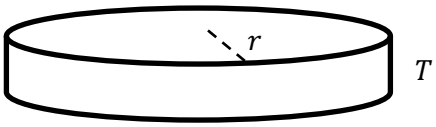
Volumes: As we stated before and as you have seen, we have been focusing on areas to help us understand properties of definite integrals. However, this idea of adding infinitesimal quantities can be applied in a number of other applications. Keep in mind our general strategy of applying a relatively simple formula on an infinitely small piece and then integrating (adding) these infinitely small quantities. Take, for example, the volume of the following box.

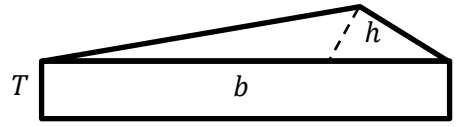


You probably remember the volume of this box as length times width times height. Let's restate the volume in this way

$$Volume\ of\ box = area\ of\ face \times thickness$$

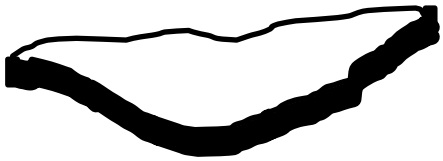
The advantage of thinking this way is that it applies to more general objects. For example, the volume of the following circular cylinder and prism follow the same formula





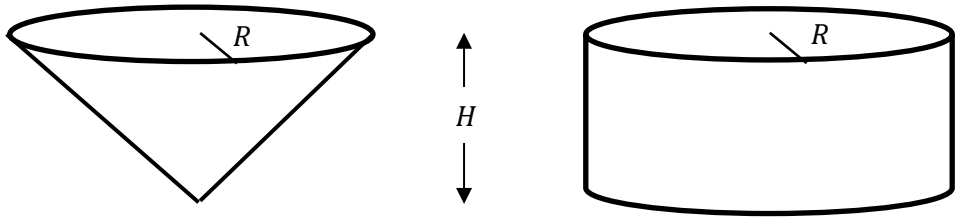
$$Volume\ of\ cylinder = area\ of\ face \times thickness = \pi r^2 T$$
$$Volume\ of\ prism = area\ of\ face \times thickness = \frac{1}{2} bhT$$

This idea even applies to "slabs" that have an irregular face.

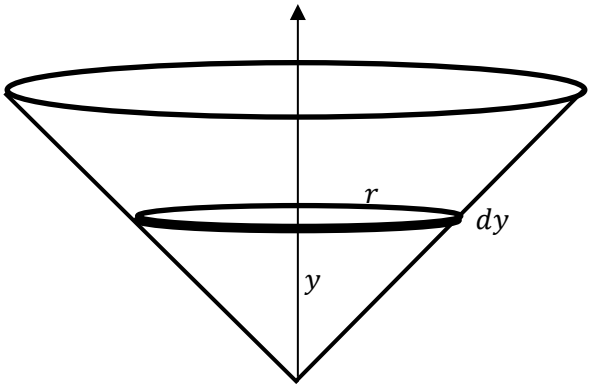


$$Volume\ of\ slab = area\ of\ face \times thickness$$

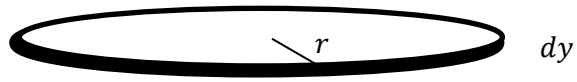
Combining this formula with our idea of dividing and integrating provides a way to compute the volumes of more general solids. For example, if you use the following cone to fill the corresponding cylinder, it seems that it takes three cones to fill the cylinder.



Let's verify this. We know that the volume of the cylinder is given by $\pi R^2 H$. We need to show that the volume of the cone is 1/3 of that. With this in mind, we will put in a vertical axis, label it y, and divide the cone into infinitely thin slices. As before, we will only draw one generic slice, but you must imagine that this cone is comprised of infinitely many such slices stacked.



Drawing this generic slice separately (remember our insistence on this), we can compute its volume.

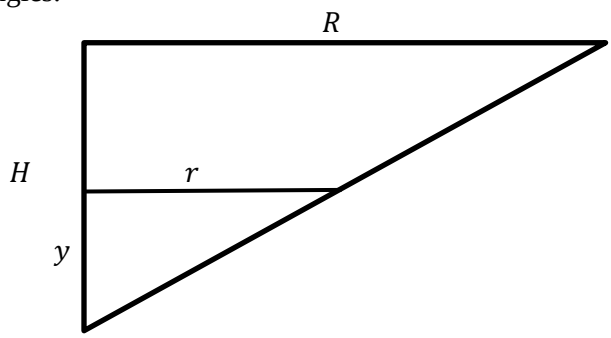


$$Volume\ of\ slice = area\ of\ face \times thickness = \pi r^2 dy$$

Integrating these volumes, we get

$$Volume\ of\ cone = \int_{y=0}^{H} \pi r^2 dy$$

What is left now is to put r in terms of y so the integral can be computed. This can be done by noticing that we have similar triangles.



This gives us the proportion

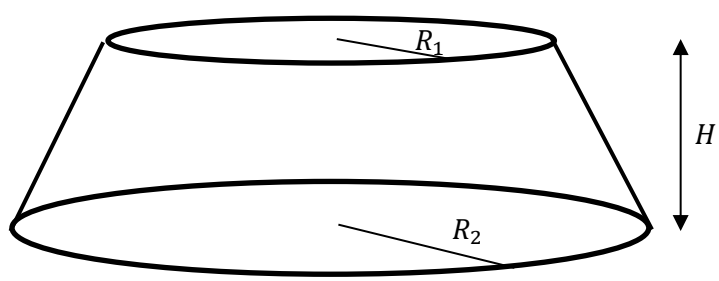
$$\frac{r}{y} = \frac{R}{H}$$

So $r = \frac{R}{H} y$.

Problem in Context: Substitute this value for r into the integral and show that the volume of the cone is, in fact, 1/3 the volume of the cylinder.

End of PIC

Problem in Context: A frustum of a cone is basically a cone with the top chopped off.

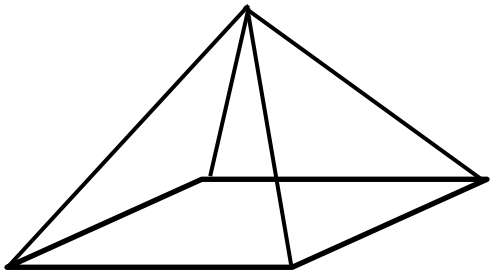


a. Use a definite integral to compute the volume of this frustum. {Remember to do the entire set up as above – we insist!]

b. Verify your answer in part a by computing the volume of the frustum without calculus. This will involve finding the volume of the cone from which the frustum was cut and subtracting the volume of the top that was cut off. To do this, you will need to find the height of that cone.

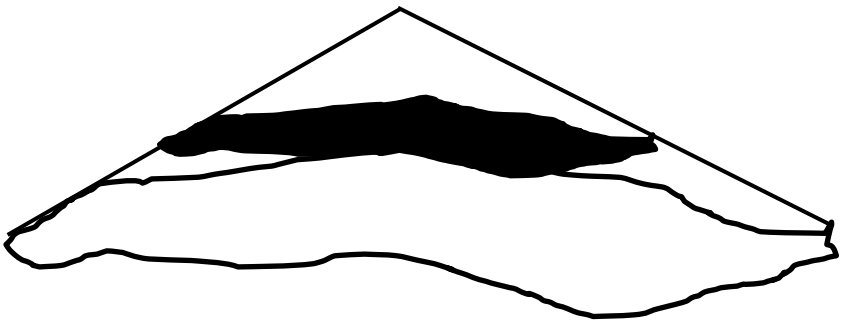
End of PIC

Problem in Context: Use the same idea as above to show that the volume of a square based pyramid is 1/3 the volume of the box with the same base and height



End of PIC

Problem in Context: Consider the following "generalized cone."



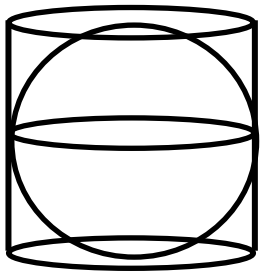
Show that the volume of this cone is 1/3 the volume of the "generalized cylinder" with the same height and same base. That is, show that the volume of this cone is $\frac{1}{3}(Area\ of\ base) \times height$

[Hint: The generic slice is similar to the base. When you set up a proportion between the areas, it is proportional to the squares of the distances from the apex.]

End of PIC

Marcus Tullius Cicero (106-43 BC) is considered to be one of the greatest orators and philosophers of the late Roman Republic. When he was quaestor (a Roman official) in Syracuse (Sicily) in 75 BC, he

set out to find the tomb of the great Greek mathematician Archimedes who died during the Roman siege of the city 137 years earlier. Local Syracusans denied any knowledge of his grave, but Cicero nevertheless discovered a tombstone overgrown with brambles and thorns. When he saw the following symbol on the stone, he knew that he found the grave of Archimedes.

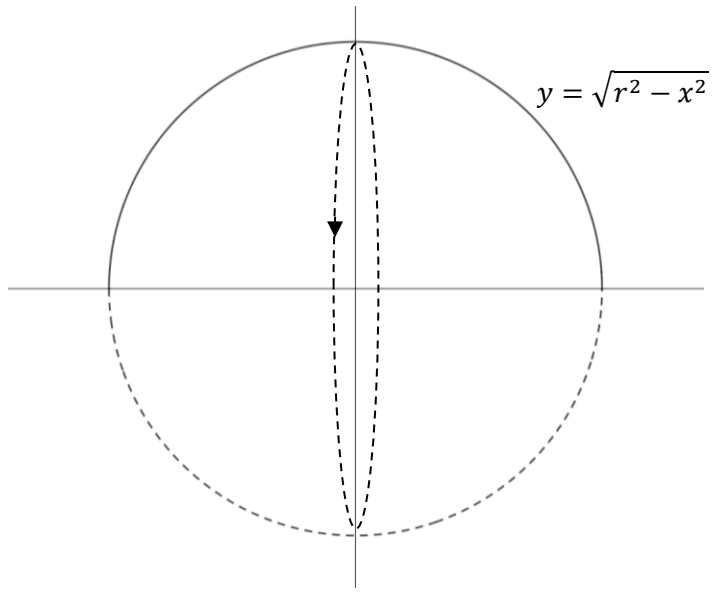


http://mathshistory.st-andrews.ac.uk/Biographies/Archimedes.html

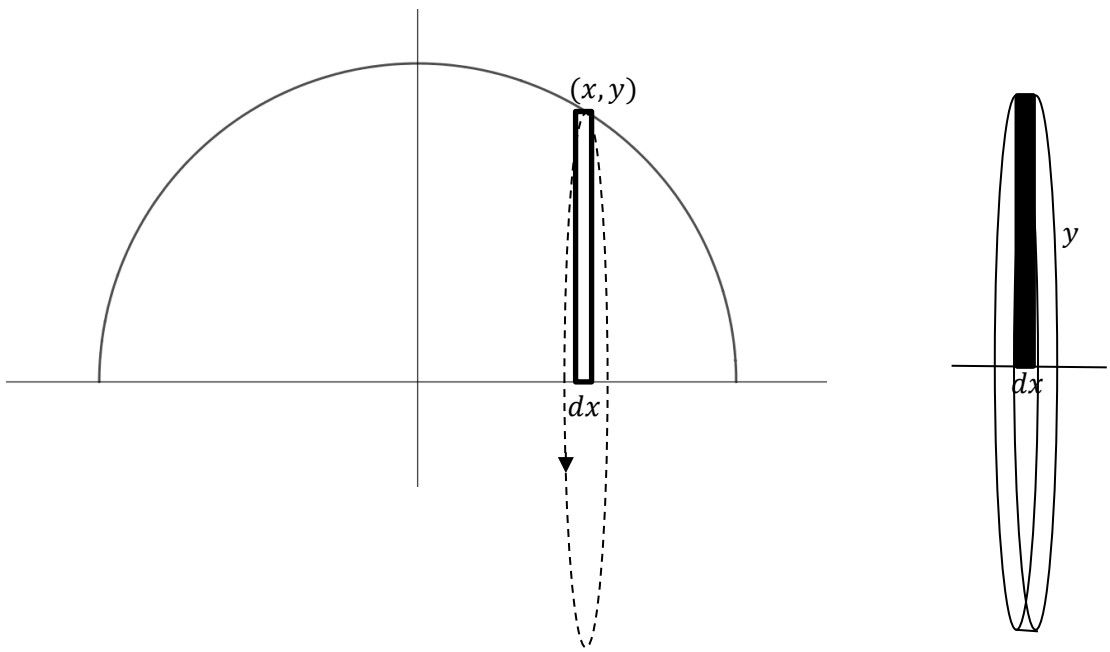
How did Cicero surmise that this was the grave of Archimedes? It seems that out of all of Archimedes works, he was proudest of his treatise *On the Sphere and the Cylinder*. In it, Archimedes proves that the volume and surface area of a sphere is 2/3 that of its circumscribed cylinder. He was so proud if this, that he instructed that this be carved in his tombstone. This leads to the modern formulas for the volume and surface area of a sphere of radius r

$$Volume = \frac{2}{3}(\pi r^2 \cdot 2r) = \frac{4}{3}\pi r^3$$
$$Surface\ area = \frac{2}{3}(2\pi r \cdot 2r + 2 \cdot \pi r^2) = 4\pi r^2$$

The way in which Archimedes discovered these formulas has led some to call him the inventor of integral calculus. This claim is arguable, but at the very least, his techniques certainly were a precursor to integral calculus. While we won't go into Archimedes' exact method, we will use integral calculus to derive these formulas for volume and surface area. We will focus of the volume now and come back to the surface area later. To compute this volume, we will consider that a sphere of radius r can be generated by revolving the curve $y = \sqrt{r^2 - x^2}$ about the x-axis.



We will draw and label a generic rectangle of width dx and height y and compute the volume of the disk generated by revolving this rectangle about the x-axis. As before, we will draw and label the disk on the side.



$$Volume\ of\ disk = \pi y^2 dx$$

$$Volume\ of\ sphere = \int_{x=-r}^{r} \pi y^2\, dx$$

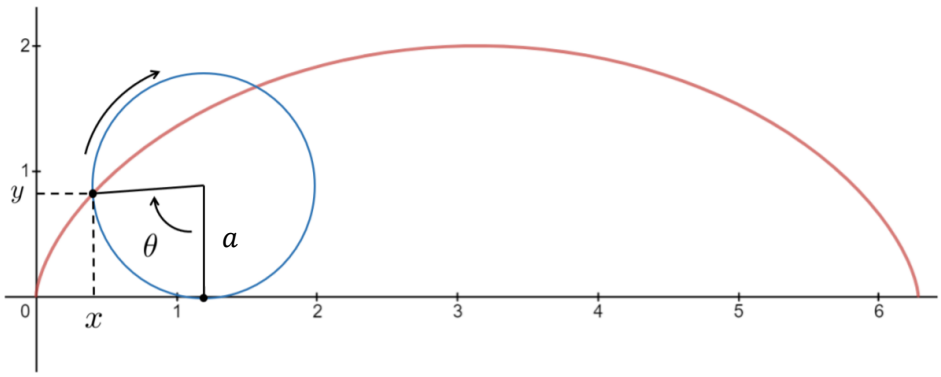
Again, we insist that you go through this process (including drawing all of the pictures) instead of trying to memorize some formula. Admit it, with the disk drawn and labeled, wasn't determining the volume of disk relatively easy. The integral only came in when we wanted to "integrate" all of the volumes of the disks together.

Problem in Context: Compute the integral above to determine that the volume of the sphere is, in fact, $\frac{4}{3}\pi r^3$.

End of PIC

Problem in Context: (I added this one in on Dec 29, 2022):

Find the volume of the solid generated by revolving one arch of the following cycloid about the x axis.

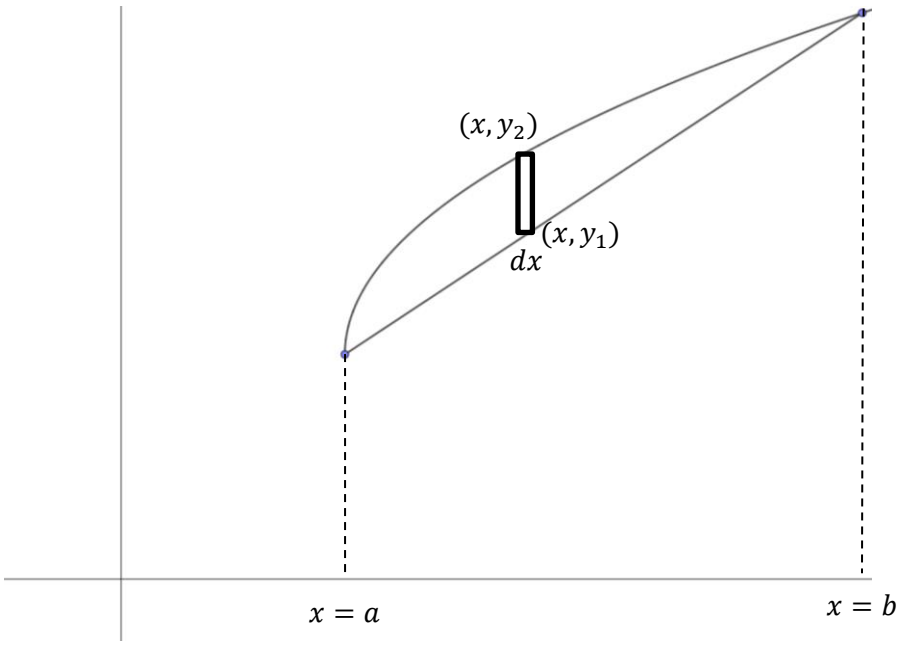


$$x = a(\theta - \sin\theta) \qquad y = a(1 - \cos\theta)$$

End of PIC

Of course, this technique can be applied to find the volume of a more general "solid of revolution."

Problem in Context: Consider the following region bounded by the curves $y = y_1(x)$ and $y = y_2(x)$ and a generic rectangle of width dx inside it.



a. Draw and label the thickness and inside and outside radii of the "washer" created by revolving this generic rectangle about the x-axis.

b. Use the result to show that the volume of the solid generated by revolving this region about the x-axis is given by

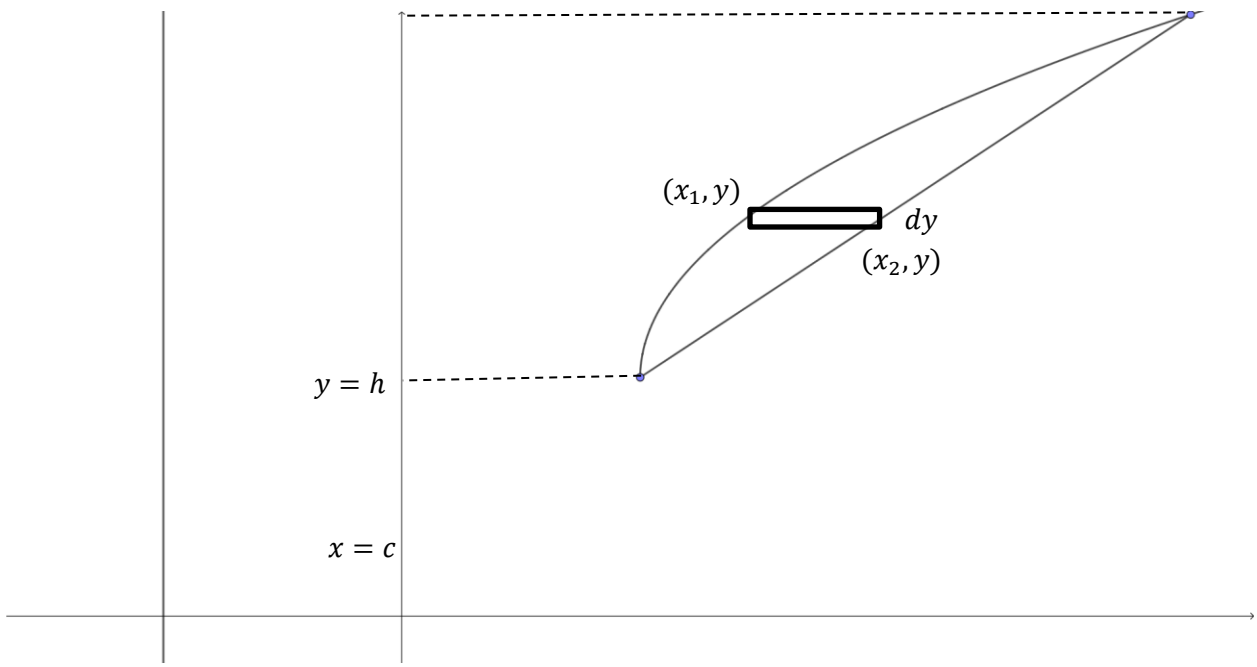
$$\pi \int_{x=a}^{b} (y_2^2 - y_1^2)dx$$

c. To emphasize that you should use the process and not memorize a formula, find the integral that will compute the volume of the solid generated by revolving this region about a horizontal line $y = c$ not passing through the region. You can assume for the sake of argument that the line lies below the region.

End of PIC

Problem in Context: Consider the following region bounded by the curves $x = x_1(y)$ and $x = x_2(y)$ and the vertical line $x = c$ to the left of this region.

$y = k$



a. Draw and label the washer created by revolving the horizontal rectangle around the line $x = c$ and use this to find a formula for the volume of this washer.

b. Integrate your answer from part a. to show that the volume of the solid formed by revolving the region about the line $x - c$ is given by

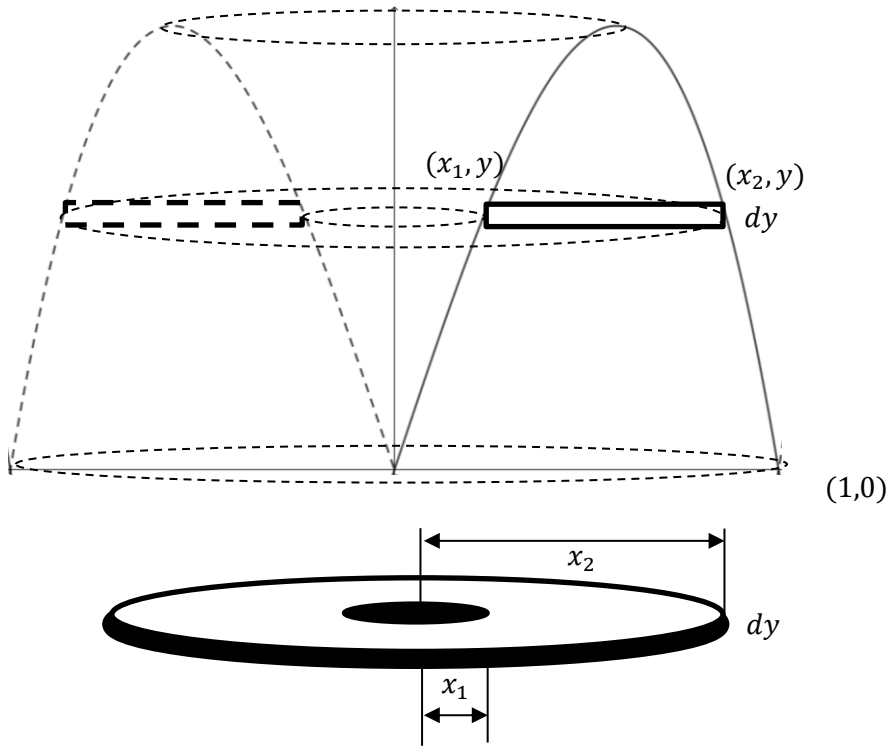
$$\pi \int_{y=h}^{k} [(x_2 - c)^2 - (x_1 - c)^2]\, dy$$

End of PIC

Practice Problems: Compute the volumes of the solids formed by generating the given region around the given lines. Make sure that in each case you first draw and label the washer (or disk) generate by revolving a generic rectangle about the line to set up your integral and write down the volume of this washer (disk). [If you do this right, you can read off the integral right from the picture. (Why make things more difficult?) Computing it is another matter, but you've done this before.]

Volumes by Shells.

As you no doubt noticed, when we had to compute the volume of a solid generated by revolving a region about the x-axis (or any horizontal line), we obtained an integral with dx in it, which meant that we would put everything in terms of x. Likewise, when we revolved about the y-axis, we had to put everything in terms of y. Sometimes this is not practical or leads to an undesirable integral. Consider the following example of the region bounded by $y = 0$ and $y = 3(x - x^3)$ revolved about the y-axis. Going through our set up (Yes, we must do it too!), we have the following generic rectangle revolved around to create a washer.

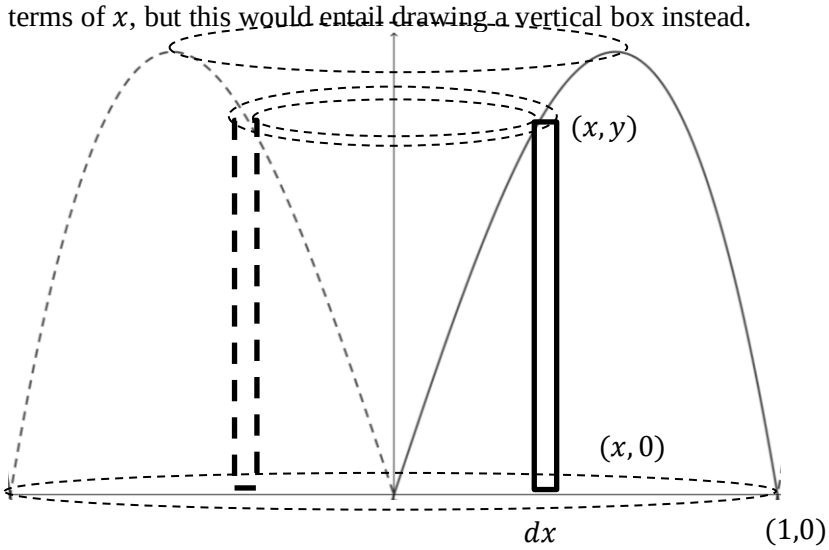


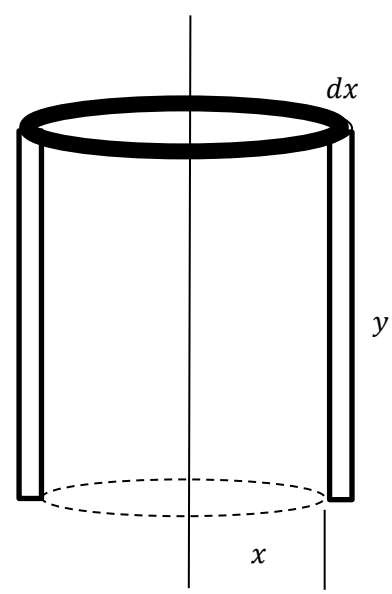
$$Volume\ of\ washer = Volme\ of\ disk - volume\ of\ hole = \pi x_2^2\ dy - \pi x_1^2\ dy = \pi(x_2^2 - x_1^2)dy$$

$$Total\ Volume = \int_{y=0}^{\max y} \pi(x_2^2 - x_1^2)\ dy$$

This set up (with the picture) was really the easy part. The harder part is putting everything in terms of y and integrating. For instance, we would need to solve $y = 3(x - x^3)$ for x in terms of y. Furthermore, we would need to figure out the maximum value of $y = 3(x - x^3)$ on the interval [0,1].

All of this is doable, but inconvenient. What would be preferable would be to leave everything in terms of x, but this would entail drawing a vertical box instead.





If we were to treat the right-hand object as a very tall washer with inside radius x, outside radius $x + dx$, and height y, we would get its volume to be

$$\pi((x + dx)^2 - x^2)y = \pi(x^2 + 2x\,dx + (dx)^2 - x^2)y = \pi(2xdx + (dx)^2)y$$

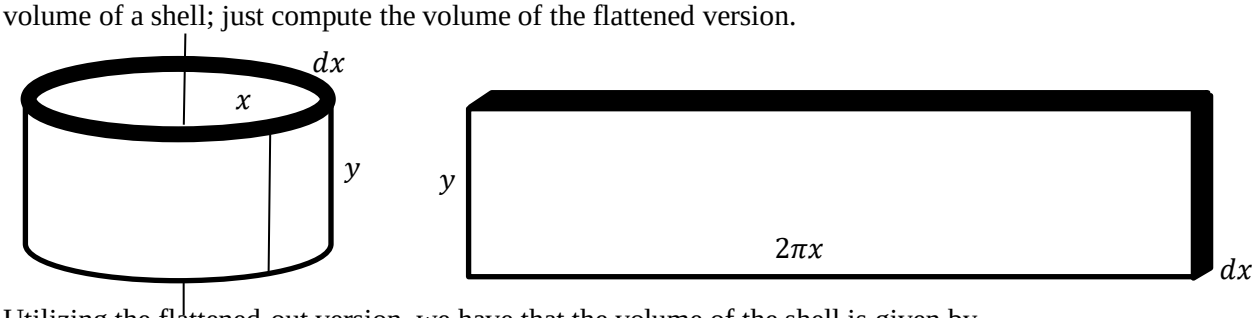
Recalling our reasoning with the product rule, we can ignore $(dx)^2$ as it is infinitely small compared to dx, so that the volume of this tall washer is

$$2\pi xy\,dx$$

which we could then integrate to obtain

$$Total\,Volume = \int_{x=0}^{1} 2\pi xy\,dx = 2\pi \int_{x=0}^{1} x(3(x - x^3)dx = 6\pi \int_{x=0}^{1} x^2 - x^4\ dx = 6\pi\left(\frac{1}{3} - \frac{1}{5}\right) = \frac{12\pi}{15}$$

If you feel as funny (or perhaps more) about ignoring the $(dx)^2$ as you did with the product rule, there is another way to look at this that might help your queasiness and provide a better way of remembering what to do in this situation. Basically, instead of calling the revolution of the generic rectangle a tall washer, we will call it a (cylindrical) shell. This sounds somewhat silly, but it really points out the difference. For a washer, the height is infinitesimal, whereas for a shell, the thickness of the wall is infinitesimal. For a physical analogy, this is the difference between an actual metal washer and a piece of metal tubing. To figure out the volume of the tubing (shell), we can slice it open and flatten it out into a rectangular piece of metal. This is not so easily done with a washer. This provides a shortcut (and device) for computing the volume of a shell; just compute the volume of the flattened version.



Utilizing the flattened-out version, we have that the volume of the shell is given by

$$Volume\ of\ shell = length \cdot height \cdot thickness = circumference \cdot height \cdot thickness = 2\pi xy\,dx$$

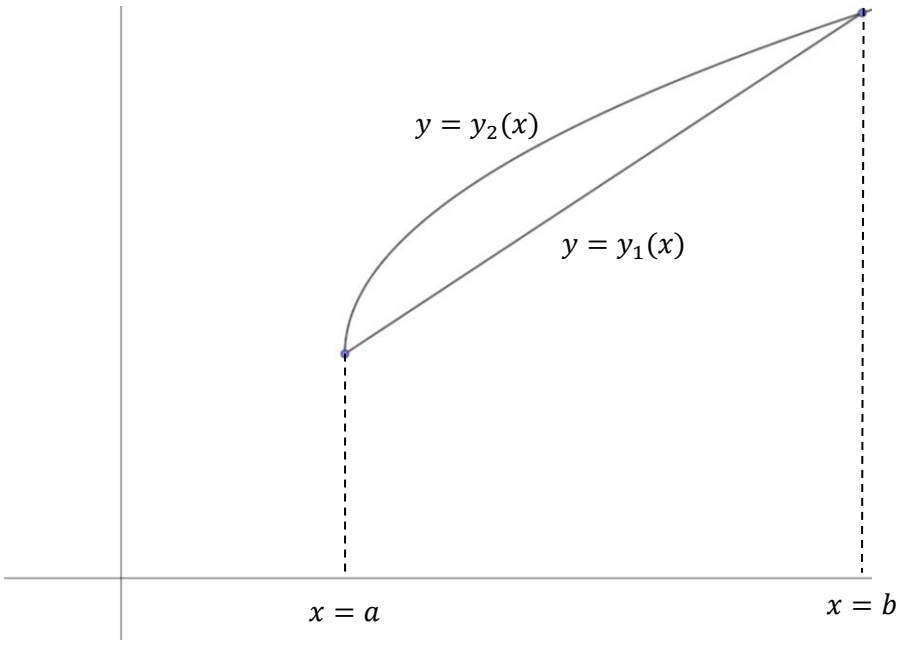
as we obtained before.

Actually, a similar formula works for any washer.

$$Volume\ of\ a\ washer = \pi(x_2^2 - x_1^2)dy = \pi(x_2 + x_1)(x_2 - x_1)dy = 2\pi\frac{(x_2 + x_1)}{2}dy(x_2 - x_1)$$
$$= average\ circumference \cdot height \cdot thickness$$

While this is interesting geometrically, it does not help with computing an integral using washers.

Problem in Context: Consider the following region bounded by the curves $y = y_1(x)$ and $y = y_2(x)$



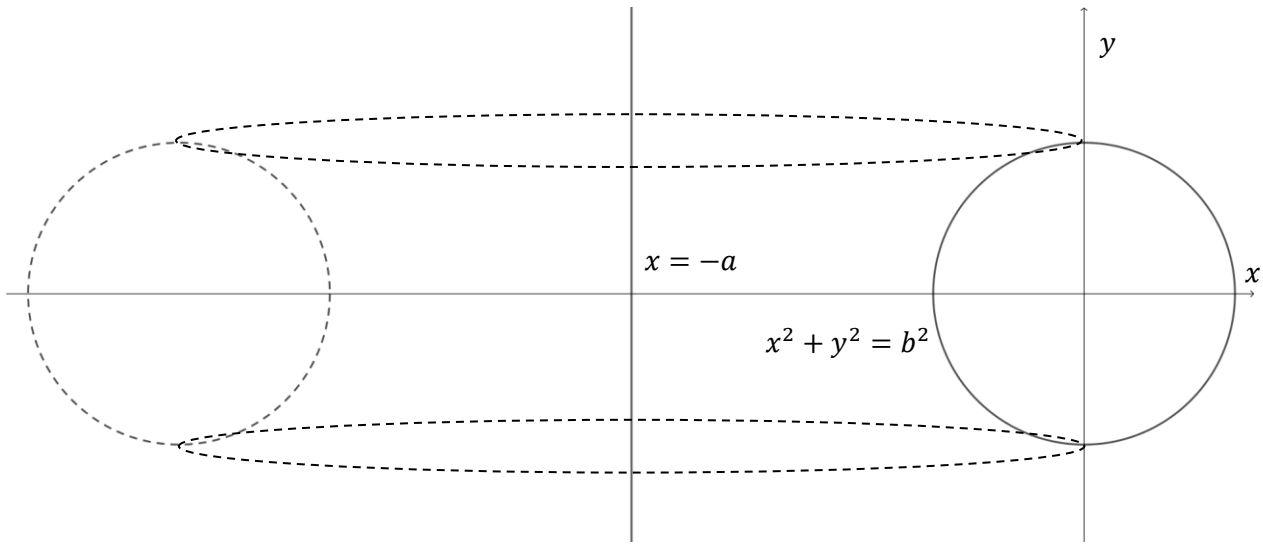
a. Draw and label a generic vertical rectangle in this region and draw and label the shell generated by revolving this rectangle about the vertical line $x = c$ to the left of the region.

b. Compute the volume of this shell and integrate it to show that the volume of the solid generated by revolving the region about the line $x = c$ is given by

$$2\pi \int_{x=a}^{b} (x - c)(y_2 - y_1)\, dx$$

End of PIC

Problem in Context: Consider a circle of radius b, whose center revolves around a line distance $a > b$ away. The "donut" formed is actually called a torus. Consider the following torus generated by revolving the circle $x^2 + y^2 = b^2$ about the line $x = -a$.

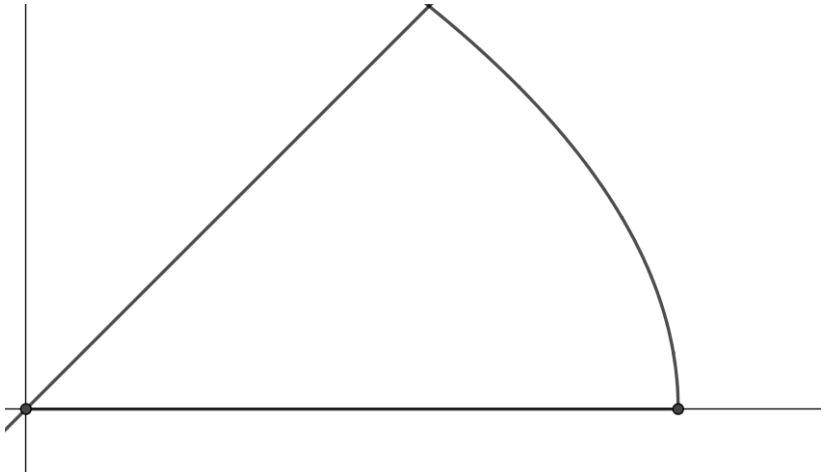


Show that the volume of this torus equals the area of the small circle times the circumference of the circle generated by revolving the center of this circle around the line. This was derived by Johannes Kepler (1571-1630) and is a special case of a theorem by Pappus of Alexandria (290-350 AD).

End of PIC

So a natural question arises, "Which should I use, washers or shells?" The answer is that you can use either one; sometimes it is more convenient to use one over the other, other times it really doesn't matter. The real question you need to ask in a particular problem is, "Is it better to put things all in terms of x or in terms of y. This will determine if you want to utilize a vertical rectangle (of width dx) or a horizontal rectangle (of width dy). This will determine whether washers or shells are more appropriate.

Example: Consider the following region bounded by $y = \sqrt{1-x}, y = x, y = 0$



If we were to draw a vertical rectangle involving dx, then any integral would, of necessity, need to be divided into two separate integrals to compute, since the y coordinate of the upper point on the rectangle changes from one curve to the other. Utilizing, a horizontal rectangle would circumvent this problem. However, we would need to be prepared to put everything in terms of y instead of x. Luckily, these two

equations don't look that bad with regard to this. We would still need to find the point of intersection, but that would have been the case with a vertical rectangle as well.

Problem in Context: Find the volumes of the solids generated by revolving this region about the x-axis and y-axis utilizing horizontal rectangles.

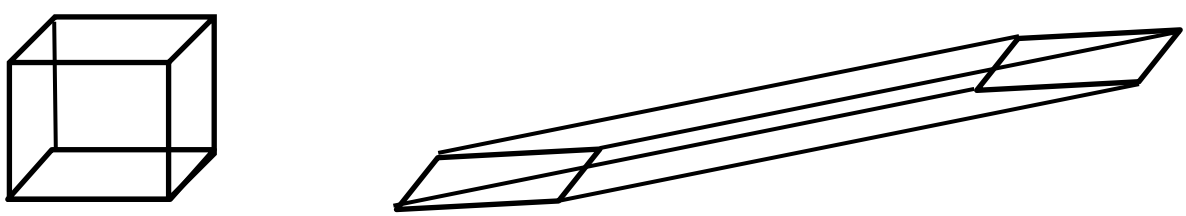
End of PIC

Practice Problems: Determine the volumes of the solids generated by revolving the following regions about the given lines. Utilize either washers or shells – whichever makes the problem easier. Make sure to include a torus.

Surface Area and Arc Length

Recall that we mentioned that in his treatise *On the Sphere and the* Cylinder, Archimedes not only showed that the sphere had 2/3 the volume of its circumscribing cylinder, but it also had 2/3 the surface area. Again, we will not duplicate how Archimedes did it, but we will verify it with calculus.

Surface area poses a bit more of a challenge that volume. This can be illustrated by the following two boxes

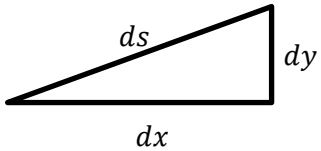


The boxes have the same base and same height, so they have the same volume. However, the slanted box clearly has more surface and could be stretched as far as you like, so the there really is no limit to how large the surface area could be made. The slant makes a difference in surface area. To see this more carefully, we will need to first develop the notion of arc length.

You have already been exposed to arc length in the differential calculus part of this book. Recall that in looking at the catenary [Section 4.3.1], we had that the hanging chain satisfied the equation

$$\frac{dy}{dx} = \frac{ws}{H}$$

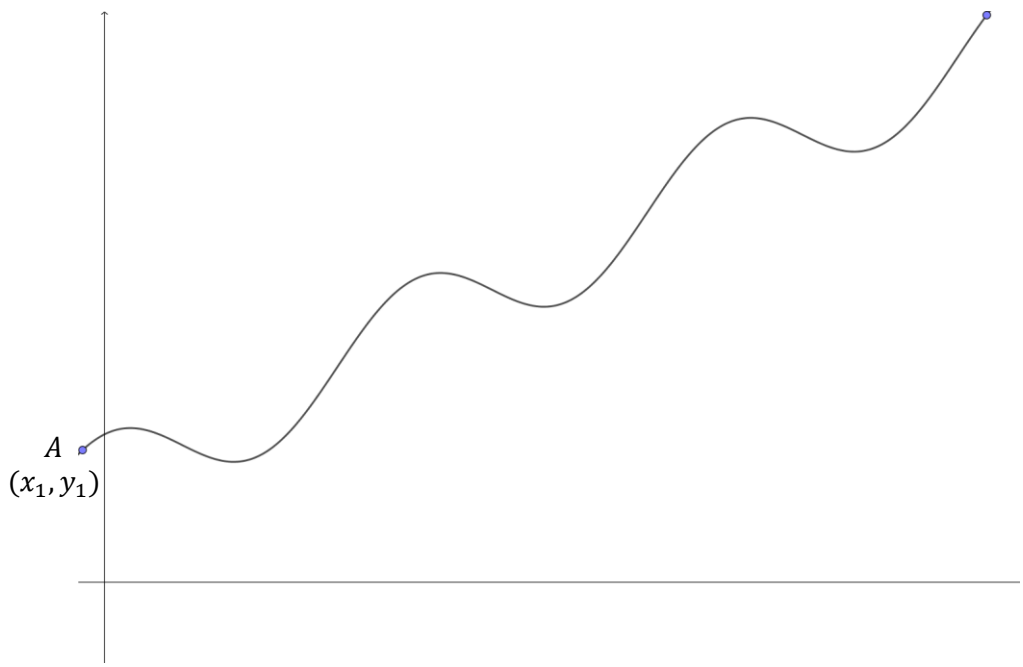
where w, H were the weight density of the chain and horizontal tension, respectively, and s is the length of the chain from the lowest point to (x, y). We then differentiated and applied the Pythagorean Theorem to the "differential" triangle



$$ds = \sqrt{(dx)^2 + (dy)^2}$$

We will exploit this to look at the arc length of a curve. Specifically, suppose we have a curve from point $A = (x_1, y_1)$ to point $B = (x_2, y_2)$ and want to compute the length of this curve s.

B
(x_2, y_2)



If we divide this curve into infinitely small segments, then we can compute its length by integrating the lengths of these segments. In other words, we have

$$s = \int_{X=A}^{B} ds$$

where X denotes a generic point (x, y) on the curve. Utilizing our infinitesimal Pythagorean Theorem, we have

$$s = \int_{X=A}^{B} ds = \int_{X=A}^{B} \sqrt{(dx)^2 + (dy)^2}$$

To actually compute this integral (via the Fundamental Theorem), we need to put this integral into a form that we can anti-differentiate. For example, we could put everything in terms of x or in terms of y yielding these two possibilities

$$s = \int_{X=A}^{B} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\ |dx| = \int_{X=A}^{B} \sqrt{\left(\frac{dx}{dy}\right)^2 + 1}\ |dy|$$

The absolute value is necessary since we are factoring out $\sqrt{(dx)^2}$ or $\sqrt{(dy)^2}$. In practice, if x is always increasing then $dx > 0$ and so the absolute value isn't necessary. The same applies to y. In many applications, this will be the case, but you should be wary that it can happen.

Problem in Context:

a. Consider the quarter circle given by $y = \sqrt{1 - x^2}$, $-\frac{1}{\sqrt{2}} \le x \le \frac{1}{\sqrt{2}}$. Since the circumference of the unit circle is 2π, then the length of this quarter circle should be $\frac{\pi}{2}$. Use an integral to verify this result. [There is a reason we didn't have you compute the length of the entire semicircle $y = \sqrt{1 - x^2}$, $-1 \le x \le 1$. If you can't see what it is, don't worry, we will come back to it later.]

b. Suppose we tried to compute the length of a quarter of the circle in the following way.

$$\int_{X=(-\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}})}^{(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}})} \sqrt{\left(\frac{dx}{dy}\right)^2 + 1}\, dy = \int_{y=\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \sqrt{\left(\frac{dx}{dy}\right)^2 + 1}\, dy = 0$$

What is the problem?

End of PIC

Again, there is real temptation to try to memorize the two formulas

$$s = \int_{X=A}^{B} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, |dx| = \int_{X=A}^{B} \sqrt{\left(\frac{dx}{dy}\right)^2 + 1}\, |dy|$$

but we urge you not to. First, why try to remember two formulas, when one just needs to remember the Pythagorean Theorem?

$$s = \int_{X=A}^{B} ds = \int_{X=A}^{B} \sqrt{(dx)^2 + (dy)^2}$$

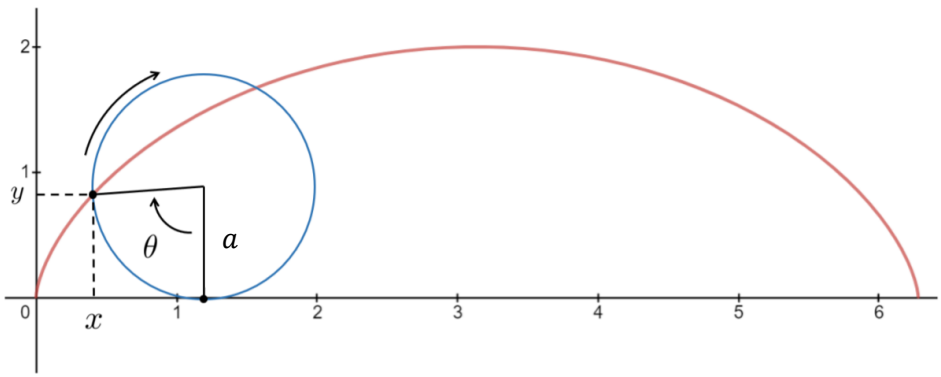
This seems much more natural. At this point, it is just a matter of determining what variable you wish to put everything in terms of. The variable doesn't even need to be x or y. For example, suppose we have $x = x(u)\ y = y(u),\ u_0 \le u \le u_1$. So we don't need to deal with absolute values, let's assume that the parameter u is always increasing so that $du > 0$. Then we could put everything in terms of u.

$$s = \int_{X=A}^{B} ds = \int_{X=A}^{B} \sqrt{(dx)^2 + (dy)^2} = \int_{u=u_0}^{u_1} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du$$

Problem in Context: Utilize the parameterization $x = \cos\theta \quad y = \sin\theta$, $0 \le \theta \le 2\pi$ to verify that the circumference of the unit circle is 2π.

End of PIC

Problem in Context: Here is another interesting fact about the cycloid. Sir Christopher Wren (best known for the buildings he designed after the great fire of London in 1666, including St. Paul's Cathedral), showed in 1658 that the length of one arch of the cycloid is 8 times the radius of the generating circle. He did this without calculus, but we will use integration to verify his result. Consider the following cycloid



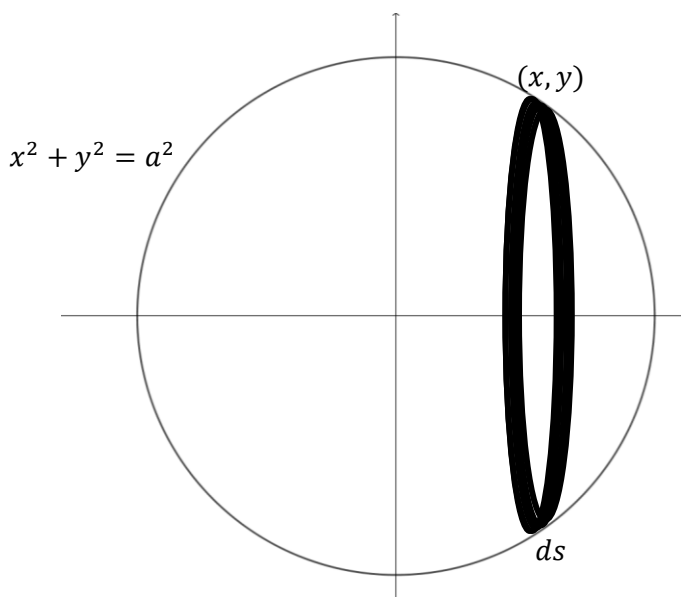
$$x = a(\theta - \sin\theta) \qquad y = a(1 - \cos\theta)$$

Show that the length of one arch of the cycloid is $8a$.

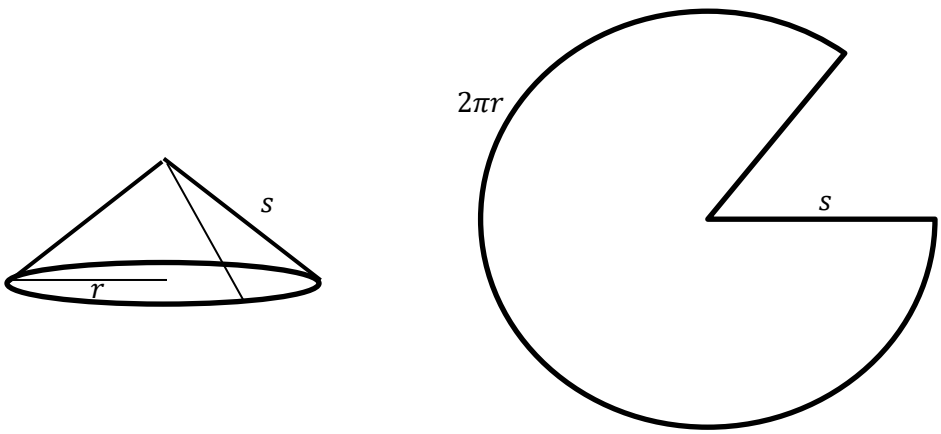
End of PIC

Practice Problems: Compute the length of the following curves. There will be examples in terms of x, y and other parameters.

Now that we have arc length in place, let's go back to the problem of computing the surface area of a sphere. Recalling that surface area needs to take into account the "slant" of the surface rather than just the height, let's actually divide the surface of the sphere into infinitely small bands.

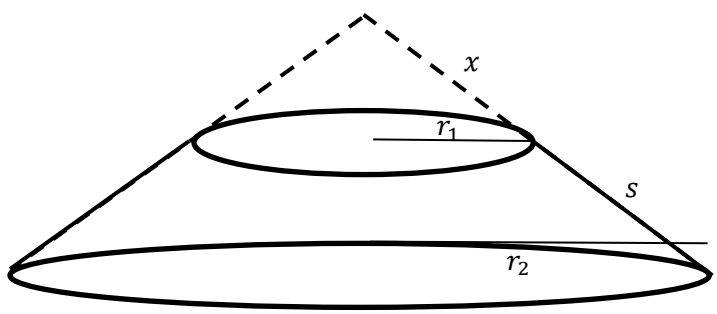


What we need to do is compute the surface area of this band (called a frustum of a cone). To do that, we need to find the surface area of a cone (without the bottom). With that in mind, let's start with a cone with base radius of r and slant length s. If we slice the cone and flatten it (much like a paper cup), then we can determine the surface area of the cone, by finding the area of the sector created by flattening it.



By dividing this sector into infinitely many "infinitely thin" triangles, and integrating the areas of these triangles, we get that the area of the sector (and thus the cone) is given by $\frac{1}{2}2\pi r \cdot s = \pi r s$.

Problem in Context: Consider the frustum cut from the following cone.

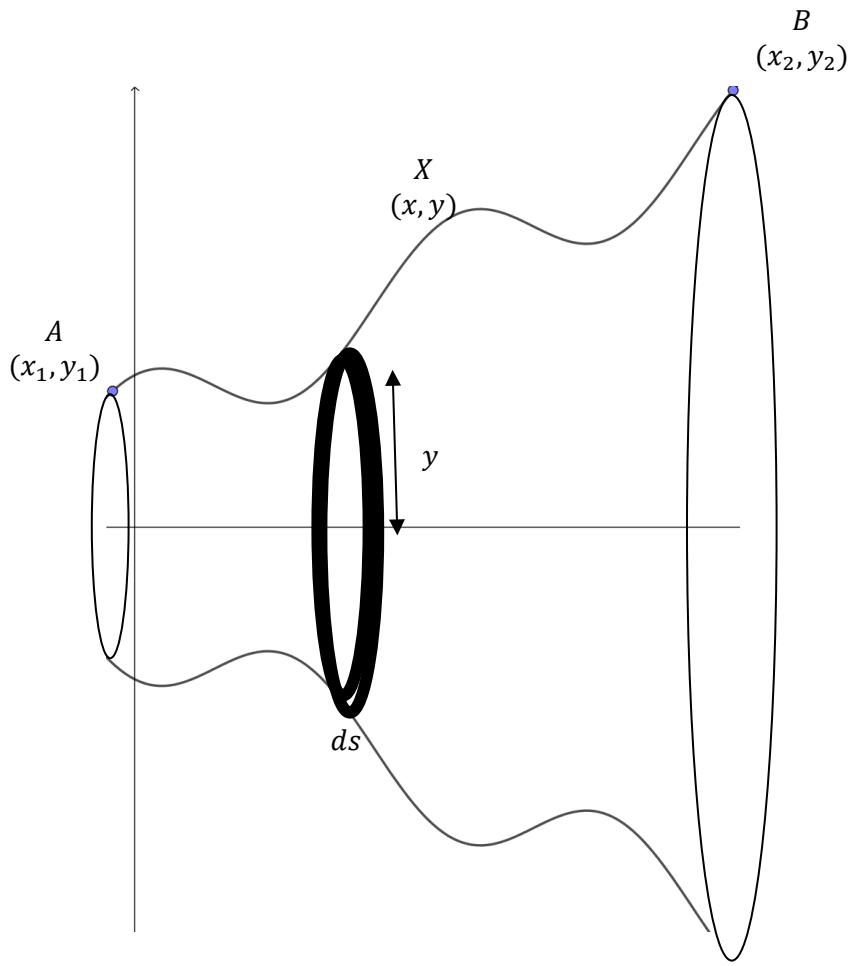


Show that the surface area of the frustum is given by

$$2\pi\left(\frac{r_1 + r_2}{2}\right) \cdot s = average\ circumference \cdot slant\ length$$

End of PIC

Armed by this, we can obtain that the surface area of a surface of revolution obtained by revolving a curve about the x-axis is given by integrating the surface areas of these infinitely thin bands.



$$Total\ Surface\ Area = \int_{X=A}^{B} 2\pi y\ ds$$

As with arc length, you are free to put this integral in terms of whichever variable you wish.

Problem in Context:

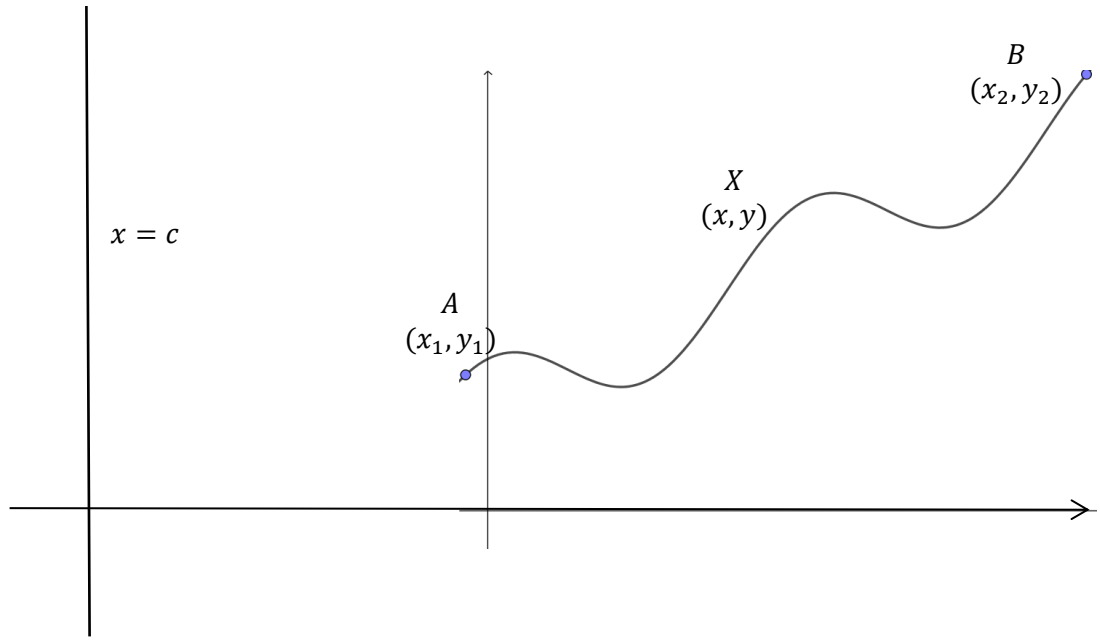
a. Use this with the equation $y = \sqrt{a^2 - x^2}, -a \le x \le a$ to show that the surface area of a sphere of radius a is $4\pi a^2$ as Archimedes indicated.

b. Use the equations $x = a\cos\theta \quad y = a\sin\theta\ ,\ \ 0 \le \theta \le \pi$ to obtain the same answer as part a.

End of PIC

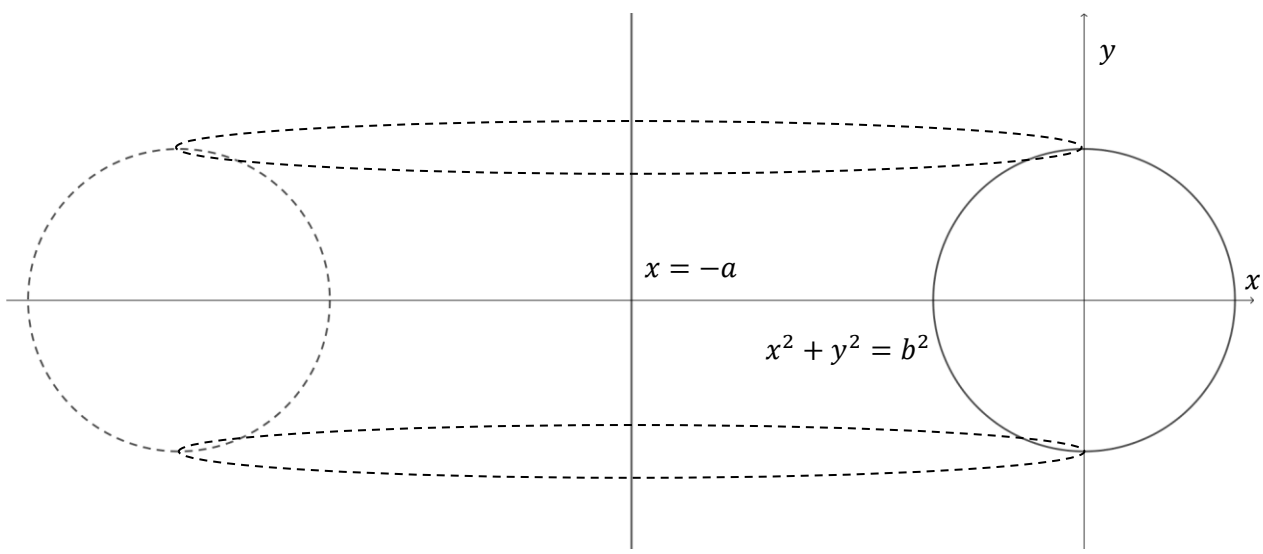
Problem in Context: Show that the surface area of the surface generated by revolving the following curve about the line $x = c$ to the left of curve is given by

$$\int_{X=A}^{B} 2\pi(x - c)ds$$



What if the line was to the right of the curve? What if it passed through the curve?

Problem in Context: Previously, PIC # XX, we showed that the volume of a torus is equal to the area of the generating circle times the circumference of the circle formed by revolving the center of this circle about axis of the torus. One might wonder if a similar result holds for the surface area of a torus; that is, is the surface area of the torus equal to the circumference of the generating circle times the circumference of the circle formed by revolving the center of this circle about the axis of the torus? To answer this, consider the following torus generated by revolving the circle $x^2 + y^2 = b^2$ about the line $x = -a$.



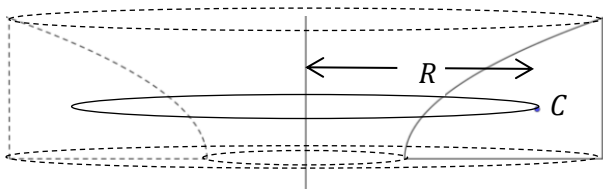
Find the surface area of this torus and use this to answer the above question. [Hint: You might find it easier using the equations $x = b\cos\theta \quad y = b\sin\theta\,,\ 0 \le \theta \le 2\pi$ instead of the equation $x^2 + y^2 = b^2$ for the generating circle.]

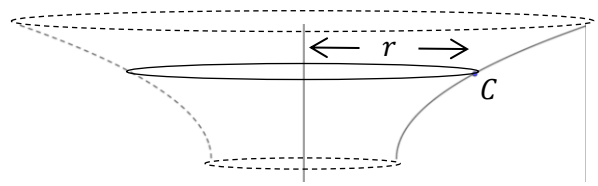
End of PIC

Practice Problems: Find the Surface areas of the following surfaces of revolution.

Center of Mass

In PIC # XX and the previous PIC, we made reference to a theorem by Pappus. One general theorem says that if a region is revolved about a line not passing through the region, then the volume of the solid of revolution is equal to the area of the region times the circumference of the circle formed by revolving the center of mass (centroid) of this region about the line. Another theorem of Pappus states that the surface area of a surface formed by revolving a curve about a line is equal to the length of the curve times the circumference of the circle formed by revolving the centroid of the curve about the line.





$Volume = 2\pi R \cdot (area\ of\ region)$ $\qquad$ $Surface\ area = 2\pi r \cdot (length\ of\ curve)$

Theorems of Pappus: C represents the centroids of the region and curve, respectively.

This played out in the two PIC's as the centroids of a disk and its boundary circle are both the center of the circle.

Disks and circles are pretty straightforward, but what about finding the center of mass of a general region. From a physical point of view, this region can represent a thin plate whose density could vary. To keep the problem geometric (and simpler), we will assume that the density is constantly 1 so that mass really is area. In this case, the center of mass is what we have been calling the centroid of the region.

Everyone probably has heard the term "center of mass." But what does this mean? Let's start this with something that everyone is familiar with: an average of grades. If we have five grades: 96, 87, 92, 83, 90, then all students know that the average, denote $\bar{X}$ (read X bar) is given by

$$\bar{X} = \frac{96 + 87 + 92 + 83 + 90}{5} = 89.6$$

What does this average mean? This means that this student's overall performance is equivalent to the performance of a student who earned 89.6 on each of the five tests. It does not take into account fluctuations among the grades, but gives an overall measure. To put this in a more physical light, if we considered a system of points each with mass 1 at these locations (96, 87, 92, 83, 90) on a number line, then collectively, this system behaves the same as a mass of 5 located at the point 89.6. It doesn't take into account the relationships among the various grades; as much as in astronomy where there is a tendency to "shrink" the mass of a planet to its center of mass, with disregard to the dynamics in that planet.

In general, if we placed a point mass of 1 at each of the positions $x_1, x_2, x_3, \ldots, x_n$, then the center of mass $\bar{x}$, should satisfy the relationship

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

$$\bar{x} = \frac{x_1 \cdot 1 + x_2 \cdot 1 + \cdots + x_n \cdot 1}{n}$$

$$center\ of\ mass = \frac{sum\ of\ position \cdot mass}{total\ mass}$$

To help with our discussions on this, we will give a name to the numerator. People have christened this the (first) moment of the system (about the origin). To give it a simple way to remember it, we write

$$moment = position \cdot mass$$
$$center\ of\ mass = \frac{total\ moment}{mass}$$

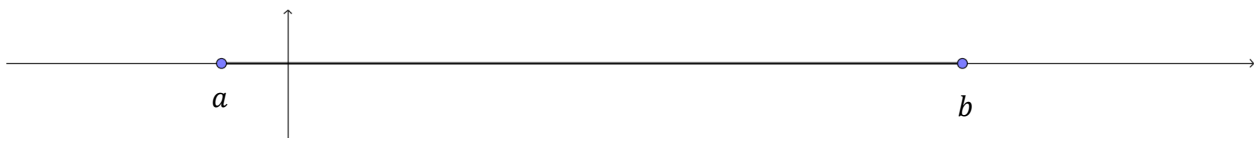
Later we will have use for the second moment about the origin, namely

$$2^{nd}\ moment = (position)^2 \cdot mass$$

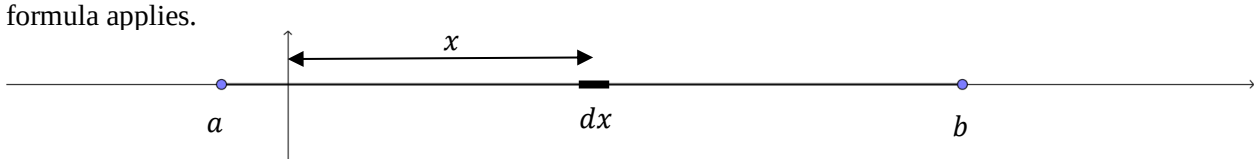
In general, one could look at the nth moment as

$$n^{th}\ moment = (position)^n \cdot mass$$

Let's apply this reasoning and our "integration" strategy to an infinite number of points, say an interval $[a, b]$.



In this case, our simple formula doesn't hold as the position varies throughout the interval. Thus we apply our integration strategy. That is, we divide the interval into infinitely small intervals where the simple formula applies.



The simple formula for moment applies to this small subinterval, so we have

$$moment\ of\ subinterval = position \cdot mass = x\ dx$$

Integrating these individual moments gives us that the total moment of the interval $[a, b]$ is

$$total\ moment = \int_{x=a}^{b} x\ dx = \left(\frac{x^2}{2}\right)\Bigg|_{x=a}^{b} = \frac{b^2 - a^2}{2}$$

Thus, we would have

$$center\ of\ mass = \frac{total\ moment}{mass} = \frac{b^2 - a^2}{2(b - a)} = \frac{a + b}{2}$$

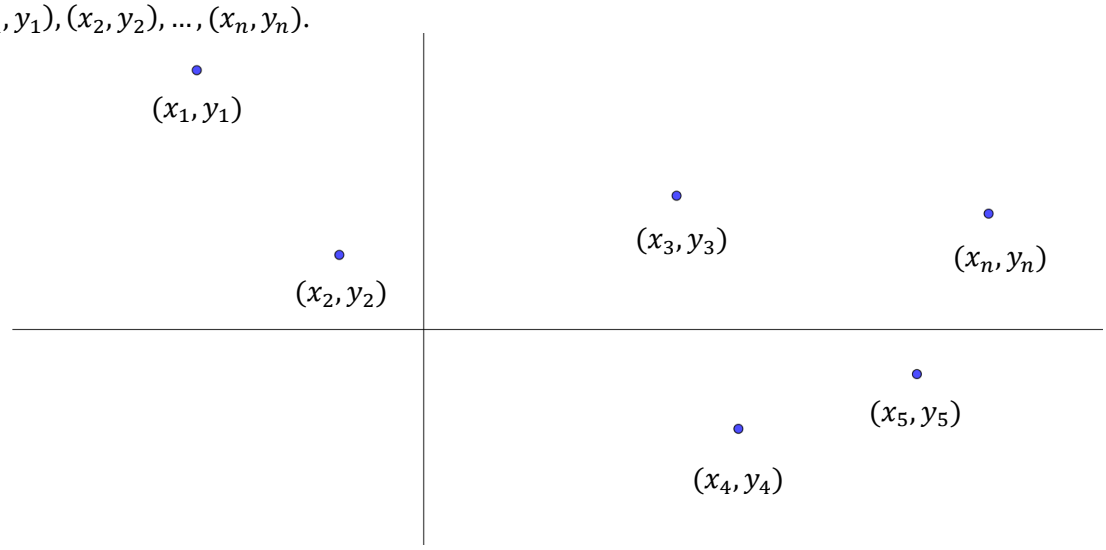
This agrees with your intuitive notion that the center of mass of the interval should be located at the midpoint of the interval.

Problem in Context: Find the center of mass of the set $A = [a_1, b_1] \cup [a_2, b_2],\ b_1 < a_2$. Is the center of mass of the set A equal to the averages of the midpoints of the two intervals? Does this make sense?

End of PIC

We did not want to give it away while you were doing the previous problem, but you will actually get what is known as a weighted average; that is, the center of mass of the two midpoints where the mass of each interval is condensed at its midpoint.

What if we ramp this up to two-dimensions. Suppose we have a number of points of mass 1 located at $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$.



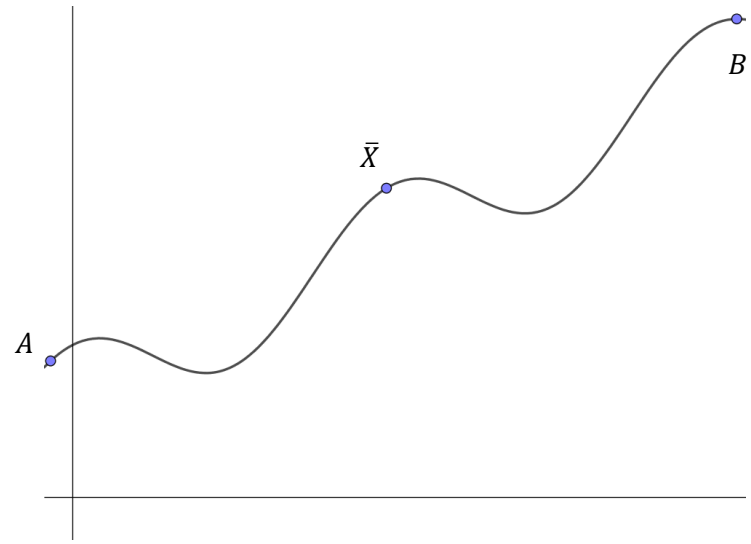
By the same reasoning as before, the center of mass $\bar{X}$ should have the property that this system as a whole should behave the same as if the entire mass n was located at $\bar{X}$. In other words, we should have

$$n \cdot \bar{X} = (x_1, y_2) \cdot 1 + (x_2, y_2) + \cdots + 1 \cdot (x_n, y_n)$$

This leads to the formulation

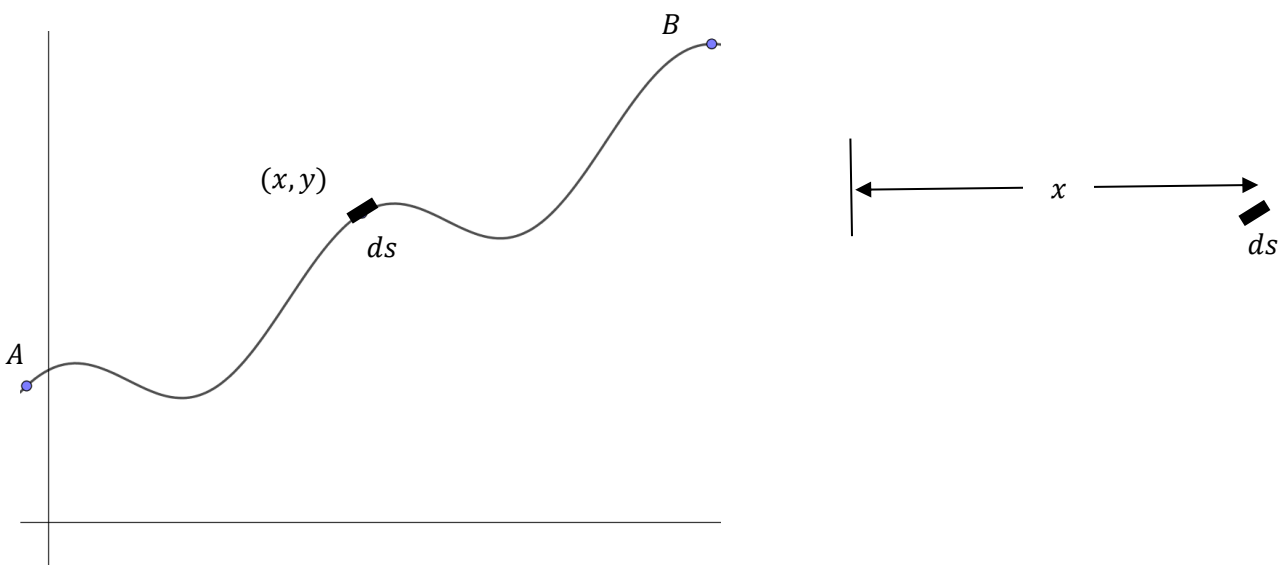
$$\bar{X} = \left(\frac{x_1 + x_2 + \cdots + x_n}{n}, \frac{y_1 + y_2 + \cdots + y_n}{n}\right) = (\bar{x}, \bar{y})$$

The thing to take away from this is that even though the center of mass, has two coordinates, we treat each coordinate separately and they are the averages of the x coordinates and y coordinates, respectively. Also, each coordinate still fits into the formula $\frac{total\ moment}{mass}$. Let's utilize this same idea to a curve of uniform density 1.



Using analogous reasoning, the center of mass of this curve will be given by $\bar{X} = (\bar{x}, \bar{y})$ where each coordinate fits into our scheme $\frac{total\ moment}{mass}$. To this end, we will look at the mass of the curve. Since the density is constantly 1, the mass is really the arc length and is given by $\int_{X=A}^{B} ds$.

To address $\bar{x}$, we will compute of the curve about the line $x = 0$. Dividing our curve into infinitely small pieces, we can use our simple formulation.



$$moment\ of\ segment\ (about\ x = 0) = location \cdot mass = x\ ds$$

$$total\ moment\ (about\ x = 0) = \int_{X=A}^{B} x\ ds$$

Problem in Context: Provide the analogous reasoning (and diagrams) to show that the total moment about the line $y = 0$ (x-axis) is given by

$$\int_{X=A}^{B} y\, ds$$

End of PIC

Putting all of this together, we have

$$\bar{x} = \frac{\int_{X=A}^{B} x\, ds}{\int_{X=A}^{B} ds} \qquad \bar{y} = \frac{\int_{X=A}^{B} y\, ds}{\int_{X=A}^{B} ds}$$

Again, we urge you to go through this process rather than trying to memorize these formulas.

Problem in Context: As a test of our reasoning, let's apply the above ideas to find the center of mass of a line segment. If this reasoning is correct, then we should get the midpoint of the line segment.

a. Find the equation of the line segment joining (x_1, y_1) and (x_2, y_2). To avoid dealing with absolute values, let's assume $x_1 < x_2$.

b. Use the above formulas to determine the coordinates of the center of mass of this line segment. Is it the midpoint of the line segment?

End of PIC

Problem in Context: By symmetry, the center of mass of a circle should be the center of the circle, but what about a semicircle? With this in mind, consider the upper semicircle of radius r centered at the origin. This can be described by the equations

$$x = r\cos\theta \qquad y = r\sin\theta\ , \qquad 0 \le \theta \le \pi$$

a. Verify that $\bar{x} = 0$ by following our process as above.

b. Compute $\bar{y}$.

End of PIC

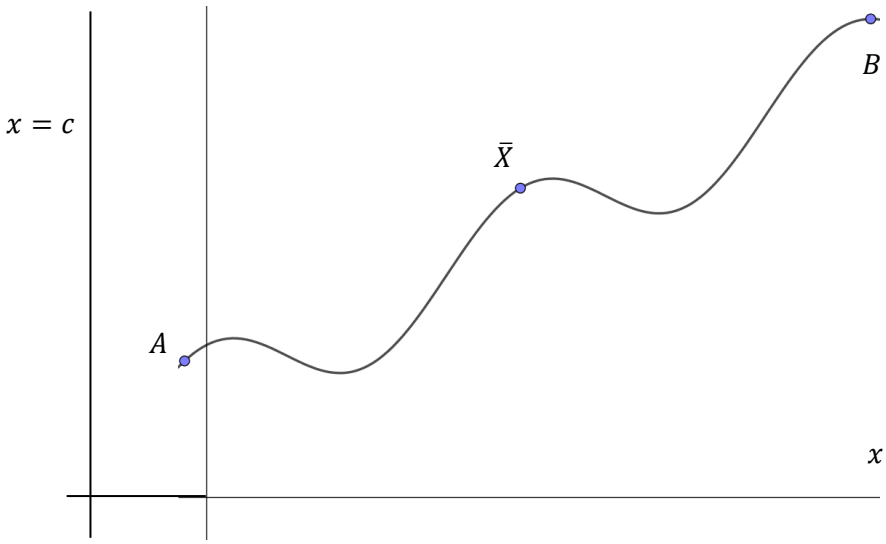
Notice that the above examples show that the center of mass of a curve need not be a point on the curve.

We are now in a position to prove Pappus' Theorem on surface area.

Problem in Context: Pappus Theorem for surface area.

Consider the following curve with center of mass $\bar{X} = (\bar{x}, \bar{y})$ revolved about a vertical line $x = c$ to the left of the curve.

y



Show that the surface area of the surface generated by revolving this curve about the line is given by

$$2\pi(\bar{x} - c) \int_{X=A}^{B} ds = (circumference\ of\ circle\ of\ radius\ \bar{x} - c) \cdot length\ of\ curve$$

End of PIC

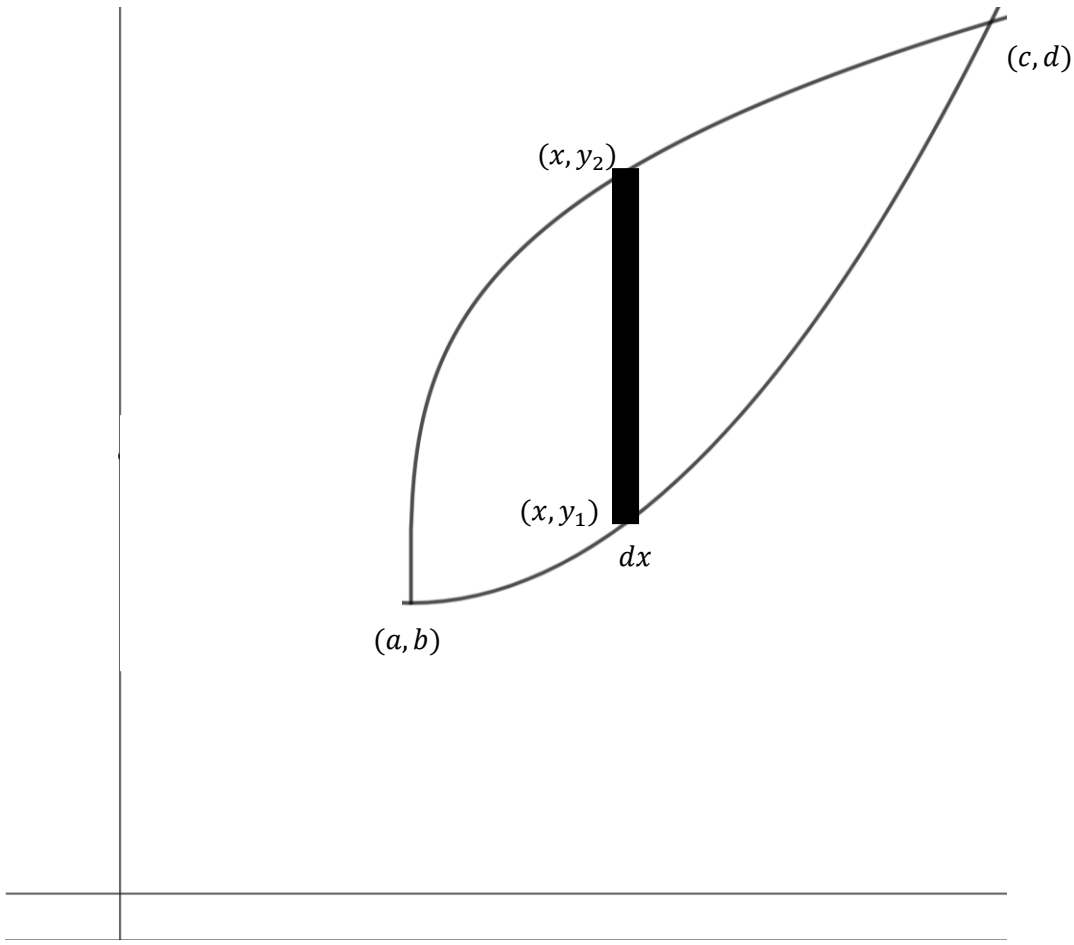
Practice Problems:

This is where work from 12/27/2022 starts

To obtain Pappus' theorem for volumes we need to expand this idea to the center of mass of a two-dimensional region. Again, the center of mass $\bar{X}$ can be split into two coordinates $(\bar{x}, \bar{y})$ which represent the averages of the x and y coordinates of all of the points in the region.
With this in mind, consider a region R in the xy plane which can be thought of as two-dimensional plate with constant density 1. We will draw this region shortly.

Again, we will utilize our strategy of dividing our region into pieces where a simple formula can be applied, and we will apply our general idea that the centroid is given by the formula $\frac{sum\ of\ location \cdot mass}{total\ mass}$. The mass is just the area, so we will focus on the numerator.

Specifically, if we want to find $\bar{x}$, we will draw a generic vertical box and determine its moment about the line $x = 0$.



The moment of this box is straightforward as every point in it is the same distance (x) from the y axis. Formally, we have

$$\begin{aligned} moment\ of\ box\ (about\ x = 0) &= (position\ of\ box) \cdot (mass\ of\ box) \\ &= x \cdot (y_2 - y_1)dx \end{aligned}$$

Integrating these moments, we get that the total moment of the region is given by

$$total\ moment = \int_{x=a}^{c} x(y_2 - y_1)dx$$

Thus, with this set up, we would have

$$\bar{x} = \frac{moment\ of\ R}{mass\ of\ R} = \frac{\int_{x=a}^{c} x(y_2 - y_1)dx}{\int_{x=a}^{c} (y_2 - y_1)dx}$$

As before, you could memorize the formulas, but it is much better to remember simple concepts such as $moment = mass \cdot position$ and apply these to a situation where things are constant. For example, we can apply the same reasoning to a horizontal box as below to derive the formula for $\bar{y}$. This is what integration is all about

Problem in Context: Use a horizontal box in the above diagram (appropriately labeled) to show that $\bar{y}$ for the region R is given by

$$\bar{y} = \frac{\int_{y=b}^{d} y(x_2 - x_1)dy}{\int_{y=b}^{d} (x_2 - x_1)dy}$$

End of PIC

Practice Problems computing centers of mass.

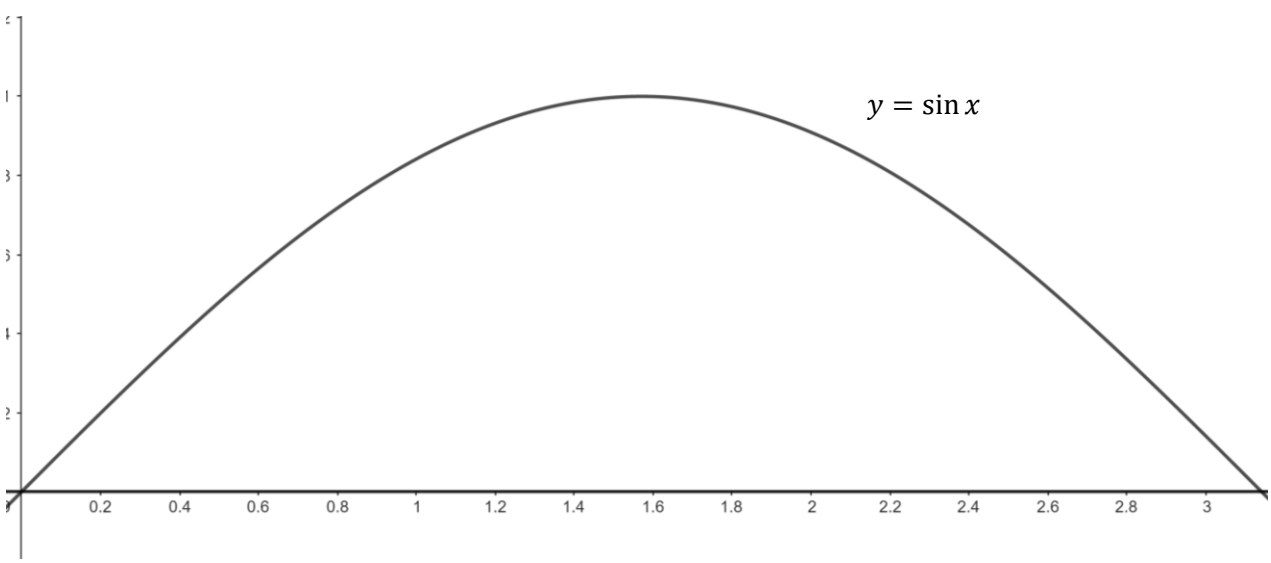
Problem in Context:

a. In everything we did so far, we assumed that the region had a constant density and we said the density was 1. Would it have made a difference if we said the density was given by some other constant δ? Explain.

b. Suppose the density δ was not constant. How would that alter the formulas for the center of mass?

End of PIC

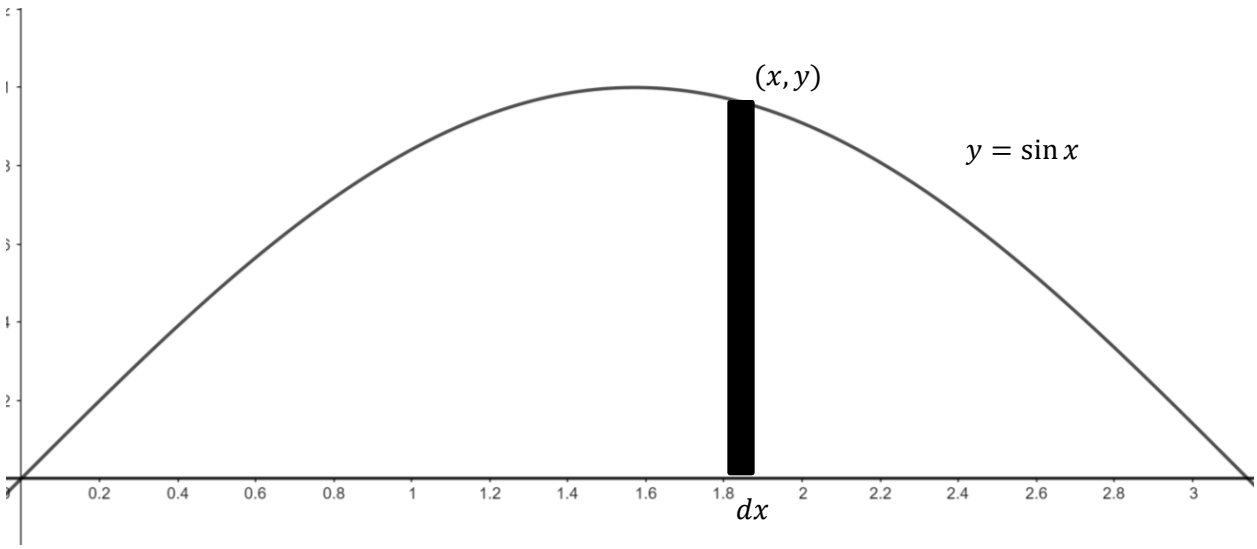
Again, you can memorize formulas, but understanding them helps you to recreate them should you forget or more appropriately, adapt the ideas should the occasion necessitate that. For example, consider the region below bounded by $y = \sin x$ and $y = 0$, $0 \leq x \leq \pi$



Assuming the density is constant, it is straightforward to see that $\bar{x} = \frac{\pi}{2}$. But what about $\bar{y}$? If we utilize the formula your derived above, we would have

$$\bar{y} = \frac{\int_{y=b}^{d} y(x_2 - x_1)dy}{\int_{y=b}^{d} (x_2 - x_1)dy}$$

Getting everything in terms of y would be problematic (but doable). It would be nice to keep everything in terms of x which would entail utilizing a vertical box. For the area in the denominator, this is not a problem and in fact is probably how you would have computed the area to begin with. For the numerator, let's put in our vertical box and see if we can compute its moment about the line $x = 0$.



The problem with computing the moment (about $x = 0$) of this vertical box is that our simple formula $mass \cdot position$ won't work as the position varies throughout the box. But wait a minute! We know what $\bar{y}$ for this box is, namely $\frac{y}{2}$. We also know what area of this box is, namely ydx. Utilizing that

$$\frac{y}{2} = \bar{y}_{box} = \frac{moment\ of\ box}{area\ of\ box}$$

so that

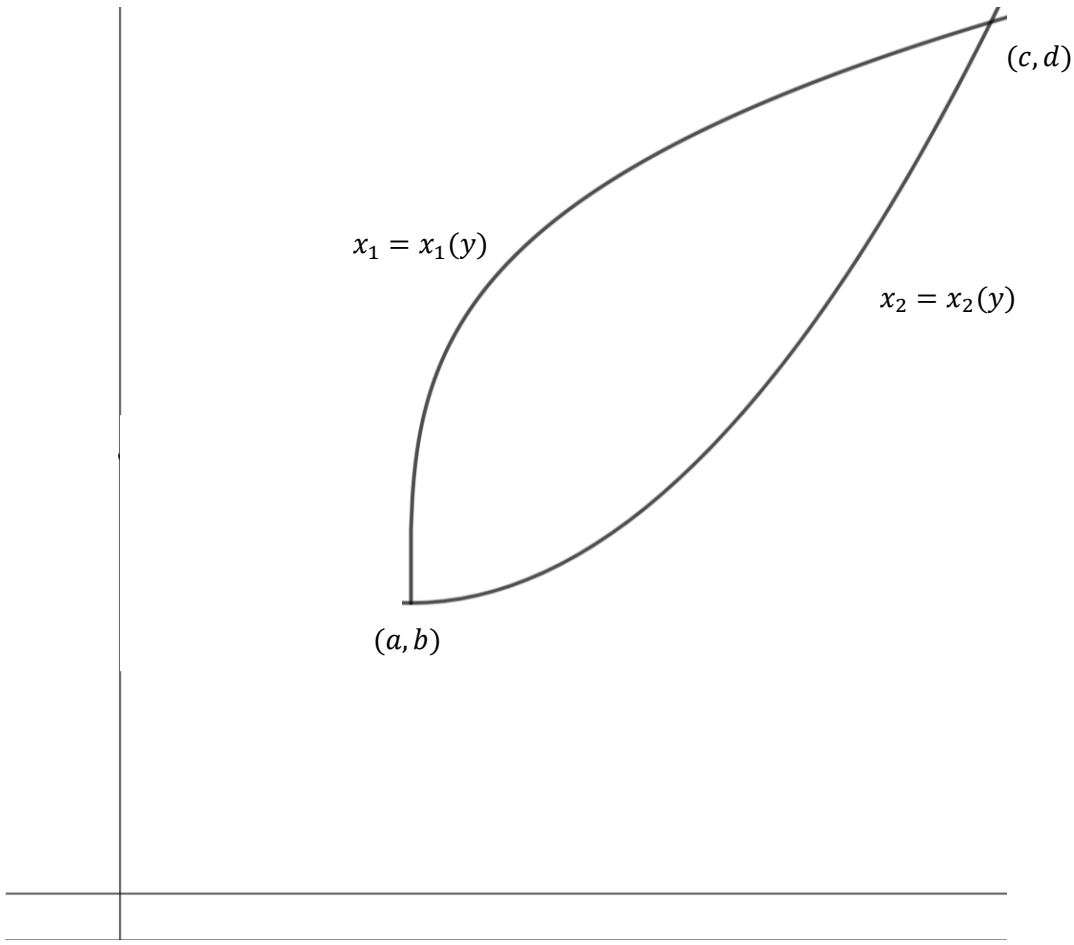
$$moment\ of\ box = \left(\frac{y}{2}\right)(ydx) = \frac{y^2}{2}dx$$

Problem in Context: Integrate this and determine the y coordinate of the center of mass of this region.

End of PIC

Problem in Context: For the following region, utilize the same sort of reasoning to show that the moment about the line $x = 0$ is given by

$$\int_{y=c}^{d} \frac{x_2^2 - x_1^2}{2}\, dy$$



End of PIC

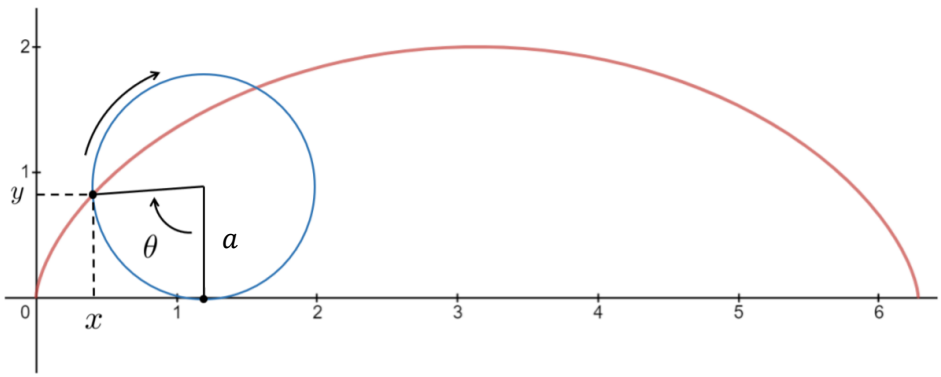
Practice Problems determining centers of mass with alternate formulation.

Problem in Context:

In PIC # XX, you were asked to find the volume of the solid generated by revolving one arch of a cycloid given by

$$x = a(\theta - \sin\theta) \qquad y = a(1 - \cos\theta)$$

around the x axis.

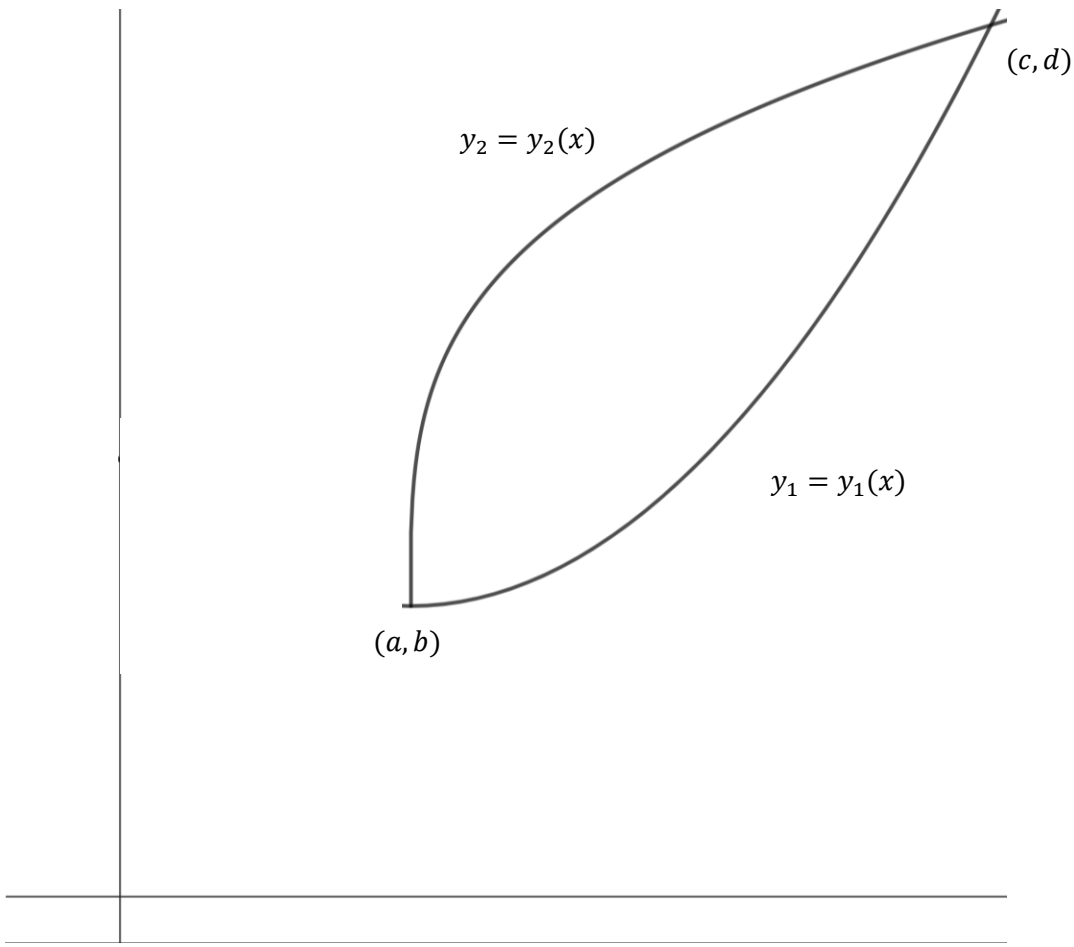


The answer to that problem was $5\pi^2 a^3$. We still haven't proven Pappus' Theorem yet, but find $\bar{y}$ of this planar region and see if Pappus' Theorem applied to this will yield the same answer.

End of PIC

Now that we have a handle on the center of mass of a two-dimensional region, we can actually make pretty quick work of Pappus' Theorem for determining the volume of a solid of revolution.

Consider the following region R lying completely to the right of the line $x = 0$.



Problem in Context: Utilize cylindrical shells to show that the volume of the solid generated by revolving this region about the line $x = 0$ is given by

$$2\pi\bar{x}_R \cdot (area\ of\ R)$$

This is Pappus' Theorem for volumes.

End of PIC

Practice Problems with Pappus: Utilizing the centroids from before?

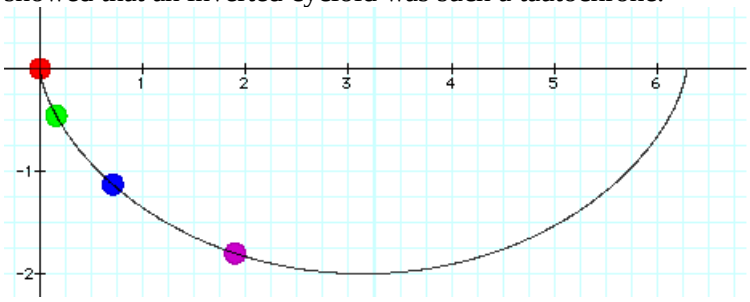
Other Applications

The Tautochrone

You may have noticed that we keep bringing up the cycloid in a number of problems involving areas, arc lengths, volumes, and centers of mass. As we said, this curve has fascinated mathematicians for a long time and many of these elegant results were cleverly obtained before the invention of calculus. We will now see how the cycloid was utilized to address a more practical problem.

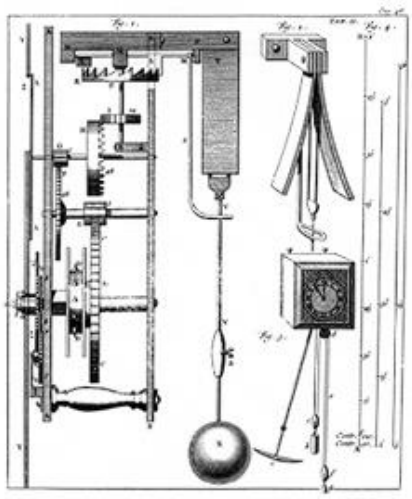
In the 1600's there was race among naval "superpowers" (Britain, France, Spain, Holland, etc.) to develop a way of measuring longitude at sea. Measuring latitude was relatively easy and could be accomplished by measuring the angle of elevation of the sun or stars. Before longitude could be measured accurately, ships would sail until they reached the correct latitude of a destination and then sailed east or west until they hit the destination. As such, there were monetary prizes awarded for anyone who develop an accurate way of measuring longitude at sea. For example, the Longitude Act, issued in Britain in 1714 offered a prize of up to £20,000 (about £5.1 million or $6.1 million in 2022 currency) for anyone who could measure longitude to an accuracy of half a degree.

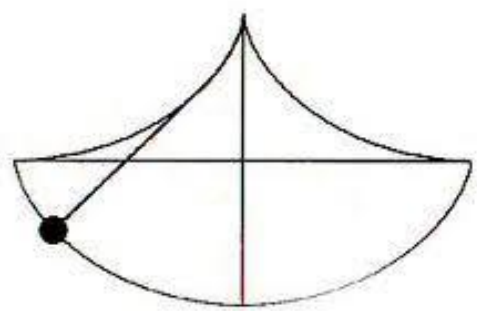
Since longitude is measured by "time" zones, then it became necessary to develop an accurate way of measuring time at sea. A regular pendulum clock, invented by the Dutch mathematician, scientist, and inventor Christiaan Huygens (1629-1695), utilized the fact that the oscillation of a pendulum is regular as long as it maintains the same amount of "swing" per oscillation. This made for an accurate timepiece on land, but it was not accurate at sea where a moving deck would make the pendulum swing at different angles and thus not have a constant period. To remedy this, Huygens developed a pendulum that would follow a tautochrone (a "same time" curve where a pendulum, exclusively under the influence of gravity, following that path would take the same amount of time to reach the bottom, no matter where it started on the curve). Huygens showed that an inverted cycloid was such a tautochrone.



A pendulum following the path of an inverted cycloid will take the same amount of time to reach the bottom no matter where it starts. Image from http://www.mathwords.com/t/t_assets/tautochrone.gif

Huygens then developed a pendulum clock which would do this. He published his work in 1673 in his book ***Horologium Oscillatorium: sive de motu pendulorum ad horologia aptato demostrationes geometricae*** (*The Pendulum Clock: or geometrical demonstrations concerning the motion of pendula as applied to clocks*). The design from that work [https://en.wikipedia.org/wiki/Horologium_Oscillatorium] is on the left. To get the pendulum to swing along a cycloidal path, Huygen's proved that if a flexible pendulum wraps around two flaps shaped like arches of a cycloid, then the bottom of the pendulum will trace a cycloid itself as seen on the left. This curve traced out is called the involute (of the cycloid) [http://rsefalicante.umh.es/LaboratorioPendulo/Pendulo2.htm].





In practice, the clock did not work any more accurately than a regular pendulum clock as it assumed the only force involved was gravity, whereas a clock at sea was subject to many more forces which could not be ignored. Subsequently, the Englishman John Harrison (1693-1776), a Yorkshire carpenter, invented a chronometer which ran on springs and proved to be very accurate.

Even still, many mathematicians cited Huygens work on this as being very elegant mathematically. Furthermore, Huygens developed his mathematics without calculus as it hadn't been invented (discovered?) yet. We will not prove that the involute of a cycloid is a cycloid, but we will utilize calculus to prove Huygens' claim that the cycloid is, in fact, a tautochrone.

To start, recall that the speed at which the pendulum is traveling is given by $v = \frac{ds}{dt}$ where s is the arc length traveled and t is time. For simplicity, we will let the radius of the circle generating the cycloid be $\frac{1}{2}$.

Problem in Context: Show that the total time it takes for the pendulum to move along the (inverted) cycloid

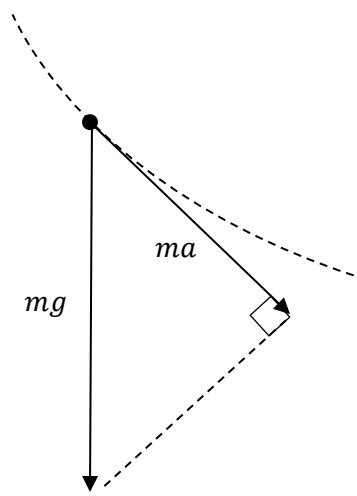
$$x = \frac{1}{2}(\theta - \sin\theta)$$
$$y = \frac{1}{2}(1 - \cos\theta)$$
$$\theta_0 \le \theta \le \pi$$

from a starting position $\theta = \theta_0$ to the bottom of the cycloid is given by

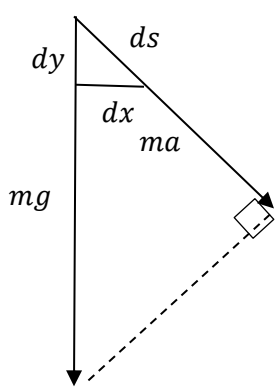
$$\int_{\theta=\theta_0}^{\pi} \frac{ds}{v} = \int_{\theta=\theta_0}^{\pi} \frac{\sin\left(\frac{\theta}{2}\right)}{v}\, d\theta$$

End of PIC.

At this point we've hit an impasse, as the speed of the pendulum v is not a constant. Gravity will cause the pendulum to speed up as it swings downward. We will assume that the pendulum does not swing too wide, not too fast, and ignore air resistance. Thus, the only force we will consider is due to gravity and we will denote that by mg, where m is the mass of the pendulum and g is the acceleration due to gravity. Below is a diagram of a pendulum following the path of a cycloid with the forces at work.



The force due to gravity is always directed downward, so only a portion of it moves the pendulum along the curve. This tangential force has a magnitude ma where $a = \frac{dv}{dt}$ is the (tangential) acceleration and is obtained by projecting the gravitational force onto the tangent line to the curve. If we draw a differential triangle, we have the following similar triangles.



Problem in Context:

a. Use the fact that the two triangles are similar to show that $gdy = ads$ and use the fact that $a = \frac{dv}{dt}$ to conclude that

$$gdy = v\,dv$$

b. Integrate both sides from $\theta = \theta_0$ to θ to show that if the pendulum starts from rest at $\big(x(\theta_0), y(\theta_0)\big)$, then the velocity at any point $(x(\theta), y(\theta))$ is given by

$$v = \sqrt{2g\big(y(\theta) - y(\theta_0)\big)}$$

End of PIC

Substituting this formula for v into our integral for the total time traveled by the pendulum, we get

$$\int_{\theta=\theta_0}^{\pi} \frac{\sin\left(\frac{\theta}{2}\right)}{\sqrt{2g\big(y(\theta) - y(\theta_0)\big)}}\, d\theta = \frac{1}{\sqrt{2g}} \int_{\theta=\theta_0}^{\pi} \frac{\sin\left(\frac{\theta}{2}\right)}{\sqrt{y(\theta) - y(\theta_0)}}\, d\theta$$

Before we proceed any further, notice that if we utilize the appropriate trigonometric identity, we get

$$y(\theta) = \frac{1}{2}(1 - \cos\theta) = \sin^2\left(\frac{\theta}{2}\right)$$

$$y(\theta_0) = \frac{1}{2}(1 - \cos\theta_0) = \sin^2\left(\frac{\theta_0}{2}\right)$$

so our formula for the total time traveled by the pendulum from $\big(x(\theta_0), y(\theta_0)\big)$ to $\left(\frac{\pi}{2}, 1\right)$ becomes

$$\frac{1}{\sqrt{2g}} \int_{\theta=\theta_0}^{\pi} \frac{\sin\left(\frac{\theta}{2}\right)}{\sqrt{\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta_0}{2}\right)}}\, d\theta = \frac{1}{\sqrt{2g}} \int_{\theta=\theta_0}^{\pi} \frac{\sin\left(\frac{\theta}{2}\right)}{\sqrt{\left(1 - \cos^2\left(\frac{\theta}{2}\right)\right) - \left(1 - \cos^2\left(\frac{\theta_0}{2}\right)\right)}}\, d\theta$$
$$= \frac{1}{\sqrt{2g}} \int_{\theta=\theta_0}^{\pi} \frac{\sin\left(\frac{\theta}{2}\right)}{\sqrt{\cos^2\left(\frac{\theta_0}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)}}\, d\theta$$

As formidable as this integral may look, bear in mind that θ_0 is a constant. Also, this looks ripe for a substitution to make it easier of the eyes. Specifically, what if we let $u = \cos\left(\frac{\theta}{2}\right)$ and $u_0 = \cos\left(\frac{\theta_0}{2}\right)$

Problem in Context: Perform this substitution and show that the time traveled by the pendulum from $\big(x(\theta_0), y(\theta_0)\big)$ to $\left(\frac{\pi}{2}, 1\right)$ is given by $\frac{\pi}{\sqrt{2g}}$ which is independent of the starting point $\big(x(\theta_0), y(\theta_0)\big)$. Thus, the cycloid really is a tautochrone.

End of PIC.

Work and Kinetic Energy

If you go back to our proof that the cycloid is a tautochrone, you will notice that we had the equation

$$gdy = v\,dv$$

If we multiply both sides by our mass m and integrate from $\theta = \theta_0$ to θ, we get

$$\int_{\theta=\theta_0}^{\theta} mgdy = \int_{\theta=\theta_0}^{\theta} mvdv$$

$$mg\big(y(\theta) - y(\theta_0)\big) = \frac{1}{2}mv^2(\theta) - \frac{1}{2}mv^2(\theta_0)$$

The left-hand side is called the work done by gravity moving an object from the height $y(\theta_0)$ to the height $y(\theta)$ and the right-hand side is called the change in kinetic energy from the beginning point $\big(x(\theta_0), y(\theta_0)\big)$ to the terminal point $\big(x(\theta), y(\theta)\big)$. So, this equation really says that

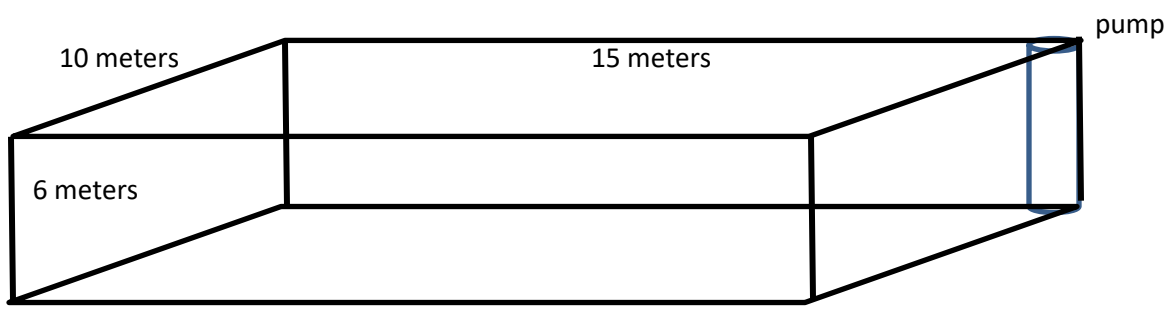
$$work = change\ in\ kinetic\ energy$$

At some point, you may have heard the expression that energy is the ability to do work. Kinematically, this is what we just showed. We will explore these two concepts in more detail to apply them to some natural questions that arise from things we did before.

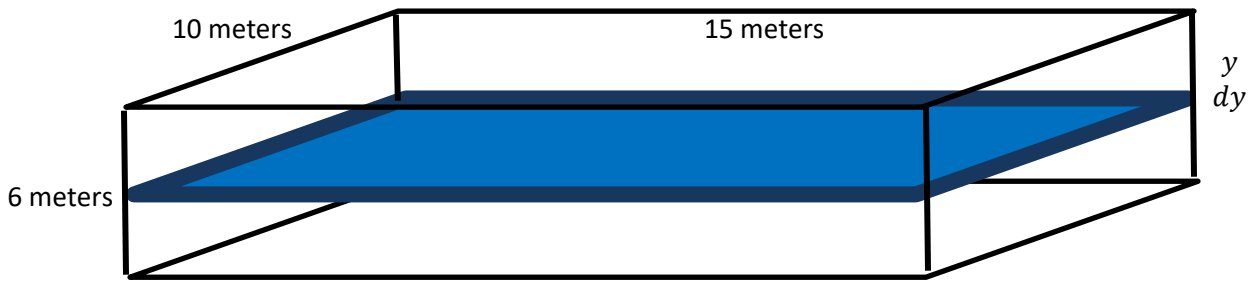
Work and Kinetic Energy:

Suppose we apply a force of 5 newtons to move an object 2 meters (in that direction), then we will have done $5 \cdot 2 = 10$ newton-meters (joules) of work. In general, if we apply a force with magnitude F newtons to move an object m meters in the direction of the force, then the amount of work done by that force is $W = F \cdot m$ newton-meters. As stated in the example above, $mg\big(y(\theta) - y(\theta_0)\big)$ was the amount of work done by gravity moving a mass from a height of $y(\theta_0)$ to a height of $y(\theta)$. Notice the x coordinate didn't matter as the direction of the force was along the y axis. But what if the force or distance is not constant?

Example: suppose we had the following inground diving pool filled with water weighing 9810 newtons/cubic meter. How long would it take a one horsepower pump at the top of the pool to empty the pool?



First things first. A one horsepower pump can do approximately 735.5 newton-meters of work per second, so we really need to find out how much work it takes to accomplish the task. If we try to use our formula $work = force \cdot distance$, then the force is pretty straightforward. It is the weight of the water. The difficulty comes from considering the distance. This varies anywhere from 0 to 6 meters. It takes very little work to pump out the water near the top where it takes much more work to pump the water that is at the bottom. This is where our integration strategy of dividing things up into pieces where our simple formula will apply comes in. Let's look at a thin slab of water below and look at the work it would take for the pump to lift that slab to the top of the pool.



Notice that we didn't put in the pump nor the tube going to the bottom of the pool. This was deliberate. You might be thinking, "Wait! If the tube reaches to the bottom of the pool, then isn't that slab of water ultimately being lifted 6 meters and not y?" Actually, if you think about it, the pump only has to lift it the last y meters as gravity will do the work of lifting it the first $6 - y$ meters. If there was no pump at all, then the water level in the tube would naturally rise to that level itself. Also, since we only have to overcome the weight of the water, which is a vertical force, then we only need to worry about the vertical distance y. With this in mind, our simple formula for work applies to lifting the slab.

$$\begin{gathered} work\ to\ lift\ slab = weight\ of\ slab \cdot distance\ to\ lift\ slab \\ = weight\ density\ of\ water \cdot volume\ of\ slab \cdot distance\ to\ lift\ slab \\ = \left(9810\,\frac{newtons}{meter^3}\right)(10 \cdot 15 \cdot dy\ meters^3)(y\ meters) \\ 1471500y\ dy\ newton\ meters \end{gathered}$$

Notice we put in the units to show that they do provide us with the correct units for work. To get the total amount of work done, we integrate these together.

$$Total\ work = \int_{y=0}^{6} 1471500y\ dy$$

Problem in Context:

a. Compute this integral to see how much work this would take and use the fact that a one horsepower pump can do 735 newton-meters of work per second to see how many hours it would take for the pump to empty the pool.

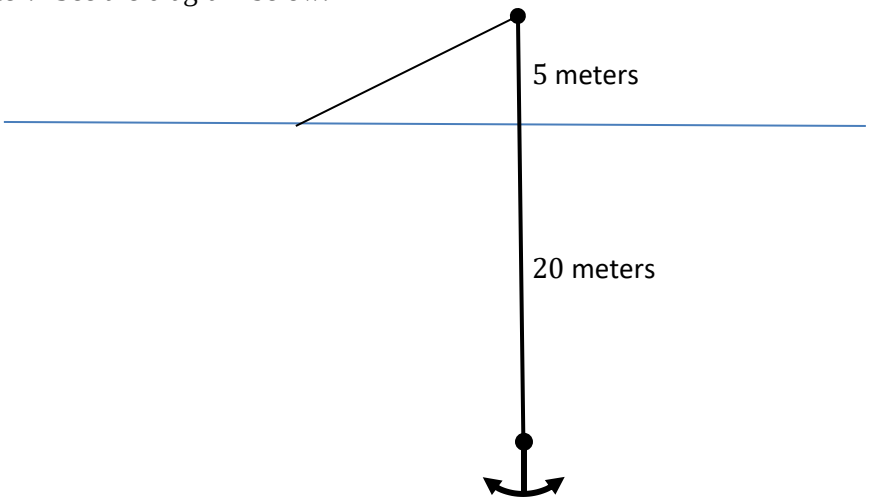
b. Take a guess: If the pool was only half full, would it take the pump half the time to empty it? Use calculus to check you guess.

End of PIC.

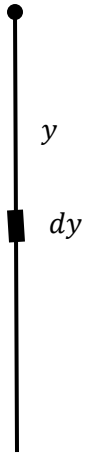
More problems pumping tanks of various shapes including one where the bottom is slanted.

In the previous example, an integral was required as the distance lifted was not constant. Some problems have the force varying.

Example: Suppose an anchor on a cruise ship weighs 200000 newtons (approximately 22.4 US tons) with a chain weighing 3000 newtons/meter (approximately 205 lb/ft) is being hauled up by a winch. How much work does it require to raise the anchor from a chain length of 20 meters to a point 5 meters above the surface of the water? See the diagram below.



We can actually approach this problem in two ways. One is reminiscent of what we did pumping the water out of the pool. We still want to apply our formula $work = force \cdot distance$. This actually works fine for the anchor as the force is the weight of the anchor and the distance is 25 meters. This would be the work required to raise the anchor alone. The chain is another matter. The lower parts of the chain need to be raised more than the upper parts. Here we will look at a small section of the chain and compute the work required to raise that. Specifically, consider the following diagram.



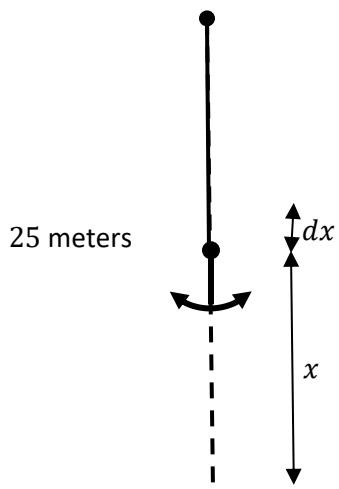
Problem in Context:

a. If dy represents an infinitely small length of chain, then compute the work involved to raise that length of chain y meters.

b. Integrate the above to find the work required to raise both the anchor and the chain. (Don't forget to add in the work to raise the anchor alone.

End of PIC.

There is another way to approach this problem which does not require examining the anchor separately. Instead of dividing the chain into small pieces where we can apply our formula for work, suppose we look at the work required to raise the anchor and chain from a distance x to a distance $x + dx$.



Since the force (combined weight of anchor and remaining chain) stays virtually constant over this interval, then we can still apply our simple formula.

$$work\ to\ raise\ the\ anchor\ and\ chain\ from\ x\ to\ x + dx = force \cdot distance$$
$$= \left(200000 + 3000(25 - x)\right) \cdot dx$$

Problem in Context: Integrate the above to compute the work done to raise the anchor and chain and compare your answer to the previous. (They should be the same.)

End of PIC

As we said the above could be done either way, but consider this following variation. Raise Bucket losing water.

- KE of disk and spin casting.
- Improper integral for length of upper semi circle. Improper integral for escape velocity